



AOS-CX 10.18.xxxx Diagnostics and Supportability Guide (6300, 6400 Switch series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

What's new in this release

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Identifying modular switch components

Applicable products

This document applies to the following products:

- HPE Aruba Networking 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A, R8S89A, R8S90A, R8S91A, R8S92A, S0E91A, SOX44A, S3L75A, S3L76A, S3L77A, S4P41A, S4P42A, S4P43A, S4P44A, S4P45A, S4P46A, S4P47A, S4P48A)
- HPE Aruba Networking 6400 Switch Series (R0X31A, R0X38B, R0X38C, R0X39B, R0X39C, R0X40B, R0X40C, R0X41A, R0X41C, R0X42A, R0X42C, R0X43A, R0X43C, R0X44A, R0X44C, R0X45A, R0X45C, R0X26A, R0X27A, JL741A, S0E48A, S0E48A #0D1, S1T83A, S1T83A #0D1)

What's new in this release

The following commands were added or modified in AOS-CX 10.18.



NOTE

For more information on new features introduced see the **[HPE Aruba Networking Airheads Broadcasting channel on Youtube](#)**.

Commands introduced or modified in 10.18

Command	Description
	The class command supports new traffic class types. The new ip-options class replaces the hypertext , ipv4-options and ipv6-options classes available in previous releases, and the new manageability class replaces the ssh and telnet classes available in previous releases.
	This command is updated to support the configuration of the new policy-blocked-flows flow monitor type.
	The output of this command is updated to include neighbor entry prefix statistics.
<u>troubleshoot aaa-mgmt</u>	Display detailed troubleshooting information for AAA management features.
<u>troubleshoot classifier</u>	Display detailed troubleshooting information for classifier policies.
<u>troubleshoot client-insight</u>	Display detailed troubleshooting information for client-insight features.
<u>troubleshoot event</u>	Display detailed troubleshooting information for event log messages.
<u>troubleshoot flow-telemetry</u>	Display detailed troubleshooting information for flow-telemetry features.
<u>troubleshoot l3</u>	Display detailed troubleshooting information for Layer-3 features, including ARP BGP, DHCP relay, DHCP server, OSPF, router discovery and VRRP.
<u>troubleshoot multicast</u>	Display detailed troubleshooting information for IGMP, MLD, PIM and PIM6 multicast features.
<u>troubleshoot port-access authentication</u>	Display detailed troubleshooting information for port-access authentication.
<u>troubleshoot proactive</u>	Enable and configure the proactive system troubleshooting mechanism.
<u>troubleshoot radius</u>	Display detailed troubleshooting information for RADIUS authentication issues.
<u>troubleshoot system</u>	Display detailed troubleshooting information for switch system-level components.
	Display detailed troubleshooting information for User-Based Tunnels (UBTs).
<u>troubleshoot tunnel</u>	Display detailed troubleshooting information for VXLAN configurations.

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: <example - text> <example - text> - example - text - example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: - For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. - For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Convention	Usage
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 6300 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on member 1.

On the HPE Aruba Networking 6400 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: member/power supply:
 - member: 1.
 - power supply: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: member/tray/fan:

- member: 1.
- tray: 1 to 4.
- fan: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: member/module:
 - member: 1.
 - member: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Debug logging

The debug logging framework provides an improved, customizable, and conditional logging framework with feature and entity based filtering options. Debug logging is a verbose, on-demand logging mechanism which customers and support can enable in order to obtain more information that will assist with troubleshooting.

Each debug logging event has both a Severity and a Module. Customers/support are required to enable a given Module in order to have those events logged. The log operation is not run when a Module is not enabled. All debug log events classified with a Severity of Error and above will always be logged. This ensures that both support and customers will be able to see these important events even when their respective debug log Module isn't enabled.



NOTE

Debug logging is disabled by default.

Note the following best practices for debugging on an HPE Aruba Networking 6300 Switch series:

- Enable debugging only for troubleshooting and in a controlled environment.
- Use debug commands with caution, as they can increase CPU utilization and affect switch stability.
- Monitor system resources while debugging to avoid crashes.
- Disable debugging once troubleshooting is complete to restore normal operations.
- Engage HPE Aruba Networking Support before enabling debug commands in production environments.

Subtopics

Debug logging commands

Debug logging commands

Subtopics

[Debug logging commands](#)

Debug logging commands

Select a command from the list in the left navigation menu.

Subtopics

[clear debug buffer](#)
[debug {all | <MODULE-NAME>}](#)
[debug db](#)
[debug destination](#)
[diag event-trap-disable](#)
[show debug](#)
[show debug buffer](#)
[show debug buffer vsf](#)
[show debug destination](#)

clear debug buffer

Syntax

```
clear debug buffer
```

Description

Clears all debug logs. Using the **show debug buffer** command will only display the logs generated after the **clear debug buffer** command.

Examples

Clearing all generated debug logs:

```
switch# show debug buffer
```

```
-----
```

```
-----
```

```
show debug buffer
```

```
-----
```

```

-----
2018-10-14:09:10:58.558710|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_CONFIG|No Port
cfg changes
2018-10-14:09:10:58.558737|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_EVENT|
lldpd_stats_run entered at time 8257199
2018-10-14:09:10:58.569317|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_CONFIG|No Port
cfg changes
2018-10-14:09:11:21.881907|hpe-sysmond|LOG_INFO|MSTR||SYSMON|SYSMON_CONFIG|
Sysmon poll interval changed to 32
switch# clear debug buffer
switch# show debug buffer
-----
show debug buffer
-----
2018-10-14:09:13:24.481407|hpe-sysmond|LOG_INFO|MSTR||SYSMON|SYSMON_CONFIG|
Sysmon poll interval changed to 51

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

debug {all | <MODULE-NAME>}

Syntax

```

debug {all | <MODULE-NAME>} [<SUBMODULE-NAME>] [severity
(emer|crit|alert|err|notice|warning|info|debug)] {port <PORT-NAME> |
vlan <VLAN-ID> | ip <IP-ADDRESS> | mac <MAC-ADDRESS> |
vrf <VRF-NAME> | instance <INSTANCE-ID>}

```

```
no debug {all | <MODULE-NAME>} [<SUBMODULE-NAME>] {port | vlan | ip | mac |  
vrf | instance}
```

Description

Enables debug logging for modules or submodules by name, with optional filtering by specific criteria.

The **no** form of this command disables debug logging.

Parameter	Description
all	Enables debug logging for all modules.
<MODULE-NAME>	Enables debug logging for a specific module. For a list of supported modules, enter the debug command followed by a space and a question mark (?).
<SUBMODULE-NAME>	Enables debug logging for a specific submodule. For a list of supported submodules, enter the debug <MODULE - NAME> command followed by a space and a question mark (?).
severity (emer crit alert err notice warning info debug)	Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is debug . Optional.
emer	Specifies storage of debug logs with a severity level of emergency only.
crit	Specifies storage of debug logs with severity level of critical and above.
alert	Specifies storage of debug logs with severity level of alert and above.
err	Specifies storage of debug logs with severity level of error and above.
notice	Specifies storage of debug logs with severity level of notice and above.
warning	Specifies storage of debug logs with severity level of warning and above.
info	Specifies storage of debug logs with severity level of info and above.

Parameter	Description
debug	Specifies storage of debug logs with severity level of debug (default).
port	Displays debug logs for the specified port, for example 1/1/1 .
vlan <VLAN-ID>	Displays debug logs for the specified VLAN. Provide a VLAN from 1 to 4094.
ip <IP-ADDRESS>	Displays debug logs for the specified IP Address.
mac <MAC-ADDRESS>	Displays debug logs for the specified MAC Address, for example A:B:C:D:E:F .
vrf <VRF-NAME>	Displays debug logs for the specified VRF.
instance <INSTANCE-ID>	Displays debug logs for the specified instance. Provide an instance ID from 1 to 255.

Examples

```
switch# debug all
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

debug db

Syntax

```
debug db {all | sub-module} [level <MINIMUM-SEVERITY>] [filter]
no debug db {all | sub-module} [level <MINIMUM-SEVERITY>] [filter]
```

Description

Enables or disables debug logging for a db module or submodules, with an option to filter by specific criteria.

The **no** form of this command disables debug logging for the db module or submodule.

Parameter	Description
all	Enables all submodules for the db log.
sub-module	Enables debug logging for supported submodules. Specify rx or tx debug logs.
filter	Specifies supported filters for the db log. Specify table , column , or client . Optional
severity (emer crit alert err notice warning info debug)	Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is debug . Optional.
emer	Specifies storage of debug logs with a severity level of emergency only.
crit	Specifies storage of debug logs with severity level of critical and above.
alert	Specifies storage of debug logs with severity level of alert and above.
err	Specifies storage of debug logs with severity level of error and above.
notice	Specifies storage of debug logs with severity level of notice and above.
warning	Specifies storage of debug logs with severity level of warning and above.
info	Specifies storage of debug logs with severity level of info and above.
debug	Specifies storage of debug logs with severity level of debug (default).

Usage

DBlog is a high performance, configuration, and state database server logging infrastructure where a user can log the transactions which are sent or received by clients to the configuration and state database server. It can be enabled through the CLI and REST, and also supports filters where a user can filter out logs on the basis of table, column, or client. It is helpful for debugging when the user wants to debug an issue with a particular client, table, or column combination. It is not enabled by default. A combination of filters can also be applied to filter out messages based on table, column, and client.

There are three submodules for the "db" module:

1. **all**: When All is enabled, no filters are applied to any of the debug logs, even if other submodules are configured with filters.
2. **tx**: If enabled, only the replies and notifications sent out for the initial and incremental updates are logged.
3. **rx**: If enabled, only the transactions sent to the configuration and state database server are logged.

The keyword **all** may be used to enable or disable debug logging for all sub-modules. Also a combination of filters can be used to filter the message types.

If the table or client filter is applied, then the messages belonging to this specific table or client will be logged. The column filter can also be applied to further filter messages on a table, providing a mechanism to filter messages on a column. The table and client filter can be used in combination or separately, but column can only be used in conjunction with table.

Examples

Configuring all submodules with severity **debug**:

```
switch# debug db all severity debug
```

Configuring the **tx** submodule with **table Interface** filter and severity **debug**:

```
switch# debug db tx table Interface severity debug
```

Configuring the **rx** submodule with **table Interface column statistics** filter and severity **debug**:

```
switch# debug db rx table Interface column statistics severity debug
```

Disabling the **rx** submodule:

```
switch# no debug db rx
```

Disabling the **tx** submodule **table Interface**:

```
switch# no debug db tx table Interface
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

debug destination

Syntax

```
debug destination {syslog | file | console | buffer} [severity (emer|crit|alert|err|notice|warning|info|debug)]  
no debug destination {syslog | file | console}
```

Description

Sets the destination for debug logs and the minimum severity level for each destination

The **no** form of this command unsets the destination for debug logs.

Parameter	Description
{syslog file console buffer}	Selects the destination to store debug logs. Required.
syslog	Specifies that the debug logs are stored in the syslog .
file	Specifies that debug logs are stored in file .
console	Specifies that debug logs are stored in console .
buffer	Specifies that debug logs are stored in buffer (default).
severity (emer crit alert err	Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is debug . Optional.

Parameter	Description
notice warning info debug)	
emer	Specifies storage of debug logs with a severity level of emergency only.
crit	Specifies storage of debug logs with severity level of critical and above.
alert	Specifies storage of debug logs with severity level of alert and above.
err	Specifies storage of debug logs with severity level of error and above.
notice	Specifies storage of debug logs with severity level of notice and above.
warning	Specifies storage of debug logs with severity level of warning and above.
info	Specifies storage of debug logs with severity level of info and above.
debug	Specifies storage of debug logs with severity level of debug (default).

Usage

Events that have a severity equal to or higher than the configured severity level are stored in the designated destination. The product defaults to **buffer** for destination and **debug** as a severity level.

Examples

```
switch# debug destination syslog severity alert
switch# debug destination console severity info
switch# debug destination file severity warning
switch# debug destination buffer severity err
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

diag event-trap-disable

Syntax

```
diag event-trap-disable <EVENT-IDS>
no diag event-trap-disable <EVENT-IDS>
```

Description

Prevents the specific event notifications from being sent as traps to the SNMP management stations.

The **no** form of this command disables this feature and reenables the notification of events to be sent as traps.

Parameter	Description
<EVENT-IDS>	Specify up to five comma separated Event IDs

Usage

To modify the traps to be disabled, such as changing the existing list of event IDs or removing an event ID from the blocked trap list, first completely remove previously configured disabled traps using the CLI command **no diag event-trap-disable <EVENT-ID>**. After that, reconfigure the new event IDs or set of event IDs for the changes to take effect.

The command to disable traps cannot be issued via network management systems such as SNMP, the REST API, HPE Aruba Networking Fabric Composer or HPE Aruba Networking Central. The configuration to disable event-traps is not persistent across reboots, and traps will be enabled again after the switch reboots.



NOTE

Do not disable default event trap IDs, as this can cause conflicts with a switch configuration with default traps enabled.

Examples

Disable and then reenables event traps 1223 and 1224.

```
switch# diagnostics
switch# diag event-trap-disable 1223,1224
switch# no diag event-trap-disable 1223,1224
```

Change the list of disabled event traps.

```
switch# diagnostics
switch# diag event-trap-disable 1223,1224
switch# no diag event-trap-disable 1223,1224
switch# diag event-trap-disable 1225,1226
```

Command History

Release	Modification
10.15.1010	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show debug

Syntax

```
show debug [vsx-peer]
```

Description

Displays the enabled debug types.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

```
switch# show debug
-----
-----
module sub_module severity vlan port      ip      mac
instance vrf
-----
-----
all      all          err      1      1/1/1    10.0.0.1  1a:2b:3c:4d:5e:6f
2        abcd
switch# show debug
-----
-----
module sub_module severity vlan port      ip      mac
instance vrf
-----
-----
all      all          err      1      1/1/1    10.0.0.1  1a:2b:3c:4d:5e:6f
2        default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show debug buffer

Syntax

```
show debug buffer [module <MODULE-NAME> | severity (emer|crit|alert|err|notice|warning|info|debug)]
```

Description

Displays debug logs stored in the specified debug buffer with optional filtering by module or severity.

Parameter	Description
<code><MODULE-NAME></code>	Filters debug logs displayed by the specified module name.
<code>severity (emer crit alert err notice warning info debug)</code>	Displays debug logs with a specified severity level. Defaults to debug . Optional.
<code>emer</code>	Displays debug logs with a severity level of emergency only.
<code>crit</code>	Displays debug logs with a severity level of critical and above.
<code>alert</code>	Displays debug logs with a severity level of alert and above.
<code>err</code>	Specifies storage of debug logs with severity level of error and above.
<code>notice</code>	Specifies storage of debug logs with severity level of notice and above.
<code>warning</code>	Displays debug logs with a severity level of warning and above.
<code>info</code>	Displays debug logs with a severity level of info and above.
<code>debug</code>	Displays debug logs with a severity level of debug (default).

Examples

```
switch# show debug buffer
-----
--
show debug buffer
-----
--
2017-03-06:06:51:15.089967|hpe-sysmond|SYSMON|SYSMON_CONFIG|LOG_INFO|Sysmon
poll interval changed to 20
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show debug buffer vsf

Syntax



NOTE

Applicable for 6300 switches only.

```
show debug buffer vsf [member <MEMBER-ID>] [{conductor | standby}]
```

Description

Displays VSF member debug logs stored in the debug buffer, with an option to filter by VSF member and role.

Parameter	Description
<MEMBER-ID>	Displays debug logs for the specified member - id. Optional. Range: 1 - 10.
conductor	Display debug logs for the VSF conductor.
standby	Display debug logs for the VSF standby.

Examples

Displaying VSF member debug logs with member-id 1:

```
switch# show debug buffer vsf member 1
-----
--
show debug buffer
-----
--
2020-12-14:07:53:17.217919|hpe-ledarbd|LOG_DEBUG|MMBR|2|LED|LED|
ledarbd_vsf_mbrs_check: Checking VSF_Member table
```

Displaying VSF member debug logs for member state conductor:

```
switch# show debug buffer vsf conductor
-----
--
show debug buffer
-----
--
2020-12-14:07:54:20.469024|hpe-ledarbd|LOG_DEBUG|CDTR|1|LED|LED|
ledarbd_pd_subsystems_check: Checking Subsystem table
```

Command History

Release	Modification
10.09	Updated parameter name for inclusive language.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

show debug destination

Syntax

```
show debug destination [vsx-peer]
```

Description

Displays the configured debug destination and severity.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

```
switch# show debug destination
-----
show debug destination
-----
CONSOLE:info
FILE:warning
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Log Rotation

Log rotation provides you with the ability to systematically rotate and archive any log files produced by the system. Log rotation reduces log space required on the switch. Log rotation rotates and compresses the log files based on size and/or period. Rotated log files are stored locally or transferred to a remote host using TFTP.

Optionally, notifications can be triggered if a log buffer percent full threshold is exceeded, giving you the opportunity to save the logs elsewhere before the buffers are rotated with the oldest data being overwritten.

Subtopics

Log file paths

About rotated log files

Changing the size of the log rotation file

Changing the time frequency for log rotation

Resetting the time frequency to daily

Identifying a remote host for receiving rotated log files

Remote transfer of rotated log files

Resetting the remote host for receiving rotated log files

Resetting the size of the log rotation file

Verifying the log rotation parameters

Log rotation troubleshooting

Log rotation commands

Log file paths

Logs stored in the following files are rotated:

- Audit logs are stored in file `/var/log/audit/audit.log`.
- Authentication logs are stored in file `/var/log/auth.log`.
- Event logs are stored in file `/var/log/event.log`.
- HTTPS server logs are stored in file `/var/log/nginx.log`.
- Security logs are stored in file `/var/log/security.log`.
- NTP logs are stored in file `/var/log/ntp.log`.
- Logs of bad login attempts are stored in `/var/log/btmp`.
- Logs of the last login sessions are stored in `/var/log/wtmp`.

About rotated log files

Rotated log files are compressed and stored locally in `/var/log/`, regardless of the remote host configuration. Rotated log files are stored with respective time extension to the granularity of hour in the format `file1-YYYYMMDDHH.gz` (for example, `messages-2015080715.gz`). Rotated log files are replaced when the number of old rotated log files exceeds three. The newly rotated log file replaces the oldest rotated log file.

TFTP, SFTP, or SCP are used to transfer rotated log files to a remote host. Only newly rotated log files are transferred to the remote host during the log rotation. Previously rotated log files are not re-transferred. After a log file is successfully transferred, it is removed from the switch.

Changing the size of the log rotation file

By default, the product rotates the log files when the maximum file size exceeds 100 MB. When the size of the log file exceeds the configured value, the rotation is triggered for that particular log file. Log rotation does not occur immediately after the maximum file size for the log file is reached since the cron job runs with an hourly periodicity.

```
logrotate maxsize <10-200 MB>
```

If you are planning to transfer the log rotation file by TFTP, set the log rotation file to no more than 32 MB.

Prerequisites

You must be in the configuration context:

```
switch(config) #
```

Changing the time frequency for log rotation

By default, the product rotates the log files daily. Enter the command at the configuration context in the CLI.

Prerequisites

You must be in the configuration context:

```
switch(config) #
```

Procedure

At the configuration context, enter:

```
logrotate period {daily | hourly | weekly | monthly }
```

daily: Rotates the log files daily. It is the default option.

hourly: Rotates the log files hourly.

weekly: Rotates the log files every week.

monthly: Rotates the log files every month.

Example command

```
switch(config)# logrotate period weekly
```

Resetting the time frequency to daily

Prerequisites

You must be in the configuration context:

```
switch(config)#
```

Procedure

At configuration context, enter the `no` form of the `logrotate period` command:

```
switch(config)# no logrotate period
```

Identifying a remote host for receiving rotated log files

You can send the rotated log files to a specified remote host Universal Resource Identifier (URI) by using the TFTP protocol. If no URI is specified, the rotated and compressed log files are stored locally in `/var/log/`. Only the TFTP protocol is supported for remote transfer, and the log rotation file cannot be more than 32 MB. Use the Linux TFTP command to transfer the file. Rotated log files are removed from the local path `/var/log/` when it is moved to TFTP server.

Prerequisites

You must be in the configuration context:

```
switch(config)#
```

Procedure

Provide the target IP address (IPv4 or IPv6) at the configuration context in the CLI:

```
switch(config)# logrotate target {tftp://A.B.C.D | tftp://X:X::X:X}
```

IPv4 Example

```
switch(config)# logrotate target tftp://192.168.1.132
```

IPv6 Example

```
switch(config)# logrotate target tftp://2001:db8:0:1::128
```

Remote transfer of rotated log files

Only the TFTP protocol is supported for remote transfer, and both IPv4 and IPv6 addresses are supported.

Only newly rotated log files are transferred to the remote host during the log rotation. Previously rotated log files are not transferred. After a file is successfully transferred, it is removed from the switch local path.

Packet level failures with TFTP are handled in the protocol itself. With each TFTP session failure, TFTP retries the file transfer three times. Retries have a timeout of five seconds.

Resetting the remote host for receiving rotated log files

Prerequisites

You must be in the configuration context:

```
switch(config) #
```

Procedure

At configuration context, enter the `no` form of the `logrotate target` command:

```
switch(config) # no logrotate target
```

Example:

```
switch(config) # logrotate target tftp://1.1.1.1
switch(config) # do show logrotate
Logrotate configurations :
Period           : daily
Maxsize          : 10MB
Target           : tftp://1.1.1.1
switch(config) # no logrotate target
switch(config) # do show logrotate
Logrotate configurations :
Period           : daily
Maxsize          : 10MB
switch(config) #
```

Resetting the size of the log rotation file

Prerequisites

You must be in the configuration context:

```
switch(config) #
```

Procedure

At configuration context, enter the `no` form of the `logrotate maxsize` command:

```
switch(config)# no logrotate maxsize
```

Verifying the log rotation parameters

At the command prompt, enter:

```
switch# show logrotate
```

Example output

```
switch# show logrotate
Logrotate configurations :
Period           : daily
Maxsize          : 10MB
switch#
```

Log rotation troubleshooting

Some common log file rotation troubleshooting items are as follows.

Subtopics

Log files not transferred remotely

Log rotation not occurring immediately after max file size

Log rotation not occurring regardless of period

Log files not transferred remotely

Symptom

Rotated log files are not transferred to a remote host.

Cause

- The remote host might not be reachable.
- The TFTP server on the remote host might not have sufficient privileges for file creation.

Action

1. Verify that the remote host is reachable.
2. Ensure that the TFTP server is configured with the required file creation permissions.
3. For example, on the TFTP-D-HPA server, change the configuration file in `/etc/default/tftpd-hpa` to include `-c` in `TFTP_OPTIONS`. (for example, `TFTP_OPTIONS="--secure -c`).

Log rotation not occurring immediately after max file size

Symptom

Log rotation does not occur immediately after the maximum file size for the log file is reached.

Cause

The log rotation checks the size of the file on the first minute of every hour. If the maximum file size is reached in the meantime, the log rotation does not occur until the next hourly check of the file size.

Action

Log rotation is working as designed. The log rotation feature is designed to check the file size on an hourly basis.

Log rotation not occurring regardless of period

Symptom

Log rotation is not happening regardless of the `period` value.

Cause

Log files are not rotated when they are empty files (the log file size is zero).

Action

Log rotation occurs when the log file size is greater than zero.

Log rotation commands

Subtopics

Log rotation commands

Select a command from the list in the left navigation menu.

Subtopics

logging threshold

logrotate maxsize

logrotate period

logrotate target

show logrotate

logging threshold

Syntax

```
logging threshold {audit-log | auth-log | commands-log | event-log |  
security-log  
                  | https-server-log} <THRESHOLD%>  
no logging threshold {audit-log | auth-log | commands-log | event-log |  
security-log  
                   | https-server-log} [<THRESHOLD%>]
```

Description

Selects the logging buffer notification threshold for the specified logging buffer. Whenever the logging buffer space consumption exceeds the selected threshold (percent of buffer capacity), a LOG_BUFFER_ALMOST_FULL event and SNMP RMON trap is triggered. This gives you the opportunity to save the logs elsewhere before the buffers are rotated with the oldest data being overwritten.

Also, a LOG_BUFFER_WRAPPED event and SNMP RMON trap is triggered if the logging buffer capacity is fully consumed and the log buffer is rotated with the oldest data being overwritten.

The **no** form of this command resets the logging buffer warning threshold to its default. All logs except **audit-log** have a default of 90 (percent) and **audit-log** has a default of 50 (percent).



NOTE

The largest REST payload that can be sent to RADIUS/TACACS servers is 1024 characters, and the maximum REST payload that can be sent to syslog servers is 3500 characters. Once this limit is exceeded, the log will display three dots (...) to indicate the the message has exceeded the character limit and is incomplete. .

Parameter	Description
audit-log	Selects the audit log.
auth-log	Selects the authentication log.
commands-log	Configure the logging threshold for commands log buffer
event-log	Selects the event log.
https-server-log	Selects the HTTPS server log.
security-log	Selects the security log.
<THRESHOLD%>	Selects the notification threshold as a percent that the selected logging buffer is full. Available percent values for all logs except audit-log: 15 30 50 70 90 100 Available percent values for audit-log: 50 100

Examples

Setting the audit log threshold:

```
switch(config)# logging threshold audit-log 100
```

Setting the authentication log threshold:

```
switch(config)# logging threshold auth-log 50
```

Setting the event log threshold:

```
switch(config)# logging threshold event-log 70
```

Setting the HTTPS server log threshold:

```
switch(config) # logging threshold https-server-log 50
```

Setting the security log threshold:

```
switch(config) # logging threshold security-log 70
```

Resetting the audit log threshold to its default of 50:

```
switch(config) # no logging threshold audit-log
```

Resetting the authentication log threshold to its default of 90:

```
switch(config) # no logging threshold auth-log
```

Resetting the event log threshold to its default of 90:

```
switch(config) # no logging threshold event-log
```

Resetting the HTTPS server log threshold to its default of 90:

```
switch(config) # no logging threshold https-server-log
```

Resetting the security log threshold to its default of 90:

```
switch(config) # no logging threshold security-log
```

Command History

Release	Modification
10.11	Introduced the commands - log parameter.
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logrotate maxsize

Syntax

```
logrotate maxsize <MAX-SIZE>
no logrotate maxsize
```

Description

Specifies the maximum allowed log file size.

A log file that exceeds either the **logrotate maxsize** or the **logrotate period** (whichever happens first), triggers rotation of the log file.

The **no** form of this command resets the size of the log file to the default (100 MB).

Parameter	Description
<MAX-SIZE>	Specifies the allowed size the log file can reach before it is compressed and stored locally or transferred to a remote host. Range: 10 to 200 MB. Default: 100 MB.

Examples

Setting the maximum log file size:

```
switch(config) # logrotate maxsize 24
```

Resetting the maximum log file size to its default of 100 MB:

```
switch(config) # no logrotate maxsize
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logrotate period

Syntax

```
logrotate period {daily | hourly | monthly | weekly}  
no logrotate period
```

Description

Sets the log file rotation time period. Defaults to daily.

A log file that exceeds either the **logrotate maxsize** or the **logrotate period** (whichever happens first), triggers rotation of the log file.

The **no** form of this command resets the log rotation period to the default of daily.

Parameter	Description
daily	Rotates log files on a daily basis (default) at 0:01.
hourly	Rotates log files every hour at the first second of the hour.
monthly	Rotates log files monthly on the first day of the month at 00:01.
weekly	Rotates log files once a week on Sunday at 00:01.

Examples

Setting a weekly period:

```
switch(config) # logrotate period weekly
```

Resetting the period to its default of daily:

```
switch(config) # no logrotate period
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logrotate target

Syntax

```
logrotate target <URI> [vrf <VRF_NAME>]  
no logrotate target [<URI>] [vrf <VRF_NAME>]
```

Description

Using TFTP, sends the rotated log files to a specified remote host identified by Universal Resource Identifier (URI).

The **no** form of this command resets the target to the default, which stores the rotated and compressed log files locally in **/var/log/**.

Command context

Parameter	Description
<URI>	Specifies the URI of the remote host. The default directory is local . tftp://{<IPv4_ADDR> IPv6_ADDR}<HOST>[/<DIRECTORY>]
<VRF_NAME>	Specifies the VRF name (Default: default).

Usage

- Rotated log files are compressed and stored locally in the path **/var/log/** regardless of the remote host configuration.

Examples

Setting an IPv4 target:

```
switch(config) # logrotate target tftp://192.168.1.132
```

Setting an IPv4 target with a directory:

```
switch(config) # logrotate target tftp://192.168.1.132/logrotate/
```

Setting an IPv4 target with the default VRF:

```
switch(config) # logrotate target tftp://192.168.1.132 vrf mgmt
```

Setting an IPv6 target with the default VRF:

```
switch(config) # logrotate target tftp://2001:db8:0:1::128 vrf default
```

Resetting the target to local:

```
switch(config) # no logrotate target
```

Command History

Release	Modification
10.09	Updated the syntax and examples.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show logrotate

Syntax

```
show logrotate [vsx-peer]
```

Description

Shows the log rotate configuration.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

```
switch# show logrotate
Logrotate configurations :
Period           : weekly
Maxsize          : 20MB
Target           : tftp://2001:db8:0:1::128 vrf mgmt
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Reboot reasons

The **show boot-history** command displays the following reboot reasons for the management module and line module:

Reboot reasons for management module

Parameter	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button.
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Console error	Console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
 NOTE VSF is not applicable for 6400 Switch series.	
VSF member renumbered	An user requested a renumber of a VSF member.
VSF switchover requested	An user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
 NOTE Not applicable for 6300 Switch series.	

Parameter	Description
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.

Uncontrolled reboots

- ops-switchd crashed
- ovsdb-server crashed
- Reset
 - Software thermal reset
 - Power on reset
 - Watchdog reset
 - CPU request reset
 - cold reset
 - Long press reset
 - Jumper reset
- switchd_agent crashed
- Power down due to hibernation and wake-up due to RTC trigger
- Power down due to hibernation and wake-up due to Button Mode trigger
- Hardware thermal reset

Reboot reasons for line module

- Line module reboot
- Line module crashed



NOTE

The line module is not applicable for 6300 Switch series.

Event Logs

Event logging logs events generated by daemons, processes, and plug-ins running within the switch software. The event logging framework captures the event logs in a system journal by updating the journal fields and meta data.

Subtopics

Showing and clearing events

Showing and clearing events

The `clear events` command is used to clear the event log of all events. The `show events` command is used to show all event logs generated by the switch since the last reboot. See the Switch system and hardware commands chapter of the Fundamentals Guide for information on these commands.

The time stamp for event log messages generated from the Service OS indicates when the event log messages were transferred to the event log after a switch boot and not when the issue occurred.

See the for information about accounting logs.

Client Filter

Event log client filter provides the ability to filter event logs for specific IP or MAC addresses. This enables the REST client to query event logs from the switch's journal while filtering for IP or MAC address values.

New keys for IP and MAC addresses are added to the switch's journal of pre-existing keys (for example ID and Category).

Subtopics

Log messages

Log messages

Log messages are generated to record various events occurring within the system. Each message contains a unique ID to represent an event and its attributes such as category, module ID or module role. The unique ID can then be used to filter for specific event types from a switch's journal.



NOTE

- REST API can filter for **all** events occurring on a specific IP or MAC address.
- REST API can filter for a **specific** event occurring on a specific IP or MAC address.
- REST API can filter for events based on a list of IPs and MACs.



CAUTION

Event log client filter does not support Go language daemons. Only C daemons are supported.

Network Configuration Validator

Network configuration validator (NCV) is a configuration troubleshooting tool that helps to detect switch configuration anomalies using a set of feature configuration templates. NCV helps to detect misconfigurations, identity incomplete or inter-dependent configurations, and mutually exclusive configurations.

NCV only displays possible warnings and does not recommend any configuration changes. NCV does not detect configuration issues based on network topologies.

Subtopics

[Showing and clearing events](#)

[Network configuration validation commands](#)

Showing and clearing events

The `clear events` command is used to clear the event log of all events. The `show events` command is used to show all event logs generated by the switch since the last reboot. See the Switch system and hardware commands chapter of the Fundamentals Guide for information on these commands.

The time stamp for event log messages generated from the Service OS indicates when the event log messages were transferred to the event log after a switch boot and not when the issue occurred.

See the for information about accounting logs.

Network configuration validation commands

Select a command from the list in the left navigation menu.

Subtopics

switch config-validator

switch config-validator

Syntax

```
switch config-validator [config <CONFIG-NAME>] [feature <feature>] [mode {consistency | vsx-sync}] [format {cli | json}]
```



NOTE

mode vsx-sync is not supported on the HPE Aruba Networking 6300 Switch series.

Description

Runs configuration validation to detect configuration anomalies.

Parameter	Description
config	Specifies configuration to be validated. The default configuration is running - config .
feature <feature>	Specifies the name of the feature to be validated. NOTE Available features vary by switch type. The HPE Aruba Networking 6300 Switch series supports vsf as an option for the feature parameter, and the 6400 Series Switch supports vsx as an option for the feature parameter.
mode	Specifies configuration validation mode. The default is consistency .

Parameter	Description
consistency	Validates feature configuration for consistency check.
vsx-sync	Validates VSX configuration synchronization between VSX peers for VSX enabled features. vsx - sync is not supported on the HPE Aruba Networking 6300 Switch series.
format	Specifies the results display format. The default is cli .

Examples

Running configuration validation with all default values. (HPE Aruba Networking 6300 Switch Series)

```
switch# switch config-validator
Line number 15: Split detect (MAD) is recommended for vsf stack.
Line number 18: VSF interface should be configured in the VSF link and the
interface should be up.
Line number 34: Configuration 'associate role <ROLE_NAME> is missing.
Line number 38: Configuration 'enable' is recommended.
Line number 43: Configuration 'enable' is recommended.
Line number 45: A group (LLDP, CDP, MAC) should be associated with only one
device profile.
```

Running configuration validation with switches for the vsx feature. (HPE Aruba Networking 6400 Switch Series)

```
switch (config)# switch config-validator config running-config feature vsx
Line number 36: Configuration `system-mac <VSX_SYSTEM_MAC>` is recommended
Line number 36: Multi chassis configuration is recommended for VSX
redundancy
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.
6400		

Cable Diagnostics

The Time-Domain Reflectometer (TDR) feature helps characterize and locate cable faults in an Ethernet cable. TDR involves showing a reflection at any impedance change within the cable when a low voltage pulse is sent into the cable. TDR measures the time between release and return of the low voltage pulse from any reflections. The distance to the reflection can be calculated by measuring the time and the transmission velocity of the pulse.

TDR or Cable Diagnostics is a port feature supported on some switches running software. TDR is used to detect cable faults on the following ports:

Table 1. Cable fault detection on supported ports types

Platforms	1GbT	5G - SmartRate	10G - SmartRate
6300	Yes	Yes	Yes
6400	Yes	Yes	Yes

TDR or Cable Diagnostics can also be run from the API.

Subtopics

[How TDR works on platforms](#)

[Cable diagnostics tests](#)

[Cable diagnostic commands](#)

How TDR works on platforms

The implementation of TDR in platforms is dependent on the physical layer chips (PHYs) that are part of the front-end network ports hardware. switches activate TDR on the PHY when a user enters the `diag cable-diagnostic` command. The switch waits for the report about TDR measurements from the PHY. The switch then reads the results and reports the values to the user.

Cable diagnostics tests



NOTE

The cable diagnostics test will bring down the link, which will take more time to complete the test.

The TDR cable diagnostic test allows an operator to test twisted pair cables for faults without physically disconnecting the cables from the switch. It helps in troubleshooting connectivity or monitoring performance on one or more switch ports.

The `diag cable-diagnostic` command can be used to run cable diagnostic tests and display the test results.

The following table provides the cable status messages and their descriptions.

Status	Meaning
good	The MDI pair is good.
open	The MDI pair is not terminated with a link partner or has an open circuit.
intra-short	The MDI pair is shorted within itself.
inter_short	The MDI pair is shorted with another pair.
high_imp	The MDI pair has high - impedance mismatch and is not guaranteed to link up.
low_imp	The MDI pair has low - impedance mismatch and is not guaranteed to link up.
unknown	The MDI pair has failed the cable diagnostic test.

The following table provides the possible cable diagnostic failure reasons for port types.

Port Type	Reasons
1GbT	Interface is busy
5G - SmartRate	Interface is busy
10G - SmartRate	Interface is busy

The following table provides the cable length accuracy for port types.

Port Type	Reasons
1GbT	When diagnostic status is "good", cable length is reported.
5G - SmartRate	When diagnostic status is "good", cable length is not reported.
10G - SmartRate	When diagnostic status is "good", cable length is not reported.

The following table provides the distance to fault accuracy for port types.

Port Type	Reasons
1GbT	When diagnostic status is not "good" or "failed", distance to fault is reported within +/- 10m.
5G - SmartRate	When diagnostic status is not "good" or "failed", distance to fault is reported within +/- 5m.
10G - SmartRate	When diagnostic status is not "good" or "failed", distance to fault is reported within +/- 5m.

Cable diagnostic commands

Select a command from the list in the left navigation menu.

Subtopics

diag cable-diagnostic

diag cable-diagnostic

Syntax

```
diag cable-diagnostic  
test <IF-NAME>
```

```
show <IF-NAME>
clear <IF-NAME>
```

Description

Provides information about the cable health after running a diagnostic test on an interface.

If you run a new cable diagnostic command when a cable diagnostic is in progress for the interface, the new cable diagnostic command fails to execute. In such a scenario, an error message is displayed.

On executing a cable diagnostic test command, it automatically clears the old test results before the new test starts.

Parameter	Description
<code><IF-NAME></code>	Specifies the name of the interface.
<code>test <IF-NAME></code>	Runs a cable diagnostic test on an interface.
<code>show <IF-NAME></code>	Displays the diagnostic test result for an interface.
<code>clear <IF-NAME></code>	Clears the cable diagnostic test results for an interface.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

The following example displays running a cable diagnostic test on interface `1/3/1`:

```
switch# diag cable-diagnostic test 1/3/1
This command will cause a loss of link on the port under test and will take
several seconds to complete.
Continue (y/n)? y
```

The following example displays the error message on executing a cable diagnostic command while the current diagnostic test is in progress:

```
switch# diag cable-diagnostic test 1/3/1
A cable diagnostic test for interface 1/3/1 is already in progress.
```

The following example displays the error message when cable diagnostic test is requested for an unsupported port:

```
switch# diag cable-diagnostic test 1/3/1
Cable diagnostic is not supported on interface 1/3/1.
```

The following examples display the cable diagnostic test result for 1GbT interface:

```
switch# diag cable-diagnostic show 1/3/1
```

Interface	Pinout	Cable Status	Impedance (Ohms)	Distance* (Meters)	MDI Mode
1/3/1 (1GbT)	1-2	good	85-115	10 +/- 10	mdi
	3-6	good	85-115	10 +/- 10	mdi
	4-5	good	85-115	5 +/- 10	mdi
	7-8	good	85-115	3 +/- 10	mdi

* Full cable length for good cables or distance to fault for faulty cables.

Cable status legend (1GbT):

Cable Status	Impedance (Ohms)	Description
good	85-115	No cable faults found
open	>115	Open circuit detected
intra-short	<85	Short circuit within the same wire pair
inter-short	<85	Short circuit with another wire pair
high-imp	>115	Cable impedance higher than expected
low-imp	<85	Cable impedance lower than expected
unknown	--	Cable test inconclusive

The following examples display the cable diagnostic test result for 5G-SmartRate interface:

```
switch# diag cable-diagnostic show 1/1/20
```

Interface	Pinout	Cable Status	Impedance (Ohms)	Distance* (Meters)	MDI Mode
1/1/20 (5G-SmartRate)	1-2	good	85-115	--	mdi
	3-6	open	>300	4 +/- 5	mdi
	4-5	open	>300	4 +/- 5	mdi
	7-8	high-imp	>115	3 +/- 5	mdi

* Full cable length for good cables or distance to fault for faulty cables.

Cable status legend (5G-SmartRate):

Cable Status	Impedance (Ohms)	Description
good	85-115	No cable faults found
open	>300	Open circuit detected
intra-short	<30	Short circuit within the same wire pair
inter-short	<30	Short circuit with another wire pair
high-imp	>115	Cable impedance higher than expected
low-imp	<85	Cable impedance lower than expected
unknown	--	Cable test inconclusive

The following example displays the error message when you execute a cable diagnostic command while the current diagnostic test is in progress:

```
switch# diag cable-diagnostic show 1/3/1
```

A cable diagnostic test for interface 1/3/1 is currently in progress.

The following example displays the error message when cable diagnostic test result is not available:

```
switch# diag cable-diagnostic show 1/3/1
```

Cable diagnostic test results for interface 1/3/1 are not available.

The following example clears the cable diagnostic test results for the specified interface:

```
switch# diag cable-diagnostic clear 1/3/1
```

The following example displays the error message when you execute a cable diagnostic command while the current diagnostic test is in progress:

```
switch# diag cable-diagnostic clear 1/3/1
```

A cable diagnostic test for interface 1/3/1 is currently in progress.



NOTE

Running a cable diagnostic test will result in a brief interruption in connectivity on all tested ports.

If a good cable is used on the SmartRate ports, the **Distance to Fault (Meters)** value is 0.

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

Supportability Copy

To effectively diagnose various issues arising at the switch, different types of data are copied out using copy commands for further analysis.

Use the **copy core-dump** command to copy the core-dump of a daemon crash.

Use the **copy show-tech** command to capture the status of the feature.

If there is feature misbehavior, use the **copy support-files feature** command to copy all feature related information for further analysis. Additionally use **copy support-log** and **copy diag-dump** to copy information that helps to analyze the internal behavior of a feature/daemon.

Use **copy command-output** to copy any **show** command's output to remote destinations or USB storage.

These files can be copied to a remote destination using sftp/tftp, additionally they can also be stored in the USB storage.

TFTP VxLAN Support

TFTP is supported over VxLAN tunnels with IPv4 or IPv6 underlay.



NOTE

TFTP over VxLAN tunnels with IPv6 underlay is only supported on the 8100 and 8360 Switch Series.

Limitations

Running-config, check-point config and startup-config will not be copied fully to destination file in TFTP server (only partial configuration will be copied) even though TFTP shows the transfer was successful. This is because fragmentation/ressembly/MTU discovery are not supported on VxLAN paths. Packets exceeding 1500 bytes are dropped when the TFTP transfer is done with default TFTP block size or default MTU size.

Workaround

Increase the MTU size (JUMBO) on all interfaces between the TFTP client and TFTP server or use a custom block size of 1375 or less for TFTP transfers.

Example of a custom blocksize configuration:

```
copy running-config tftp://72.1.1.100;blocksize=1374/runv4 cli vrf vrf1
copy running-config tftp://[20:2:100];blocksize=1374/runv6 cli vrf vrf1
```

Subtopics

Supportability copy commands

Supportability copy commands

Select a command from the list in the left navigation menu.

Subtopics

[copy checkpoint](#)

[copy command-output](#)

[copy core-dump \[<MEMBER/SLOT>\] daemon](#)

copy core-dump [<MEMBER/SLOT>] kernel
copy core-dump [<MEMBER/SLOT>] kernel <STORAGE-URL>
copy core-dump vsf member daemon
copy core-dump vsf member kernel
copy diag-dump feature <FEATURE>
copy diag-dump local-file
copy diag-dump vsf member local-file
copy <IMAGE>
copy running-config
copy show-tech feature
copy show-tech local-file
copy show-tech vsf member local-file
copy startup-config
copy support-files
copy support-files local-file
copy support-files vsf member
copy support-log
copy support-log vsf member

copy checkpoint

Syntax

```
copy checkpoint <CHECKPOINT-NAME> {<STORAGE-URL> | <REMOTE-URL>}
```

Description

Copies the checkpoint using TFTP, SFTP, SCP, or USB.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the checkpoint name.
{<STORAGE-URL> <REMOTE-URL>}	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<STORAGE-URL>	Specifies the USB to copy command output. Syntax: {usb};/ <FILE>

Parameter	Description
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: <code>{tftp://} {<IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE></code> <code>{sftp:// scp:// <USER> @} {<IP> <HOST> }[: <PORT>] / <FILE></code>

Examples

Copying checkpoint chpt to a remote URL:

```
switch# copy checkpoint chpt scp://root@10.0.1.1/config vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

copy command-output

Syntax

```
copy command-output "<COMMAND>" {<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>] }
```

Description

Copies the specified command output using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
<code><COMMAND></code>	Specifies the command from which you want to obtain its output. Required. Users with auditor rights can specify these two commands only: show accounting log show events
<code>{ <STORAGE-URL> <REMOTE-URL> [vrf <VRF-NAME>] }</code>	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: {tftp://} <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @} <IP> <HOST> }[: <PORT>]/ <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.

Examples

Copying the output from the **show events** command to a remote URL:

```
switch# copy command-output "show events" tftp://10.100.0.12/file
```

Copying the output from the **show tech** command to a remote URL with a VRF named mgmt:

```
switch# copy command-output "show tech" scp://user@10.100.0.12/file vrf mgmt
```

Copying the output from the **show tech** command to a remote URL with a VRF named mgmt:

```
switch# copy command-output "show tech" tftp://10.100.0.12/file vrf mgmt
```

Copying the output from the **show events** command to a file named **events** on a USB drive:

```
switch# copy command-output "show events" usb:/events
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

copy core-dump [<MEMBER/SLOT>] daemon

Syntax

```
copy core-dump [<MEMBER/SLOT>] daemon <DAEMON-NAME>[:<INSTANCE-ID>] <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies the core-dump from the specified daemon using TFTP, SFTP, SCP, or USB.

Parameter	Description
<code><MEMBER/SLOT></code>	Specifies the slot ID on an 8400 or 6400 switch. Required. Syntax: Slot number for line (1/1 - 1/4, 1/7 - 1/10) MM(1/5 or 1/6)
<code><DAEMON-NAME></code>	Specifies the name of the daemon. Required.

Parameter	Description
[: <INSTANCE-ID>]	Specifies the instance of the daemon core dump. Optional.
<REMOTE_URL>	Specifies the remote destination URL. Required. The syntax of the URL is the following: Syntax: • {ftp:// scp:// ssh:// rsync:// http:// https:// ftp:// ssh:// rsync:// http:// https:// } { <IP> <HOST> } [: <PORT>] [; blocksize= <VAL>] / <FILE> • {ftp:// scp:// ssh:// rsync:// http:// https:// } { <IP> <HOST> } [: <PORT>] / <FILE>
vrf <VRF-NAME>	Specifies the VRF name. If no VRF name is provided, the VRF named default is used. Optional.

Examples

Copying the core dump from daemon **ops-vland** to a remote URL with a VRF named **mgmt**:

```
switch# copy core-dump daemon ops-vland sftp://abc@10.0.14.211/
vland_coredump.xz vrf mgmt
```

Copying the core dump from daemon **ops-vland** to a remote URL with a VRF named **mgmt**:

```
switch# copy core-dump daemon ops-vland scp://abc@10.0.14.211/
vland_coredump.xz vrf mgmt
```

Copying the core dump from daemon **ops-switchd** to a USB drive:

```
switch# copy core-dump daemon ops-switchd usb:/switchd
```

Copying the core dump with slot ID 1/1 from daemon **hpe-sysmond** to a remote URL:

```
switch# copy core-dump 1/1 daemon hpe-sysmond sftp://abc@10.0.14.206/
core.hpe-sysmond.xz vrf mgmt
```

Copying the core dump from the **hpe-config** process to a USB drive:

```
switch# copy core-dump daemon hpe-config usb:/config_core
```

Command History

Release	Modification
10.08	Added SCP support.

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6400	Manager (#)	Administrators or local user group members with execution rights for this command.

copy core-dump [<MEMBER/SLOT>] kernel

Syntax

```
copy core-dump [<MEMBER/SLOT>] kernel <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies a kernel core dump using TFTP, SFTP, or SCP.

Parameter	Description
<MEMBER/SLOT>	Specifies the slot ID on an 8400 or 6400 switch. Required. Syntax: Slot number for line (1/1 - 1/4 , 1/7 - 1/10) MM(1/5 or 1/6)
<REMOTE-URL>	Specifies the URL to copy the command output. Required. Syntax: {tftp://} <IP> <HOST> }[: <PORT>][:blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @} <IP> <HOST> }[: <PORT>] / <FILE>
vrf <VRF-NAME>	Specifies the VRF name. The default VRF name is default. Optional.

Examples

Copying the kernel core dump to the URL:

```
switch# copy core-dump kernel tftp://10.100.0.12/kernel_dump.tar.gz
```

Copying the kernel core dump to the URL with the VRF named mgmt:

```
switch# copy core-dump kernel tftp://10.100.0.12/kernel_dump.tar.gz vrf mgmt
```

Copying the kernel core dump from slot ID 1/1 to the URL with the VRF named mgmt:

```
switch# copy core-dump 1/1 kernel sftp://abc@10.0.14.206/kernel_dump.tar.gz  
vrf mgmt
```

Copying the kernel core dump from slot ID 1/1 to the URL with the VRF named mgmt:

```
switch# copy core-dump 1/1 kernel scp://abc@10.0.14.206/kernel_dump.tar.gz  
vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6400	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy core-dump [<MEMBER/SLOT>] kernel <STORAGE-URL>
```

Syntax

```
copy core-dump [<MEMBER/SLOT>] kernel <STORAGE-URL>
```

Description

Copies the kernel core dump to a USB drive.

Parameter	Description
<code><MEMBER/SLOT></code>	Specifies the slot ID. Required. Syntax: Slot number for line (1/1 - 1/4, 1/7 - 1/10) MM(1/5 or 1/6)
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Required. Syntax: {usb}:/ <FILE>

Examples

Copying the kernel core dump to a USB drive:

```
switch# copy core-dump kernel usb:/kernel.tar.gz
```



NOTE

For more information on features that use this command, refer to the for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6400	Manager (#)	Administrators or local user group members with execution rights for this command.

copy core-dump vsf member daemon

Syntax

Applicable for 6300 switches only.

```
copy core-dump vsf member <MEMBER-ID>
    daemon [<DAEMON-NAME> | <DAEMON-NAME>:<INSTANCE-ID>]
<REMOTE-URL> [vrf <VRF-NAME>]
copy core-dump vsf member <MEMBER-ID>
    daemon [<DAEMON-NAME> | <DAEMON-NAME>:<INSTANCE-ID>]
<STORAGE-URL>
```

Description

Copies the core-dump from the specified daemon using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
vsf member <MEMBER-ID>	Specifies the member - id of the VSF member. Required.
<DAEMON-NAME>	Specifies the name of the daemon. Required.
[:<INSTANCE-ID>]	Specifies the instance of the daemon core dump. Optional.
<REMOTE_URL>	Specifies the remote destination URL. Required. The syntax of the URL is the following: Syntax: • {tftp://}{ <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> • {sftp:// scp:// <USER> @}{ <IP> <HOST> }[: <PORT>] / <FILE>
vrf <VRF-NAME>	Specifies the VRF name. If no VRF name is provided, the VRF named default is used. Optional.
<STORAGE-URL>	Specifies the USB to copy command output. Required. Syntax: {usb}:/ <FILE>

Examples

Copying the core dump from daemon hpe-sysmond to a remote URL with a VRF named mgmt:

```
switch# copy core-dump vsf member 1 daemon hpe-sysmond sftp://  
abc@10.0.14.206/sysmon.xz vrf mgmt
```

Copying the core dump from daemon hpe-sysmond to a remote URL with a VRF named mgmt:

```
switch# copy core-dump vsf member 2 daemon hpe-sysmond scp://  
user@10.0.14.206/sysmon.xz vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

copy core-dump vsf member kernel

Syntax

Applicable for 6300 switches only.

```
copy core-dump vsf member <MEMBER-ID> kernel <REMOTE-URL> [vrf <VRF-NAME>]  
copy core-dump vsf member <MEMBER-ID> kernel <STORAGE-URL>
```

Description

Copies the kernel core-dump using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
<code><MEMBER-ID></code>	Specifies the member - id of the VSF member. Required.
<code><REMOTE_URL></code>	Specifies the remote destination URL. Required. The syntax of the URL is the following: Syntax: {ftp://}{ <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @}{ <IP> <HOST> }[: <PORT>] / <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. If no VRF name is provided, the VRF named default is used. Optional.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Required. Syntax: {usb}:/ <FILE>

Examples

Copying the kernel core dump to the URL with a VRF named mgmt:

```
switch# copy core-dump vsf member 3 kernel sftp://abc@10.0.14.206/
kernel.tar.gz vrf mgmt
```

Copying the kernel core dump to the URL with a VRF named mgmt:

```
switch# copy core-dump vsf member 3 kernel scp://abc@10.0.14.206/
kernel.tar.gz vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

copy diag-dump feature <FEATURE>

Syntax

```
copy diag-dump feature <FEATURE> {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies the specified diagnostic information using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
<FEATURE>	The name of a feature, for example aaa or vrrp . Required.
{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL> }	Select either the remote URL or the storage URL for the destination of the copied command output. Required.
<REMOTE-URL>	Specifies the remote destination URL. Required. The syntax of the URL is the following: Syntax: {tftp://} { <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @} { <IP> <HOST> }[: <PORT>] / <FILE>
vrf <VRF-NAME>	Specifies the VRF name. If no VRF name is provided, the VRF named default is used. Optional.
	Specifies the USB to copy command output. Required. Syntax: {usb} : /<FILE>

Parameter	Description
<STORAGE-URL>	

Examples

Copying the output from the aaa feature to a remote URL with a specified VRF:

```
switch# copy diag-dump feature aaa tftp://10.100.0.12/diagdump.txt vrf mgmt
```

Copying the output from the aaa feature to a remote URL with a specified VRF:

```
switch# copy diag-dump feature aaa scp://user@10.100.0.12/diagdump.txt vrf mgmt
```

Copying the output from the vrrp feature to a USB drive:

```
switch# copy diag-dump feature vrrp usb:/diagdump.txt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy diag-dump local-file

Syntax

```
copy diag-dump local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies the diagnostic information stored in a local file using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
<code>{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL>}</code>	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: <code>{tftp://}{ <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE></code> <code>{sftp:// scp:// <USER> @}{ <IP> <HOST> }[: <PORT>] / <FILE></code>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: <code>{usb}:/ <FILE></code>

Usage

The **copy diag-dump local-file** command can be used only after the information is captured. Run the **diag-dump <FEATURE-NAME> basic local-file** command before you enter the **copy diag-dump local-file** command to capture the diagnostic information for the specified feature into the local file.

Examples

Copying the output from the local file to a remote URL:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file tftp://10.100.0.12/diagdump.txt
```

Copying the output from the local file to a remote URL:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file scp://user@10.100.0.12/diagdump.txt
```

Copying the output from the local file to a USB drive:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file usb:/diagdump.txt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy diag-dump vsf member local-file

Syntax

Applicable for 6300 switches only.

```
copy diag-dump vsf member <MEMBER-ID> local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies the diagnostic information stored in a local file using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
vsf member <MEMBER-ID>	Specifies the member - id of the VSF member. Required.
{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL>}	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<REMOTE-URL>	Specifies the URL to copy the command output. Syntax: {tftp://}{<IP> <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

Parameter	Description
	{sftp:// scp:// <USER> @{ <IP> <HOST> }: <PORT> / <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>

Usage

The **copy diag-dump local-file** command can be used only after the information is captured. Run the **diag-dump <FEATURE-NAME> basic local-file** command before you enter the **copy diag-dump local-file** command to capture the diagnostic information for the specified feature into the local file.

Examples

Copying the output from the local file to a remote URL:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump vsf member 2 local-file scp://user@10.100.0.12/
diagdump.txt
```

Copying the output from the local file to a remote URL:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump vsf member 2 local-file tftp://10.100.0.12/
diagdump.txt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <IMAGE>

Syntax

```
copy <IMAGE> { <STORAGE-URL> | <REMOTE-URL> } <FILE-NAME> [vrf <VRF-NAME>]
```

Description

Copies the image using TFTP, SFTP, SCP, or USB.

Parameter	Description
<IMAGE>	Specifies the image.
{ <STORAGE-URL> <REMOTE-URL> }	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<STORAGE-URL>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>
<REMOTE-URL>	Specifies the URL to copy the command output. Syntax: {tftp:// sftp:// scp://} <IP> <HOST>]: <PORT>];blocksize= <VAL>] / <FILE> {sftp:// scp://} <USER> @ { <IP> <HOST>]: <PORT>]/ <FILE>
	Specifies the file name.

Parameter	Description
<FILE-NAME>	
vrf <VRF-NAME>	Specifies the VRF name. The default VRF name is default. Optional.

Examples

Copying the image to a remote URL:

```
switch# copy scp://root@20.0.1.1/primary.swi primary vrf mgmt
```

Copying the secondary image to a remote URL:

```
switch# copy secondary scp://root@20.0.1.1/primary.swi vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

copy running-config

Syntax

```
copy running-config {<STORAGE-URL> | <REMOTE-URL>}/config <CONFIG-NAME>
[vrf <VRF-NAME>]
```

Description

Copies the running configuration using TFTP, SFTP, SCP, or USB.

Parameter	Description
<code>{<STORAGE-URL> <REMOTE-URL>}</code>	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: <code>{tftp://}{<IP> <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE></code> <code>{sftp:// scp:// <USER> @}{<IP> <HOST> }[:<PORT>]/<FILE></code>
<code>config <CONFIG-NAME></code>	Specifies the running configuration.
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.
<code>overwrite</code>	The configuration is only applied to the running configuration if all commands succeed without errors.

Usage

By default, the CLI configuration contained in the remote file is applied on top of the running configuration, and all the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the failure is logged in the event log and the next line in the CLI configuration is processed.

When using the optional **overwrite** parameter, the configuration is only applied to running configuration if all the commands succeeded without errors. If errors are present, the existing running-config will remain intact. You can then inspect the event logs to check and fix the errors and retry the operation. If more than 20 errors are present, the operation stops processing any further commands, and will display the errors at that point in the event logs. For more information on this feature, refer to the video on the [HPE Aruba Networking Airheads Broadcasting Channel](#).

Examples

Copying the running configuration to a remote URL:

```
switch# copy running-config scp://root@10.0.1.1/config cli vrf mgmt
```

Command History

Release	Modification
10.15	The overwrite parameter is introduced.
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

copy show-tech feature

Syntax

```
copy show-tech feature <FEATURE> {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies show tech output using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL>}	Select either the remote URL or the storage URL for the destination of the copied command output. Required.
<REMOTE-URL>	Specifies the URL to copy the command output. Required. Syntax:

Parameter	Description
	{tftp://}{ <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @}{ <IP> <HOST> }[: <PORT>] / <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Required. Syntax: {usb}:/ <FILE>

Example

Copying show tech output of the **aaa** feature using SCP:

```
switch# copy show-tech feature aaa scp://user@10.0.0.12/file.txt vrf mgmt
```

Copying show tech output of the `config` feature using SFTP on the `mgmt` VRF:

```
switch# copy show-tech feature config sftp://root@10.0.0.1/tech.txt vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy show-tech local-file

Syntax

```
copy show-tech local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies show tech output stored in a local file.

Parameter	Description
{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL>] }	Select either the remote URL or the storage URL for the destination of the copied command output. Required.
<REMOTE-URL>	Specifies the URL to copy the command output. Syntax: • {tftp://} <IP> <HOST> }; <PORT>]; blocksize= <VAL>] / <FILE> • {sftp:// scp:// <USER> @} <IP> <HOST> }; <PORT>]/ <FILE>
vrf <VRF-NAME>	Specifies the VRF name. The default VRF name is default. Optional.
<STORAGE-URL>	Specifies the USB to copy command output. Syntax: {usb};/ <FILE>

Usage

Before entering the **copy show-tech local-file** command, run the **show tech** command with the **local-file** parameter for the specified feature.

Examples

Copying the output to a remote URL:

```
switch# copy show-tech local-file tftp://10.100.0.12/file.txt
```

Copying the output to a remote URL:

```
switch# copy show-tech local-file scp://user@10.100.0.12/file.txt
```

Copying the output to a remote URL with a VRF:

```
switch# copy show-tech local-file tftp://10.100.0.12/file.txt vrf mgmt
```

Copying the output to a USB:

```
switch# copy show-tech local-file usb:/file
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy show-tech vsf member local-file

Syntax

Applicable for 6300 switches only.

```
copy show-tech vsf member <MEMBER-ID> local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies show tech output stored in a local file.

Parameter	Description
vsf member <MEMBER-ID>	Specifies the member - id of the VSF member. Required.
{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL> }	Select either the remote URL or the storage URL for the destination of the copied command output. Required.

Parameter	Description
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: {tftp://} <IP> <HOST> }]: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @} <IP> <HOST> }]: <PORT>]/ <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>

Usage

Before entering the `copy show-tech local-file` command, run the `show tech` command with the `local-file` parameter for the specified feature.

Examples

Copying the output to a remote URL with a VRF:

```
switch# copy show-tech vsf member 2 local-file tftp://10.100.0.12/
showtech.txt vrf mgmt
```

Copying the output to a USB:

```
switch# copy show-tech vsf member 2 local-file usb:/file
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

copy startup-config

Syntax

```
copy startup-config {<STORAGE-URL> | <REMOTE-URL>}/config <CONFIG-NAME>
[vrf <VRF-NAME>]
```

Description

Copies the running configuration using TFTP, SFTP, SCP, or USB.

Parameter	Description
{ <STORAGE-URL> <REMOTE-URL> }	Select either the storage URL or the remote URL for the destination of the copied command output. Required.
<STORAGE-URL>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>
<REMOTE-URL>	Specifies the URL to copy the command output. Syntax: {tftp://} <IP> <HOST>]: <PORT>];blocksize= <VAL>] / <FILE> {sftp:// scp://} <USER> @{ <IP> <HOST>]: <PORT>]/ <FILE>
config <CONFIG-NAME>	Specifies the startup configuration.
vrf <VRF-NAME>	Specifies the VRF name. The default VRF name is default. Optional.

Examples

Copying the startup configuration to a remote URL:

```
switch# copy startup-config scp://root@10.0.1.1/config json vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

copy support-files

Syntax

```
copy support-files
  <REMOTE-URL> [vrf <VRF-NAME>]
  <STORAGE-URL>
all <REMOTE-URL> [vrf <VRF-NAME>]
all <STORAGE-URL>
feature <FEATURE-NAME>
  <STORAGE-URL>
previous-boot <REMOTE-URL> [vrf <VRF-NAME>]
previous-boot <STORAGE-URL>
all-mms <REMOTE_URL> [ vrf <VRF_NAME> ]
```

For the 6400 switch only:

```
module <SLOT-ID>
  <REMOTE-URL> [vrf <VRF-NAME>]
```

```

module <SLOT-ID>
    <STORAGE-URL>
standby <REMOTE-URL> [vrf <VRF-NAME>]

```

For the 6300 switch only:

```

vsf member <MEMBER-ID> <REMOTE-URL> {vrf <VRF-NAME>}
vsf member <MEMBER-ID>
    <STORAGE-URL>

```

Description

Copies a set of support files to a compressed file in tar.gz format using TFTP, SFTP, SCP, or USB or to a directory over SFTP, USB, or local.



NOTE

Do not press **Control + Z** at the password prompt while copying support files, as this will prevent the console from executing other commands.



NOTE

This command does not support TFTP transfer on 6300 switches.

Parameter	Description
<FEATURE-NAME>	The feature name, for example, <code>aaa</code> .
{<REMOTE-URL> [vrf <VRF-NAME>] <STORAGE-URL> }	Select either the remote URL or the storage URL for the destination of the copied command output. Required.
<REMOTE-URL>	Specifies the URL to copy the command output. Syntax: {tftp://} { <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @} { <IP> <HOST> }[: <PORT>] / <FILE>
vrf <VRF-NAME>	Specifies the VRF name. The default VRF name is default. Optional.
	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>

Parameter	Description
<code><STORAGE-URL></code>	
<code><MEMBER-ID></code>	The member ID in the VSF stack. Range 1 - 10.
<code><SLOT-ID></code>	Specifies the slot ID on 6400 switches. Optional. Syntax: Slot number for line (1/1 - 1/4, 1/7 - 1/10) MM(1/5 o r 1/6)
<code>all-mms <REMOTE_URL> [vrf <VRF_NAME>]</code>	Copies both active and standby support files.

Usage

If feature name is not provided, the command collects generic system-specific support information. If a feature name is provided, the command collects feature-specific support information.



NOTE

In order to collect data from standby and member in a VSF stack, the command will prompt for the local user password once.



NOTE

In order to collect data from the standby 6400 switch, the command will prompt for the local user password once.

Examples

Copying the support files to a remote URL:

```
switch# copy support-files tftp://10.100.0.12/file.tar.gz
```

Copying the support files of the **lldp** feature to a remote URL with a specified VRF:

```
switch# copy support-files feature lldp tftp://10.100.0.12/file.tar.gz vrf  
mgmt
```

Copying the support files from the previous boot to a remote URL with a specified VRF:

```
switch# copy support-files previous-boot scp://user@10.0.14.206/file.tar.gz  
vrf mgmt
```

Copying the support files to a USB:

```
switch# copy support-files usb:/file.tar.gz
```

Copying the files from a module to a remote URL with a specified VRF on an 8400 or 6400 switch:

```
switch# copy support-files module 1/1 tftp://10.100.0.12/file.tar.gz vrf
mgmt
```

Copying the files from a standby module to a remote URL with a specified VRF on an 8400 or 6400 switch:

```
switch# copy support-files standby sftp://root@10.0.14.216/file.tar.gz vrf
mgmt
```

Copying all the support files to a remote URL:

```
switch# copy support-files all sftp://root@10.0.14.216/file.tar.gz vrf mgmt
```

Copying the support files of the **config** feature to a USB:

```
switch# copy support-files feature config usb:/file.tar.gz
```

Copying both active and standby support files:

```
switch# copy support-files all-mms sftp://root@10.0.14.216/
support_files_dir_path/ vrf mgmt
```



NOTE

Copying support files as a directory will not be allowed when system memory is insufficient. In the event this type of copy operation is attempted with low system memory, the command-line interface may display the error message: **Failed to collect support files due to insufficient system memory.**

Command History

Release	Modification
10.18	Added <code>all-mms</code> parameter.
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy support-files local-file

Syntax

```
copy support-files [feature <FEATURE-NAME> | previous-boot | all | module  
<SLOT-ID> | standby  
| vsf member <MEMBER-ID>  
] local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```



NOTE

The **module** and **standby** are supported only on 6400 switch. The **vsf member** is supported only on 6300 switch.

Description

Stores a set of support files as a compressed file in the switch locally and copies the preserved support files to a directory using TFTP, SFTP, SCP, USB, or local. . You can store only one copy of the support file locally. When you store a new support file, it overwrites the existing support file



NOTE

Do not press **Control + Z** at the password prompt while copying support files, as this will prevent the console from executing other commands.

Parameter	Description
<FEATURE-NAME>	Specifies the feature for the support files.
	Specifies the module slot number identifier for the support files. Range: 1/1 - 1/4, 1/7 - 1/10

Parameter	Description
<SLOT-ID>	
<MEMBER-ID>	Specifies the VSF member identifier for the support files. Range : 1 - 10
<REMOTE-URL>	Specifies the URL to copy the support files.
<STORAGE-URL>	Specifies the USB to copy the support files.
<VRF-NAME>	Specifies the VRF name. The default VRF name is default.

Usage

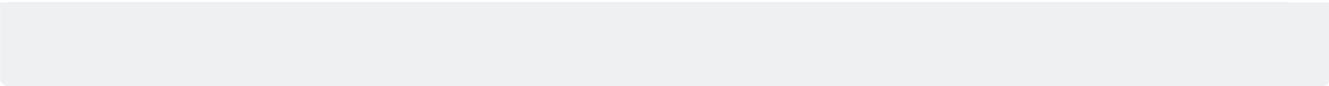
If the copy of the support files to the destination fails, an alternate option is prompted to store the collected data in the local file. This helps us to retry the copy process using **copy support-files local-file** *<REMOTE-URL/STORAGE-URL>* without the need of regenerating the file.

Examples

Copying support file to the local file:

```
switch# copy support-files local-file
switch# copy support-files feature lldp local-file
switch# copy support-files previous-boot local-file
switch# copy support-files all local-file
The operation to copy all support files could take a while to complete.
Do you want to continue (y/n)?
switch# copy support-files module 1/1 local-file

switch# copy support-files standby local-file
switch# copy support-files vsf member 7 local-file
```



Copying local support file to a remote URL and storage URL:

```
switch# copy support-files local-file usb:/support_files_dir_path/  
switch# copy support-files local-file scp://root@10.0.14.206//  
support_files_dir_path/abc.tar.gz vrf mgmt
```



NOTE

Copying support files as a directory will not be allowed when system memory is insufficient. In the event this type of copy operation is attempted with low system memory, the command-line interface may display the error message: **Failed to collect support files due to insufficient system memory.**

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy support-files vsf member

Syntax

Applicable for 6300 switches only.

```
copy support-files vsf member <MEMBER-ID> {<REMOTE-URL> [vrf <VRF-NAME>] |  
<STORAGE-URL>}
```

Description

Copies a set of support files using TFTP, SFTP, SCP, USB, or local.



NOTE

Do not press **Control + Z** at the password prompt while copying support files, as this will prevent the console from executing other commands.

Parameter	Description
<code><MEMBER-ID></code>	Specified the member - id of the VSF member. Required.
<code>{<REMOTE-URL> [vrf <VRF-NAME> <STORAGE-URL>]}</code>	Select either the remote URL or the storage URL for the destination of the copied command output. Required.
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: • {tftp://}{ <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> • {sftp:// scp:// <USER> @}{ <IP> <HOST> }[: <PORT>] / <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. The default VRF name is default. Optional.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>

Usage

If feature name is not provided, the command collects generic system-specific support information. If a feature name is provided, the command collects feature-specific support information.

Examples

Copying the support files to a USB:

```
switch# copy support-files vsf member 2 usb:/file.tar.gz
```

Copying all the support files to a remote URL with a specified VRF:

```
switch# copy support-files vsf member 2 scp://user@10.100.0.12/file.tar.gz/  
vrf mgmt
```

Copying all the support files to a remote URL with a specified VRF:

```
switch# copy support-files vsf member 2 sftp://user@10.100.0.12/  
support_files_dir_path/ vrf mgmt
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

copy support-log

Syntax

```
copy support-log <DAEMON-NAME> [<MEMBER/SLOT>] {<STORAGE-URL> | <REMOTE-  
URL> [vrf <VRF-NAME>]}
```

Description

Copies the specified support log for a daemon TFTP, SFTP, SCP, USB, or local.

Parameter	Description
	Specifies the slot ID on an 8400 or 6400 switch. Optional. Syntax: Slot number for line (1/1 - 1/4, 1/7 - 1/10) MM(1/5 or 1/6)

Parameter	Description
<MEMBER/SLOT>	
<DAEMON-NAME>	Specifies the name of the daemon. Required.
{<STORAGE-URL> <REMOTE-URL> [vrf <VRF-NAME>]}	Selects either the storage URL or the remote URL for the destination of the copied command output. Required.
<STORAGE-URL>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>
<REMOTE-URL>	Specifies the URL to copy the command output. Syntax: {tftp://}{ <IP> <HOST> }[: <PORT>][;blocksize= <VAL>] / <FILE> {sftp:// scp:// <USER> @}{ <IP> <HOST> }[: <PORT>]/ <FILE>
vrf <VRF-NAME>	Specifies the VRF name. If no VRF name is provided, the VRF named default is used. Optional.

Usage

Fast log is a high performance, per-daemon binary logging infrastructure used to debug daemon level issues by precisely capturing the per daemon/module/functionalities debug traces in real time. Fast log, also referred to as support logs, helps users to understand the feature internals and its specific happenings. The fast logs from one daemon are not overwritten by other daemon logs because fast logs are captured as part of a daemon core dump. Fast logs are enabled by default.

Examples

Copying the support log from the daemon hpe-fand to a remote URL:

```
switch# copy support-log hpe-fand tftp://10.100.0.12/file
```

Copying the support log from the daemon fand to a remote URL with a VRF named mgmt:

```
switch# copy support-log fand scp://user@10.100.0.12/file vrf mgmt
```

Copying the support log from the daemon hpe-fand to a remote URL with a VRF named mgmt:

```
switch# copy support-log hpe-fand tftp://10.100.0.12/file vrf mgmt
```

Copying the support log from the daemon hpe-fand to a USB:

```
switch# copy support-log hpe-fand usb:/support-log
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy support-log vsf member

Syntax

Applicable for 6300 switches only.

```
copy support-log vsf member <MEMBER-ID>
    <DAEMON-NAME> {<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>]}
```

Description

Copies the specified support log for a daemon using TFTP, SFTP, SCP, USB, or local.

Parameter	Description
<MEMBER-ID>	Specifies the member - id of the VSF member. Required.
	Specifies the name of the daemon. Required.

Parameter	Description
<code><DAEMON-NAME></code>	
<code>{ <STORAGE-URL> <REMOTE-URL> [vrf <VRF-NAME>] }</code>	Selects either the storage URL or the remote URL for the destination of the copied command output. Required.
<code><STORAGE-URL></code>	Specifies the USB to copy command output. Syntax: {usb}:/ <FILE>
<code><REMOTE-URL></code>	Specifies the URL to copy the command output. Syntax: {tftp://} <IP> <HOST> [: <PORT>];blocksize= <VAL>] / <FILE> {sftp:// scp://} <USER> @{ <IP> <HOST> } [: <PORT>] / <FILE>
<code>vrf <VRF-NAME></code>	Specifies the VRF name. If no VRF name is provided, the VRF named default is used. Optional.

Usage

Fast log is a high performance, per-daemon binary logging infrastructure used to debug daemon level issues by precisely capturing the per daemon/module/functionalities debug traces in real time. Fast log, also referred to as support logs, helps users to understand the feature internals and its specific happenings. The fast logs from one daemon are not overwritten by other daemon logs because fast logs are captured as part of a daemon core dump. Fast logs are enabled by default.

Examples

Copying the support log from the daemon hpe-fand to a remote URL with a VRF named mgmt:

```
switch# copy support-log vsf member 2 hpe-fand tftp://10.100.0.12/file vrf mgmt
```

Copying the support log from the daemon hpe-fand to a remote URL with a VRF named mgmt:

```
switch# copy support-log vsf member 2 hpe-fand scp://user@10.100.0.12/file vrf mgmt
```

Copying the support log from the daemon hpe-fand to a USB:

```
switch# copy support-log vsf member 2 hpe-fand usb:/support-log
```

Command History

Release	Modification
10.08	Added SCP support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

Traceroute

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

Traceroute over VXLAN

Traceroute and traceroute6 are supported over VXLAN from VTEP to VTEP, VTEP to host, and host to VTEP over L2 VNI/L3 VNI. A unique IP on VTEP should be used as the traceroute source and destination.

Both source and destination VTEPs require 10.8 or later for this feature to work. Traceroute and traceroute6 over VXLAN cannot be used to track the underlay hops between VTEPs.



NOTE

Traceroute and traceroute6 are supported on all platforms with non-VXLAN.



NOTE

Traceroute and traceroute6 are supported on all platforms with VXLAN IPv4 underlay support.

Traceroute and traceroute6 are supported on 6300, 6400, 8100, and 8360 with VXLAN IPv6 underlay.

Subtopics

Traceroute commands

Traceroute commands

Select a command from the list in the left navigation menu.

Subtopics

traceroute

traceroute6

traceroute

Syntax

```
traceroute {<IPV4-ADDR> | <HOSTNAME>} [ip-option loosesourceroute <IPV4-ADDR>] [dstport <NUMBER> | maxttl <NUMBER> | minttl <NUMBER> | probes <NUMBER> | timeout <TIME>] [vrf <VRF-NAME>] source {<IPV4-ADDR> | <IFNAME>}
```



NOTE

Traceroute over VXLAN with ip-option loosesourceroute on L3VNI is not supported.

Description

Uses traceroute for the specified IPv4 address or hostname with or without optional parameters.

Parameter	Description
IPv4-address <IPV4-ADDR>	Specifies the IPv4 address.
hostname	Specifies the hostname of the device to traceroute.
ip-option	Specifies the IP option.
loosesourceroute <IPV4-ADDR>	Specifies the route for loose source record route. Enter one or more intermediate router IP addresses separated by ',' for loose source routing.
dstport <NUMBER>	Specifies the destination port, <1 - 34000>. Default: 33434
maxttl <NUMBER>	Specifies the maximum number of hops to reach the destination, <1 - 255>. Default: 30
minttl <NUMBER>	Specifies the Minimum number of hops to reach the destination, <1 - 255>. Default: 1
probes <NUMBER>	Specifies the number of probes, <1 - 5>. Default: 3
timeout <TIME>	Specifies the traceroute timeout in seconds, <1 - 60>. Default: 3 seconds
vrf <VRF-NAME>	Specifies the virtual routing and forwarding (VRF) to use .
source {<IPV4-ADDR> <IFNAME>}	Specifies the source IPv4 address or interface name.

Usage

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

Examples

switch#

traceroute 10.0.10.1

```
traceroute to 10.0.10.1 (10.0.10.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 10.0.40.2
0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms
0.002ms switch#
```

traceroute localhost

traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 127.0.0.1 0.018ms 0.006ms 0.003ms switch#

traceroute 10.0.10.1 maxttl 20

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 3 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms switch#

traceroute 10.0.10.1 minttl 1

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms switch#

traceroute 10.0.10.1 dstport 33434

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms switch#

traceroute 10.0.10.1 probes 2

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 2 probes 1 10.0.40.2 0.002ms 0.002ms 2 10.0.30.1 0.002ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms switch#

traceroute 10.0.10.1 timeout 5

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 5 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms switch#

traceroute localhost vrf red

traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 127.0.0.1 0.003ms 0.002ms 0.001ms switch#

traceroute localhost mgmt

traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 127.0.0.1 0.018ms 0.006ms 0.003ms switch#

traceroute 10.0.10.1 maxttl 20 timeout 5 minttl 1 probes 3 dstport 33434

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 5 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms switch#

traceroute 10.0.10.1 ip-option loosesourceroute 10.0.40.2

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms switch#

traceroute 10.0.10.1 ip-option loosesourceroute 10.0.40.2 maxttl 20 timeout 5 minttl 1 probes 3 dstport 33434

traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 5 sec. timeout, 3 probes 1 10.0.40.2 0.002ms 0.002ms 0.001ms 2 10.0.30.1 0.002ms 0.001ms 0.001ms 3 10.0.10.1 0.001ms 0.002ms 0.002ms

```
switch# traceroute 10.0.0.2 source 10.0.0.1
traceroute to 10.0.0.2 (10.0.0.2), 30 hops max
```

```

1    10.0.0.2  0.299ms  0.155ms  0.115ms
switch# traceroute 10.0.0.2 source 1/1/1
traceroute to 10.0.0.2 (10.0.0.2), 30 hops max
1    10.0.0.2  0.479ms  0.222ms  0.171ms

```

Command History

Release	Modification
10.08	Added <code>source IP address</code> and <code>source interface name</code> parameters.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

traceroute6

Syntax

```

traceroute6 {<IPv6-ADDR> | <HOSTNAME>} [dstport <NUMBER> | maxttl <NUMBER>
| probes <NUMBER> | timeout <TIME>] [vrf <VRF-NAME>] source {<IPv6-ADDR> |
<IFNAME>}

```

Description

Uses traceroute for the specified IPv6 address or hostname with or without optional parameters.

Parameter	Description
IPv6-address <code><IPv6-ADDR></code>	Specifies the IPv6 address.
hostname	Specifies the hostname of the device to traceroute.

Parameter	Description
<code>dstport <NUMBER></code>	Specifies the destination port, <1 - 34000>. Default: 33434
<code>maxttl <NUMBER></code>	Specifies the maximum number of hops to reach the destination, <1 - 255>. Default: 30
<code>probes <NUMBER></code>	Specifies the number of probes, <1 - 5>. Default: 3
<code>timeout <TIME></code>	Specifies the traceroute timeout in seconds, <1 - 60>. Default: 3 seconds
<code>vrf <VRF-NAME></code>	Specifies the virtual routing and forwarding (VRF) to use, <VRF - NAME>.
<code>source {<IPV6-ADDR> <IFNAME>}</code>	Specifies the source IPv6 address or interface name.

Usage

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

Examples

switch#

traceroute6 0:0::0:1

traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.117 ms 0.032 ms 0.021 ms switch#

traceroute6 localhost

traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.089 ms 0.03 ms 0.014 ms switch#

traceroute6 0:0::0:1 maxttl 30

traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.117 ms 0.032 ms 0.021 ms switch#

traceroute6 0:0::0:1 dsrport 33434

traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.117 ms 0.032 ms 0.021 ms switch#

traceroute6 0:0::0:1 probes 2

traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 2 probes, 24 byte packets 1 localhost (::1)
0.117 ms 0.032 ms switch#

traceroute6 0:0::0:1 timeout 3

tracert to 0:0:0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.117 ms 0.032 ms 0.021 ms switch#

tracert6 localhost vrf red

tracert to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.077 ms 0.051 ms 0.054 ms switch#

tracert6 localhost mgmt

tracert to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.089 ms 0.03 ms 0.014 ms switch#

tracert6 0:0:0:1 maxttl 30 timeout 3 probes 3 dstport 33434

tracert to 0:0:0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets 1 localhost (::1)
0.117 ms 0.032 ms 0.021 ms

```
switch# tracert6 2001::2 source 2001::1
tracert to 2001::2 (2001::2) from 2001::1, 30 hops max, 3 sec. timeout,
3 probes, 24 byte packets
1 2001::2 (2001::2) 0.4331 ms 0.3186 ms 0.1874 ms
switch# tracert6 2001::2 source 1/1/1
tracert to 2001::2 (2001::2) from 2001::1, 30 hops max, 3 sec. timeout,
3 probes, 24 byte packets
1 2001::2 (2001::2) 0.6145 ms 0.4165 ms 0.1620 ms
```

Command History

Release	Modification
10.08	Added source IP address and source interface name parameters.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Ping

The ping (Packet Internet Groper) command is a common method for troubleshooting the accessibility of devices. It uses Internet Control Message Protocol (ICMP) echo requests and ICMP echo replies to determine if the other device is alive. It also measures the amount of time the request takes to receive a reply from the specified destination. The ping command is mostly used to verify IP connectivity between two endpoints which could be switch to switch, host to host, or host to switch. The reply packet tells if the host received the ping and the amount of time it took to return the packet.

Ping over VXLAN

Ping and ping6 are supported over VXLAN (with both IPv4 and IPv6 underlay) from VTEP to VTEP, VTEP to host, and host to VTEP over L2 VNI/L3 VNI. A unique IP on VTEP should be used as the source and destination. Both source and destination VTEPs require 10.8 or later for this feature to work. Ping and ping6 over VXLAN with IPv4 underlay is supported on all platforms with VXLAN support.



NOTE

Ping and ping6 over VXLAN with IPv6 underlay is supported on 6300, 6400, 8100, and 8360 Switch Series. IPv6 underlay is supported starting from 10.12.1000 version.

Ping with a large datagram size will not work as fragmentation, reassembly, and MTU discovery on the VXLAN paths are not supported. The default MTU size for an underlay port is 1500. When a packet is encapsulated via VXLAN, a VXLAN header of 50 bytes is added. With the default MTU size set on ports, packets larger than 1422 size is not expected to go over the VXLAN tunnel and is dropped. To use larger datagram size packets, the MTU must be increased.

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Ping commands

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Ping commands

Select a command from the list in the left navigation menu.

Subtopics

ping

Syntax

```
ping <IPv4-ADDR> | <hostname> [data-fill <pattern> | datagram-size <size> |
    interval <time> | repetitions <number> | timeout <time> | tos <number> |
    ip-option {include-timestamp | include-timestamp-and-address | record-
route} |
    vrf <vrfname> | do-not-fragment][source {IPv4-ADDR | IFNAME}]
```



NOTE

Ping on VXLAN with ip-option such as include-timestamp-and-address, include-timestamp and record-route is not supported.

Description

Pings the specified IPv4 address or hostname with or without optional parameters.

Parameter	Description
ping <IPv4-ADDR>	Selects the IPv4 address to ping.
<HOSTNAME>	Selects the hostname to ping. Range: 1 - 256 characters
data-fill <PATTERN>	Specifies the data pattern in hexadecimal digits to send. A maximum of 16 "pad" bytes can be specified to fill out the ICMP packet. Default: AB
datagram-size <SIZE>	Specifies the ping datagram size. Range: 0 - 65399, default: 100.
interval <TIME>	Specifies the interval between successive ping requests in seconds. Range: 1 - 60 seconds, default: 1 second.
repetitions <NUMBER>	Specifies the number of packets to send. Range: 1 - 10000 packets, default: Five packets.

Parameter	Description
<code>timeout <TIME></code>	Specifies the ping timeout in seconds. Range: 1 - 60 seconds, default: 2 seconds.
<code>tos <NUMBER></code>	Specifies the IP Type of Service to be used in Ping request. Range: 0 - 255
<code>ip-option {include-timestamp include-timestamp-and-address record-route}</code>	Specifies an IP option (record - route or timestamp option).
<code>include-timestamp</code>	Specifies the intermediate router time stamp.
<code>include-timestamp-and-address</code>	Specifies the intermediate router time stamp and IP address.
<code>record-route</code>	Specifies the intermediate router addresses.
<code>vrf <VRF-NAME></code>	Specifies the virtual routing and forwarding (VRF) to use. When VRF option is not given, the default VRF is used.
<code>source {IPv4-ADDR IFNAME}</code>	Specifies the source IPv4 address or interface to use.
<code>do-not-fragment</code>	Specifies the do - not - fragment (DF) bit in IP header of the Ping packet. This option does not allow the packet to be fragmented when it has to go through a segment with a smaller maximum transmission unit (MTU).

Examples

Pinging an IPv4 address:

```
switch# ping 10.0.0.0
PING 10.0.0.0 (10.0.0.0) 100(128) bytes of data.
108 bytes from 10.0.0.0: icmp_seq=1 ttl=64 time=0.035 ms
108 bytes from 10.0.0.0: icmp_seq=2 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=3 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=4 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=5 ttl=64 time=0.033 ms
--- 10.0.0.0 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.033/0.034/0.035/0.000 ms
```

Pinging the localhost:

```
switch# ping localhost
PING localhost (127.0.0.1) 100(128) bytes of data.
108 bytes from localhost (127.0.0.1): icmp_seq=1 ttl=64 time=0.060 ms
108 bytes from localhost (127.0.0.1): icmp_seq=2 ttl=64 time=0.035 ms
108 bytes from localhost (127.0.0.1): icmp_seq=3 ttl=64 time=0.043 ms
108 bytes from localhost (127.0.0.1): icmp_seq=4 ttl=64 time=0.041 ms
108 bytes from localhost (127.0.0.1): icmp_seq=5 ttl=64 time=0.034 ms
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.034/0.042/0.060/0.011 ms
```

Pinging a server with a data pattern:

```
switch# ping 10.0.0.2 data-fill 1234123412341234acde123456789012
PATTERN: 0x1234123412341234acde123456789012
PING 10.0.0.2 (10.0.0.2) 100(128) bytes of data.
108 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.207 ms
108 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.187 ms
108 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.225 ms
108 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.197 ms
108 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.210 ms
--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.187/0.205/0.225/0.015 ms
```

Pinging a server with a datagram size:

```
switch# ping 10.0.0.0 datagram-size 200
PING 10.0.0.0 (10.0.0.0) 200(228) bytes of data.
208 bytes from 10.0.0.0: icmp_seq=1 ttl=64 time=0.202 ms
208 bytes from 10.0.0.0: icmp_seq=2 ttl=64 time=0.194 ms
208 bytes from 10.0.0.0: icmp_seq=3 ttl=64 time=0.201 ms
208 bytes from 10.0.0.0: icmp_seq=4 ttl=64 time=0.200 ms
208 bytes from 10.0.0.0: icmp_seq=5 ttl=64 time=0.186 ms
--- 10.0.0.0 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.186/0.196/0.202/0.016 ms
```

Pinging a server with an interval specified:

```
switch# ping 9.0.0.2 interval 2
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.199 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.192 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.208 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.182 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.194 ms
--- 9.0.0.2 ping statistics ---
```

```
5 packets transmitted, 5 received, 0% packet loss, time 7999ms
rtt min/avg/max/mdev = 0.182/0.195/0.208/0.008 ms
```

Pinging a server with a specified number of packets to send:

```
switch# ping 9.0.0.2 repetitions 10
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.213 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.204 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.201 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.184 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.202 ms
108 bytes from 9.0.0.2: icmp_seq=6 ttl=64 time=0.184 ms
108 bytes from 9.0.0.2: icmp_seq=7 ttl=64 time=0.193 ms
108 bytes from 9.0.0.2: icmp_seq=8 ttl=64 time=0.196 ms
108 bytes from 9.0.0.2: icmp_seq=9 ttl=64 time=0.193 ms
108 bytes from 9.0.0.2: icmp_seq=10 ttl=64 time=0.200 ms
--- 9.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 8999ms
rtt min/avg/max/mdev = 0.184/0.197/0.213/0.008 ms
```

Pinging a server with a specified timeout:

```
switch# ping 9.0.0.2 timeout 3
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.175 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.192 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.190 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.181 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.197 ms
--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.175/0.187/0.197/0.007 ms
```

Pinging a server with the specified IP Type of Service:

```
switch# ping 9.0.0.2 tos 2
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.033 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.031 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.034 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.031 ms
--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.031/0.032/0.034/0.006 ms
```

Pinging a local host with the specified VRF.

```

switch# ping localhost vrf red
PING localhost (127.0.0.1) 100(128) bytes of data.
108 bytes from localhost (127.0.0.1): icmp_seq=1 ttl=64 time=0.048 ms
108 bytes from localhost (127.0.0.1): icmp_seq=2 ttl=64 time=0.052 ms
108 bytes from localhost (127.0.0.1): icmp_seq=3 ttl=64 time=0.044 ms
108 bytes from localhost (127.0.0.1): icmp_seq=4 ttl=64 time=0.036 ms
108 bytes from localhost (127.0.0.1): icmp_seq=5 ttl=64 time=0.055 ms
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.036/0.047/0.055/0.006 ms

```

Pinging the localhost with the default VRF:

```

switch# ping localhost vrf mgmt
PING localhost (127.0.0.1) 100(128) bytes of data.
108 bytes from localhost (127.0.0.1): icmp_seq=1 ttl=64 time=0.085 ms
108 bytes from localhost (127.0.0.1): icmp_seq=2 ttl=64 time=0.057 ms
108 bytes from localhost (127.0.0.1): icmp_seq=3 ttl=64 time=0.047 ms
108 bytes from localhost (127.0.0.1): icmp_seq=4 ttl=64 time=0.038 ms
108 bytes from localhost (127.0.0.1): icmp_seq=5 ttl=64 time=0.059 ms
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.038/0.057/0.085/0.016 ms

```

Pinging a server with the intermediate router time stamp:

```

switch# ping 9.0.0.2 ip-option include-timestamp
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.031 ms
TS:      59909005 absolute
          0
          0
          0
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
TS:      59910005 absolute
          0
          0
          0
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.038 ms
TS:      59911005 absolute
          0
          0
          0
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.035 ms
TS:      59912005 absolute
          0
          0

```

```

0
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.037 ms
TS:      59913005 absolute
0
0
0
--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.031/0.035/0.038/0.002 ms

```

Pinging a server with the intermediate router time stamp and address:

```

switch# ping 9.0.0.2 ip-option include-timestamp-and-address
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.030 ms
TS:      9.0.0.2 60007355 absolute
        9.0.0.2 0
        9.0.0.2 0
        9.0.0.2 0
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.037 ms
TS:      9.0.0.2 60008355 absolute
        9.0.0.2 0
        9.0.0.2 0
        9.0.0.2 0
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.037 ms
TS:      9.0.0.2 60009355 absolute
        9.0.0.2 0
        9.0.0.2 0
        9.0.0.2 0
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.038 ms
TS:      9.0.0.2 60010355 absolute
        9.0.0.2 0
        9.0.0.2 0
        9.0.0.2 0
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.039 ms
TS:      9.0.0.2 60011355 absolute
        9.0.0.2 0
        9.0.0.2 0
        9.0.0.2 0
--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.030/0.036/0.039/0.005 ms

```

Pinging a server with the intermediate router address:

```

switch# ping 9.0.0.2 ip-option record-route
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.

```

```

108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.034 ms
RR:      9.0.0.2
        9.0.0.2
        9.0.0.2
        9.0.0.2
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.038 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.036 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.037 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.035 ms (same route)
--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.034/0.036/0.038/0.001 ms

```

Pinging a server with do-not-fragment:

```

switch# ping 192.168.1.8 datagram-size 2000 do-not-fragment
PING 192.168.1.8 (192.168.1.8) 2000(2028) bytes of data.
2008 bytes from 192.168.1.8: icmp_seq=1 ttl=64 time=0.721 ms
2008 bytes from 192.168.1.8: icmp_seq=2 ttl=64 time=0.792 ms
2008 bytes from 192.168.1.8: icmp_seq=3 ttl=64 time=0.857 ms
2008 bytes from 192.168.1.8: icmp_seq=4 ttl=64 time=0.833 ms
2008 bytes from 192.168.1.8: icmp_seq=5 ttl=64 time=0.836 ms
--- 192.168.1.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4056ms
rtt min/avg/max/mdev = 0.721/0.807/0.857/0.048 ms

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ping6

Syntax

```
ping6 {<IPv6-ADDR> | <HOSTNAME>} [data-fill <PATTERN> | datagram-size  
<SIZE> |  
    interval <TIME> | repetitions <NUMBER> | timeout <TIME>  
    | vrrp <VRID>  
    |  
    vrf <VRF-NAME> | source <IPv6-ADDR> | <IFNAME>]
```

Description

Pings the specified IPv6 address or hostname with or without optional parameters. The VRRP option is provided to self-ping the configured link-local address on the VRRP group.

Parameter	Description
IPv6-ADDR	Selects the IPv6 address to ping.
HOSTNAME	Selects the hostname to ping. Range: 1 - 256 characters
data-fill <PATTERN>	Specifies the data pattern in hexadecimal digits to send. A maximum of 16 "pad" bytes can be specified to fill out the ICMP packet. Default: AB
datagram-size <SIZE>	Specifies the ping datagram size. Range: 0 - 65399, default: 100.
interval <TIME>	Specifies the interval between successive ping requests in seconds. Range: 1 - 60 seconds, default: 1 second.
repetitions <NUMBER>	Specifies the number of packets to send. Range: 1 - 10000 packets, default: Five packets.
timeout <TIME>	Specifies the ping timeout in seconds. Range: 1 - 60 seconds, default: 2 seconds.
vrrp <VRID>	Specifies the VRRP group ID.
vrf <VRF-NAME>	Specifies the virtual routing and forwarding (VRF) to use. When this option is not provided, the default VRF is used.
source <IPv6-ADDR> <IFNAME>	Specifies the source IPv6 address or interface to use.

Examples

Pinging an IPv6 address:

```
switch# ping6 2020::2
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.386 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.235 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.249 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.240 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.252 ms
--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.235/0.272/0.386/0.059 ms
```

Pinging the localhost:

```
switch# ping6 localhost
PING localhost(localhost) 100 data bytes
108 bytes from localhost: icmp_seq=1 ttl=64 time=0.093 ms
108 bytes from localhost: icmp_seq=2 ttl=64 time=0.051 ms
108 bytes from localhost: icmp_seq=3 ttl=64 time=0.055 ms
108 bytes from localhost: icmp_seq=4 ttl=64 time=0.046 ms
108 bytes from localhost: icmp_seq=5 ttl=64 time=0.048 ms
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.046/0.058/0.093/0.019 ms
```

Pinging a server with a data pattern:

```
switch# ping6 2020::2 data-fill ab
PATTERN: 0xab
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.038 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.074 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.077 ms
--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.038/0.068/0.077/0.015 ms
```

Pinging a server with a datagram size:

```
switch# ping6 2020::2 datagram-size 200
PING 2020::2(2020::2) 200 data bytes
208 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.037 ms
208 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.076 ms
208 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
```

```
208 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.077 ms
208 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.066 ms
--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.037/0.066/0.077/0.016 ms
```

Pinging a server with an interval specified:

```
switch# ping6 2020::2 interval 5
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.043 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.074 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.075 ms
--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 19999ms
rtt min/avg/max/mdev = 0.043/0.068/0.075/0.014 ms
```

Pinging a server with a specified number of packets to send:

```
switch# ping6 2020::2 repetitions 6
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.039 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.070 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.071 ms
108 bytes from 2020::2: icmp_seq=6 ttl=64 time=0.078 ms
--- 2020::2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 4999ms
rtt min/avg/max/mdev = 0.039/0.068/0.078/0.015 ms
```

Pinging a local host with the specified VRF.

```
switch# ping6 localhost vrf red
PING localhost(localhost) 100 data bytes
108 bytes from localhost: icmp_seq=1 ttl=64 time=0.038 ms
108 bytes from localhost: icmp_seq=2 ttl=64 time=0.050 ms
108 bytes from localhost: icmp_seq=3 ttl=64 time=0.039 ms
108 bytes from localhost: icmp_seq=4 ttl=64 time=0.040 ms
108 bytes from localhost: icmp_seq=5 ttl=64 time=0.027 ms
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4001ms
rtt min/avg/max/mdev = 0.027/0.038/0.050/0.010 ms
```

Pinging the localhost with the default VRF:

```
switch# ping6 localhost vrf mgmt
PING localhost(localhost) 100 data bytes
```

```
108 bytes from localhost: icmp_seq=1 ttl=64 time=0.032 ms
108 bytes from localhost: icmp_seq=2 ttl=64 time=0.022 ms
108 bytes from localhost: icmp_seq=3 ttl=64 time=0.040 ms
108 bytes from localhost: icmp_seq=4 ttl=64 time=0.022 ms
108 bytes from localhost: icmp_seq=5 ttl=64 time=0.046 ms
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.022/0.032/0.046/0.010 ms
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Troubleshooting

Subtopics

[Operation not permitted](#)

[Network is unreachable](#)

[Destination host unreachable](#)

Operation not permitted

Symptom

The switch displays an `operation not permitted` message when a user attempts to send a ping request.

Example:

```
switch# ping 100.1.2.10
PING 100.1.2.10 (100.1.2.10) 100(128) bytes of data
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
--- 100.1.2.10 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4000ms
```

Cause

When an ACL is applied to the Control Plane, sending a ping request may be denied. If the ping packet matches a drop entry in the ACL, applying a Control Plane may block traffic sent from the switch CLI ping command.

When this situation occurs, the following error message is displayed: `ping: sendmsg: Operation not permitted`. The message indicates that the ICMP echo request packet has not been sent and is blocked by the Control Plane ACL.

When this message is not displayed, the ping request packet has been sent correctly. A ping failure in this case represents a failure to receive the ICMP echo reply packet.

Action

1. Modify the ACL to allow the ping traffic.
2. Unapply the ACL from egress (8400/8320/8325/9300/9300S switches) or Control Plane.
3. Ping a destination which is not matched by the ACL. For example, if the ACL is blocking traffic based on destination IP. Depending on the ACL content, this might not always be possible like when the ACL blocks all ICMP packets.

Network is unreachable

Symptom

User receives a "network is unreachable" message on sending a ping request.

Cause

The ping packet did not get sent, because the switch cannot find an interface with a route that leads to the destination for one of the following reasons:

- A configuration error, such as an interface having an incorrect IP address or subnet defined.
- DHCP having failed to assign an address at all.

- The user meant to ping out the management vrf, but forgot to add `vrf mgmt` to the ping command.

Action

Adjust the switch configuration to ensure that a route to the destination network exists.

Destination host unreachable

Symptom

User receives a `Destination host unreachable` message on sending a ping request.

Cause

This issue typically indicates that the host is down or otherwise not returning ICMP echo requests. It is also possible that an intermediate network hop is dropping the packets.

Action

Investigate whether an intermediate hop is not returning pings by using the **tracert** command. Check the intermediate hop, and then the endpoint. If the destination is another switch, it is possible that Ingress ACLs on that switch are blocking ping packets. In such cases, the configuration option on the destination switch should be examined.

Troubleshooting

Subtopics

Operation not permitted

Network is unreachable

Destination host unreachable

Operation not permitted

Symptom

The switch displays an `operation not permitted` message when a user attempts to send a ping request.

Example:

```
switch# ping 100.1.2.10
PING 100.1.2.10 (100.1.2.10) 100(128) bytes of data
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
--- 100.1.2.10 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4000ms
```

Cause

When an ACL is applied to the Control Plane, sending a ping request may be denied. If the ping packet matches a drop entry in the ACL, applying a Control Plane may block traffic sent from the switch CLI ping command.

When this situation occurs, the following error message is displayed: `ping: sendmsg: Operation not permitted`. The message indicates that the ICMP echo request packet has not been sent and is blocked by the Control Plane ACL.

When this message is not displayed, the ping request packet has been sent correctly. A ping failure in this case represents a failure to receive the ICMP echo reply packet.

Action

1. Modify the ACL to allow the ping traffic.
2. Unapply the ACL from egress (8400/8320/8325/9300/9300S switches) or Control Plane.
3. Ping a destination which is not matched by the ACL. For example, if the ACL is blocking traffic based on destination IP. Depending on the ACL content, this might not always be possible like when the ACL blocks all ICMP packets.

Network is unreachable

Symptom

User receives a "network is unreachable" message on sending a ping request.

Cause

The ping packet did not get sent, because the switch cannot find an interface with a route that leads to the destination for one of the following reasons:

- A configuration error, such as an interface having an incorrect IP address or subnet defined.
- DHCP having failed to assign an address at all.

- The user meant to ping out the management vrf, but forgot to add `vrf mgmt` to the ping command.

Action

Adjust the switch configuration to ensure that a route to the destination network exists.

Destination host unreachable

Symptom

User receives a `Destination host unreachable` message on sending a ping request.

Cause

This issue typically indicates that the host is down or otherwise not returning ICMP echo requests. It is also possible that an intermediate network hop is dropping the packets.

Action

Investigate whether an intermediate hop is not returning pings by using the **tracert** command. Check the intermediate hop, and then the endpoint. If the destination is another switch, it is possible that Ingress ACLs on that switch are blocking ping packets. In such cases, the configuration option on the destination switch should be examined.

Using classifier policies for traffic capture and analysis

AOS-CX can use a classifier policy to troubleshooting network issues by mirroring packets for capture and analysis.

The process to filter mirrored traffic requires hardware resources, and this process may compete for these resources with other configured features, including existing Classifier Policies and Access Control Lists (ACLs). Prior to configuring and installing a policy for troubleshooting purposes, issue the **show policy** commands and **show resources** commands to determine what existing features are consuming hardware resources on the switch. This will also help to ensure that configurations created for troubleshooting are not overriding existing configurations. If a switch already has a classifier policy applied to one context (applied globally, interface, VLAN), consider adding entries to that policy, temporarily unapplying the existing policy, or applying the troubleshooting policy to a different context.



NOTE

Classifier Policies cannot capture traffic on the out-of-band management port. If the port is out-of-band, its packets do not enter or leave through the switch ASIC, which is required for mirroring operations.

Step one: create a traffic class

The following example creates a traffic class to match traffic to be mirrored for future evaluation. In this example:

- TCP and UDP protocols should include entries for both source and destination port matches in order to mirror traffic either to or from that port number.
- traffic utilizes HTTPS and therefore will match that entry.
- In this example, the **count** keyword is included so that hit counts can be monitored to verify that traffic is matching the class entries.
- The last entry (**ignore any any any count**) is included to count packets of all other traffic types passing through the device, and to confirm other traffic is reaching the policy for evaluation, and not being mirrored.
- Sequence numbers (for example, 10, 20, 30) are not mandatory when creating class entries, and will be auto-generated if not manually specified.
- Comment lines are optional for functionality but included for clarity.

```
switch(config)# class ip support-mirror
 10 comment OSPF protocol
 10 match ospf any any count
 20 comment GRE protocol
 20 match gre any any count
 30 comment BGP dst port
 30 match tcp any any eq bgp count
 40 comment BGP src port
 40 match tcp any eq bgp any count
 50 comment VxLAN dst port
 50 match udp any any eq vxlan count
 60 comment VxLAN src port
 60 match udp any eq vxlan any count
 70 comment RADIUS authentication dst port
 70 match udp any any eq radius count
 80 comment RADIUS authentication src port
 80 match udp any eq radius any count
 90 comment HTTPS dst port
 90 match tcp any any eq https count
100 comment HTTPS src port
100 match tcp any eq https any count
110 comment HTTP dst port
110 match tcp any any eq http count
120 comment HTTP src port
120 match tcp any eq http any count
```

```

130 comment ICMP Echo (Ping)
130 match icmp any any icmp-type echo count
140 comment ICMP Echo (Ping) Reply
140 match icmp any any icmp-type echo-reply count
150 comment Count all other traffic
150 ignore any any any count
exit

```

Step two: create a policy

Create a policy that uses the class created in the previous step and sends matching traffic to mirror session 1:

```

switch(config)# policy support-mirror
10 class ip support-mirror action mirror 1
exit

```

Step three: apply the policy

If you do not know which interface or VLAN the relevant traffic uses to enter the switch, apply the policy globally. Alternatively, you can apply the policy to one or more specific contexts. Note that while it is possible to apply policies to multiple interfaces, interface types, and directions, each application consumes hardware resources and may not be successful if resources are exhausted.

Apply the policy globally in the ingress (in) direction:

```

apply policy support-mirror in

```



NOTE

AOS-CX does not allow you to apply a policy globally in the egress direction; apply a policy to a specific interface, VLAN, or LAG if egress traffic is required.

Apply a policy to a specific physical interface in the ingress (in) direction:

```

switch (config)# interface 1/1/1
switch(config-if)# apply policy support-mirror in

```

Apply a policy to a specific physical interface in Egress (out) direction:

```

switch (config)#interface 1/1/1
switch(config-if)# apply policy support-mirror out

```

Apply a policy to a specific LAG in Ingress (in) direction:

```

switch (config)# interface lag 1
switch(config-lag-if)# apply policy support-mirror in

```

Apply a policy to a specific LAG in Egress (out) direction:

```

switch (config)# interface lag 1
switch(config-lag-if)# apply policy support-mirror out

```

Apply a policy to a specific VLAN in Ingress (in) direction:

```
switch (config)# vlan 100
switch(config-if-vlan)# apply policy support-mirror in
```

Apply a policy to a specific VLAN in Egress (out) direction:

```
switch (config)# vlan 100
switch(config-if-vlan)# apply policy support-mirror out
```

Step four: confirm policy Installation

The output of the **show class** and **show policy** commands should include the configuration that was configured in the previous steps. The output must **not** include any lines starting with an exclamation point (!), such as:

```
! policy support-mirror user configuration does not match active
configuration.
```

A message that starts with exclamation point (!), indicates a policy installation failure, possibly due to hardware resource limitations.

Step five: confirm policy resource consumption

Next, confirm that the ingress global policy lookup process is consuming TCAM entries. If the switch is a chassis, each module should show resources consumed, as this policy is installed on the ASIC in each line card. The following example shows the output of a **show resources** command issued on a 6300 switch:

```
6300(config)# show resources
Resource Usage:
Mod  Description
Resource                                Used Reserved    Free
-----
--
1/1  Ingress Global Policy Lookup
Ingress TCAM Entries                    35      2048
Total
Ingress TCAM Entries                    35      2048    18432
Egress TCAM Entries                     0         0     8192
Ingress Lookups                         1         8
Ingress Flex Lookups                    0         1
Egress Lookups                          0         4
Ingress Policers                        0      2047
Egress Policers                         0      2047
```

Hardware resources will be consumed on each module that has a policy applied to an interface. Ingress and egress policies consume separate hardware resources. The following example shows the output of a **show resources** command issued on a 6300 switch with a policy applied to a physical interface in both **in** and **out** directions:

```
6300(config)# show resources
Resource Usage:
```

Mod	Description	Used	Reserved	Free
Resource				

--				
1/1	Ingress Port Policy Lookup			
	Ingress TCAM Entries	37	2048	
	Egress Port Policy Lookup			
	Egress TCAM Entries	37	2048	
	Total			
	Ingress TCAM Entries	37	2048	18432
	Egress TCAM Entries	37	2048	6144
	Ingress Lookups	1		8
	Ingress Flex Lookups	0		1
	Egress Lookups	1		3
	Ingress Policers	0		2047
	Egress Policers	0		2047

Step six: configure a mirror session

In this example, the mirror session will be configured to send traffic to the switch CPU for capture:

```
switch(config)# mirror session 1
switch(config-mirror-1) destination cpu
switch(config-mirror-1) enable
```

You may also choose to mirror packets to an external capture host, such as a workstation running Wireshark for live packet analysis and further filtering. See .

Step seven: start packet capture

Start a packet capture using TShark.

```
switch# diag utilities tshark
```

Traffic should be captured and dumped to screen. The following example displays partial TShark output when a ping packet is sent through the switch:

```
switch# diag utilities tshark
Inspecting traffic mirrored to the CPU until Ctrl-C is entered.
Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on
interface MirrorRxNet, id 0
Interface id: 0 (MirrorRxNet)
Interface name: MirrorRxNet
Encapsulation type: Ethernet (1)
Arrival Time: Jul 18, 2023 22:45:21.213862080 UTC
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1689720321.213862080 seconds
[Time delta from previous captured frame: 0.000000000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
```

```
[Time since reference or first frame: 0.000000000 seconds]
Frame Number: 1
Frame Length: 98 bytes (784 bits)
Capture Length: 98 bytes (784 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
...
```

Step eight: capture packets to a file or mirror it to a host

Once basic policy configuration is confirmed as working as expected, you can use the **diag utilities tshark file** command to capture the TShark output to a file.

```
switch# diag utilities tshark file
```



NOTE

This command will store packets to a circular file; only the most recent 32MB of traffic will be captured.

Use the following command to copy a pcap file to a remote device:

```
switch# copy tshark-pcap ?
REMOTE_URL  URL syntax - sftp://USER@[IP|HOST][:PORT]/FILE or
tftp://[IP|HOST][:PORT][:blocksize=VAL]/FILE
```

If you do not want to capture the TShark output to a file, you can mirror it to an external capture host. In the following example, packets are mirrored to external capture host (such as a workstation with WireShark) is connected to interface 1/1/2:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/2
switch(config-mirror-1)# enable
```

Step nine: check packet hit counts

Use the **show policy hitcounts** command to confirm that the policy is evaluating traffic.

```
switch(config)# show policy hitcounts
```

In the following example, no ICMP echo reply packets are captured, as the policy globally is applied to ingress traffic only. Remember, AOS-CX does not support applying policies globally in the

egress direction; You must apply a policy to a specific context (an interface, VLAN, or LAG) to monitor egress traffic.

```
switch# show policy hitcounts support-mirror
Statistics for Policy support-mirror:
global (in):
Matched Packets  Configuration
10 class ip support-mirror action mirror 1
```

```

0 10 match ospf any any count
0 20 match gre any any count
0 30 match tcp any any eq bgp count
0 40 match tcp any eq bgp any count
0 50 match udp any any eq vxlan count
0 60 match udp any eq vxlan any count
0 70 match udp any any eq radius count
0 80 match udp any eq radius any count
0 90 match tcp any any eq https count
0 100 match tcp any eq https any count
0 110 match tcp any any eq http count
0 120 match tcp any eq http any count
5 130 match icmp any any icmp-type echo count
0 140 match icmp any any icmp-type echo-reply count
0 150 ignore any any any count

```

In this example, hit counts will be shown separately for each application of a policy and will reflect the direction of the traffic (that is, ICMP echo packets entering the switch and ICMP echo reply packets leaving the switch.)

```

6300(config-if)# show policy hitcounts support-mirror
Statistics for Policy support-mirror:
Interface 1/1/1 (in):
Matched Packets  Configuration
    0    10 class ip support-mirror action mirror 1
    0    10 match ospf any any count
    0    20 match gre any any count
    0    30 match tcp any any eq bgp count
    0    40 match tcp any eq bgp any count
    0    50 match udp any any eq vxlan count
    0    60 match udp any eq vxlan any count
    0    70 match udp any any eq radius count
    0    80 match udp any eq radius any count
    0    90 match tcp any any eq https count
    0   100 match tcp any eq https any count
    0   110 match tcp any any eq http count
    0   120 match tcp any eq http any count
    0   130 match icmp any any icmp-type echo count
    0   140 match icmp any any icmp-type echo-reply count
    0   150 ignore any any any count
Interface 1/1/1 (out):
Matched Packets  Configuration
    10 class ip support-mirror action mirror 1
    0    10 match ospf any any count
    0    20 match gre any any count

```

```

0 30 match tcp any any eq bgp count
0 40 match tcp any eq bgp any count
0 50 match udp any any eq vxlan count
0 60 match udp any eq vxlan any count
0 70 match udp any any eq radius count
0 80 match udp any eq radius any count
0 90 match tcp any any eq https count
0 100 match tcp any eq https any count
0 110 match tcp any any eq http count
0 120 match tcp any eq http any count
0 130 match icmp any any icmp-type echo count
5 140 match icmp any any icmp-type echo-reply count
0 150 ignore any any any count

```

Packet forwarding information

The packet forwarding information feature shows the forwarding information of a packet given packet header details and ingress information. This helps validate system configuration without actual traffic flowing into the system. In the case of the packet egressing out of LAG or ECMP, this feature also shows the physical interface the packet is egressed out on, based on the load balance setting configured in the system.

Subtopics

Packet forwarding information commands

Packet forwarding information commands

Select a command from the list in the left navigation menu.

Subtopics

show forwarding-info

show forwarding-info

Syntax

```

show forwarding-info mac ingress-interface <IFNAME> source-mac-address <MAC-
ADDR>
    destination-mac-address <MAC-ADDR> [vlan <VLAN-ID>] [timeout <TIMEOUT>]

```

```

[ethertype <ETHERTYPE>]
show forwarding-info mac-ip ingress-interface <IFNAME> source-mac-address
<MAC-ADDR>
    destination-mac-address <MAC-ADDR>
    {source-ip-address <IP-ADDR> destination-ip-address <IP-ADDR>} |
    source-ipv6-address <IPV6-ADDR> destination-ipv6-address <IPV6-ADDR>}
    [vlan <VLAN-ID>] [transport-protocol <TR-PROT-NUM>]
    [source-l4-port <L4-PORT> destination-l4-port <L4-PORT>]
    [vrf <VRF-NAME>] [timeout <TIMEOUT>][ethertype <ETHERTYPE>]
show forwarding-info ip
    {source-ip-address <IP-ADDR> destination-ip-address <IP-ADDR>} |
    source-ipv6-address <IPV6-ADDR> destination-ipv6-address <IPV6-ADDR>}
    [ingress-interface <IFNAME>][transport-protocol <TR-PROT-NUM>]
    [source-l4-port <L4-PORT> destination-l4-port <L4-PORT>]
    [vrf <VRF-NAME>] [timeout <TIMEOUT>]

```

Description

Shows the forwarding information based on current system configurations and hardware states for forwarding lookups. Given the user packet information, this command shows the egress physical interface of the packet. L3 hash mode and L4 hash mode are supported for LAG.

Parameter	Description
ingress-interface <IFNAME>	Specifies the ingress interface (for example: 1/1/1).
source-mac-address <MAC-ADDR>	Specifies the source MAC address. Format: AA:BB:CC:DD:EE:FF .
destination-mac-address <MAC-ADDR>	Specifies the destination MAC address. Format: AA:BB:CC:DD:EE:FF .
vlan <VLAN-ID>	Specifies the egress VLAN. Default 1. Range: 1 to 4094.
timeout <TIMEOUT>	Specifies the response timeout in seconds. Default: 3. Range: 1 to 60.
ethertype <ETHERTYPE>	(For 5420, 6300/6400, and 8360 Switch series) Specify the ethertype value to determine the egress interface.
source-ip-address <IP-ADDR>	Specifies the source IPv4 address. Format: A.B.C.D

Parameter	Description
<code>destination-ip-address <IP-ADDR>]</code>	Specifies the destination IPv4 address. Format: A.B.C.D
<code>source-ipv6-address <IPV6-ADDR></code>	Specifies the source IPv6 address. Format: X:X::X:X
<code>destination-ipv6-address <IPV6-ADDR></code>	Specifies the destination IPv6 address. Format: X:X::X:X
<code>transport-protocol <TR-PROT-NUM></code>	Specifies the transport protocol number. Range 1 to 255. For example, use 6 for TCP, 17 for UDP, 1 for ICMP, 2 for IGMP.
<code>source-l4-port <L4-PORT></code>	Specifies the L4 source port. Range 1 to 65535.
<code>destination-l4-port <L4-PORT></code>	Specifies the L4 destination port. Range 1 to 65535.
<code>vrf <VRF-NAME></code>	Specifies the egress VRF name. Default: default .

Usage

The following limitations need to be considered:

- The forwarding-information feature is not applicable for broadcast, multicast, and unknown packets.
- Sub-interfaces and tunnel interfaces (VXLAN and MPLS) in both ingress and egress are not supported. However, as of 10.15.1000, VXLAN is supported as an egress interface on 6300 and 8360 switches. In 10.17, VXLAN is supported as an egress interface on 5420, 6200, 6400, and 8100 switches.



NOTE

It is recommended to use only the mac-ip option of the forwarding info command in VXLAN scenarios as it ensures that all dependent parameters are included and that an accurate output is generated.

- Ingress interfaces are limited to the physical interfaces.
- The **vlan** and **vrf** parameters must be used for packet forwarding information wherever applicable. If these parameters are not specified, their indicated defaults are used.

- The **mac** and **mac-ip** parameters are not supported for LAGs in L2 hash mode for both bridged and routed traffic.
- When a LAG is in L3 hash mode, the L2 data is not used for hashing.
- For bridged traffic, the **mac** or **mac-ip** parameters must be used. For routed traffic, the **ip** or **mac-ip** parameters must be used.
- For bridged traffic, ensure that the VLAN membership of the ingress port parameter matches the value of the VLAN parameter.
- When using L3 hashing, the **show forwarding-info mac** form of this command is not applicable.
- When a LAG is in L3 hashing mode it will only hash L3 data. As a result, the L2 data displayed in the output of the **show forwarding-info mac-ip** command is ignored.
- This command does not display data for unsupported egress interfaces, such as a tunnel or sub-interface. The egress interface must be a physical port or a LAG.
- When ISL redirection occurs for a packet, the forwarding information does not show the correct egress interface in the VSX MLAG. Note that VXLAN forwarding information for VSX is not supported.
- The forwarding information output of this command does not reflect the PBR policy for the destination route.

Examples

Showing forwarding information when LAG is in L3 hash mode (the default):

```

    interface lag 1
    no shutdown
    no routing
    vlan access 1
    exit
show forwarding-info mac ingress-interface 1/1/4 source-mac-address
    00:00:00:00:00:01 destination-mac-address 00:00:00:00:00:02 vlan 1
Ingress-interface: 1/1/4
Source mac-address: 00:00:00:00:00:01
Destination mac-address: 00:00:00:00:00:02
VLAN: 1
Forwarding info lookup needs IP inputs when lag is in L3 hash mode.
```

Showing forwarding information when LAG is in L4 hash mode:

```

    interface lag 1
    no shutdown
    no routing
```

```

vlan access 1
hash l4-src-dst
exit
show forwarding-info mac ingress-interface 1/1/4 source-mac-address
    00:00:00:00:00:01 destination-mac-address 00:00:00:00:00:02 vlan 1
Ingress-interface: 1/1/4
Source mac-address: 00:00:00:00:00:01
Destination mac-address: 00:00:00:00:00:02
VLAN: 1
Egress interface: lag1 -> 1/1/1

```

Showing forwarding information when LAG is in L2 hash mode:

```

interface lag 1
no shutdown
no routing
vlan access 1
hash l2-src-dst
exit
show forwarding-info mac ingress-interface 1/1/4 source-mac-address
    00:00:00:00:00:01 destination-mac-address 00:00:00:00:00:02 vlan 1
Ingress-interface: 1/1/4
Source mac-address: 00:00:00:00:00:01
Destination mac-address: 00:00:00:00:00:02
VLAN: 1
Forwarding info lookup on lag port is unsupported when lag is in L2 hash
mode.

```

Showing forwarding information when the egress interface is a physical port:

```

show mac-address-table
MAC age-time          : 300 seconds
Number of MAC addresses : 1
MAC Address           VLAN      Type                Port
-----
00:00:00:00:00:02     1       static             lag1
00:00:00:00:00:03     1       static             1/1/5
show forwarding-info mac ingress-interface 1/1/4 source-mac-address
    00:00:00:00:00:01 destination-mac-address 00:00:00:00:00:03 vlan 1
Ingress-interface: 1/1/4
Source mac-address: 00:00:00:00:00:01
Destination mac-address: 00:00:00:00:00:03
VLAN: 1
Egress interface: 1/1/5

```

Showing forwarding information for when the ingress interface is ROP:

```

show forwarding-info mac-ip ingress-interface 1/1/14
source-mac-address 00:11:01:00:00:01 destination-mac-address
b8:d4:e7:dd:d3:00
source-ip-address 101.0.0.2 destination-ip-address 201.0.0.2 vrf default
Ingress-interface: 1/1/14
Source mac-address: 00:11:01:00:00:01
Destination mac-address: b8:d4:e7:dd:d3:00
VLAN: 1
VRF: default
Source IP: 101.0.0.2
Destination IP: 201.0.0.2
L2 Warning: Port is routing enabled. Please use the 'show forwarding-info
ip' command.
Egress interface: ECMP 1/1/17
Showing forwarding information with ROP and SVI:

```

```

interface 1/1/4
no routing
no shutdown
vlan trunk native 1
vlan trunk allowed all
interface 1/1/6
no shutdown
ip address 10.10.10.2/24
show ip route
Displaying ipv4 routes selected for forwarding
Origin Codes: C - connected, S - static, L - local
               R - RIP, B - BGP, O - OSPF, D - DHCP
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
               IA - OSPF internal area, E1 - OSPF external type 1
               E2 - OSPF external type 2
VRF: default
Prefix          Nexthop      Interface  VRF(egress)  Origin/
Distance/      Age
Type          Metric
-----
-----
10.10.10.0/24   -          1/1/6      -             C
[0/0]          -
10.10.10.2/32   -          1/1/6      -             L
[0/0]          -
show forwarding-info ip source-ip-address 10.10.10.1 destination-ip-address
10.10.10.4

```

```

Source IP: 10.10.10.1
Destination IP: 10.10.10.4
Egress interface: 1/1/6
interface 1/1/4
    no routing
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/7
    no routing
    no shutdown
    vlan access 2
interface vlan 2
    ip address 2.2.2.2/24
interface 1/1/4
    no routing
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/7
    no routing
    no shutdown
    vlan access 2
interface vlan 2
    ip address 2.2.2.2/24
show ip route
Displaying ipv4 routes selected for forwarding
Origin Codes: C - connected, S - static, L - local
               R - RIP, B - BGP, O - OSPF, D - DHCP
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
               IA - OSPF internal area, E1 - OSPF external type 1
               E2 - OSPF external type 2
VRF: default
Prefix          Nexthop    Interface  VRF(egress)  Origin/
Distance/      Age
                                           Type        Metric
-----
-----
2.2.2.0/24      -          vlan2      -             C
[0/0]          -
2.2.2.2/32      -          vlan2      -             L
[0/0]          -
show forwarding-info ip source-ip-address 2.2.2.1 destination-ip-address
2.2.2.4
Source IP: 2.2.2.1

```

Destination IP: 2.2.2.4
Egress interface: vlan2 -> 1/1/7

Showing forwarding information in L4 hash mode with ROP LAG:

```
switch# show forwarding-info mac-ip ingress-interface 1/1/1
source-mac-address 00:00:00:00:01 destination-mac-address 00:00:00:00:00:02
source-ip-address 10.10.10.10 destination-ip-address 10.10.10.11
transport-protocol 6 source-l4-port 1234 destination-l4-port 5678 vrf
default
Ingress interface: 1/1/1
Source MAC address: 00:00:00:00:00:01
Destination MAC address: 00:00:00:00:00:02
VLAN: 1
VRF: default
Source IP: 10.10.10.10
Destination IP: 10.10.10.11
Protocol: TCP(6)
Source L4 Port: 1234
Destination L4 Port: 5678
Egress interface: lag1 -> 1/1/2
```

Showing forwarding information in ECMP hash mode with ROP LAG:

```
switch# show forwarding-info mac-ip ingress-interface 1/1/1
source-mac-address 00:00:00:00:01 destination-mac-address 00:00:00:00:00:02
source-ip-address 10.10.10.10 destination-ip-address 200.0.0.1
transport-protocol 6 source-l4-port 1234 destination-l4-port 5678 vrf
default
Ingress interface: 1/1/1
Source MAC address: 00:00:00:00:00:01
Destination MAC address: 00:00:00:00:00:02
VLAN: 1
VRF: default
Source IP: 10.10.10.10
Destination IP: 200.0.0.1
Protocol: TCP(6)
Source L4 Port: 1234
Destination L4 Port: 5678
Egress interface: ECMP lag1 -> 1/1/2
```

Attempting to show forwarding information but with the request timed out:

```
switch# show forwarding-info mac-ip ingress-interface 1/1/1
source-mac-address 00:00:00:00:01 destination-mac-address 00:00:00:00:00:02
```

```

source-ip-address 10.10.10.10 destination-ip-address 200.0.0.1
transport-protocol 6 source-l4-port 1234 destination-l4-port 5678 vrf
default
Ingress interface: 1/1/1
Source MAC address: 00:00:00:00:00:01
Destination MAC address: 00:00:00:00:00:02
VLAN: 1
VRF: default
Source IP: 10.10.10.10
Destination IP: 200.0.0.1
Protocol: TCP(6)
Source L4 Port: 1234
Destination L4 Port: 5678
Request timed out

```

Showing forwarding information on a 6300 Switch series that includes the inner destination and inner source MAC addresses:

```

switch# show forwarding-info mac-ip ingress-interface 1/1/26
source-mac-address 00:11:22:33:44:55 destination-mac-address
00:50:56:96:13:b6
source-ip-address 10.1.30.11 destination-ip-address 10.1.50.21 vlan 10
vrf test1
Ingress-interface: 1/1/26
Source mac-address: 00:11:22:33:44:55
Destination mac-address: 00:50:56:96:13:b6
VLAN: 10
VRF: test1
Source IP: 10.1.30.11
Destination IP: 10.1.50.21
Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:00:12:12:12:12
Inner Source MAC: 00:50:56:96:13:b6
VNI: 1010
VNI Type: l3_vni
Tunnel Source IP: 10.0.0.1
Tunnel Destination IP: 10.0.0.2
UDP Source Port: 51432
UDP Dest Port: 4789
Tunnel Egress Path:

```

Nexthop	L3 Interface	L2 Interface	Destination MAC
192.168.11.10	1/1/35	1/1/35	b8:d4:e7:dd:e2:00

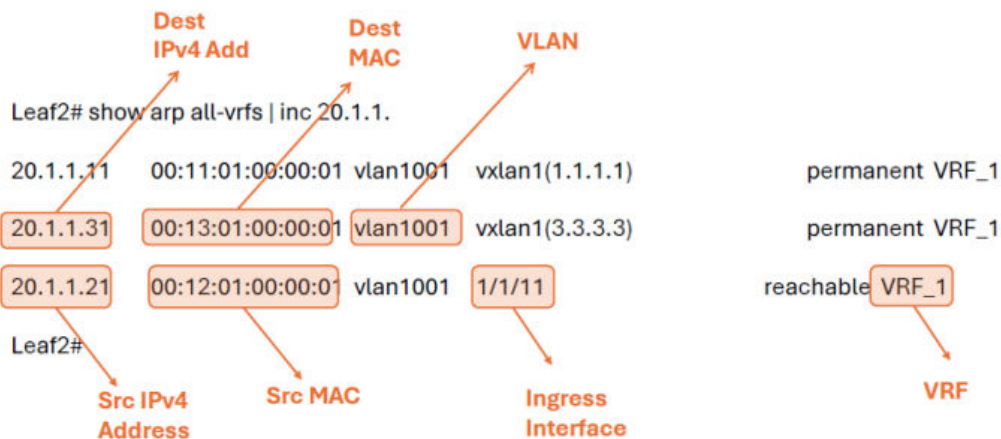
Showing forwarding information on a 6300/6400 or 8360 switch series with output that includes an ethertype and egress interface:

```
switch# show forwarding-info mac ingress-interface 1/3/47
source-mac-address 00:00:00:00:01 destination-mac-address 00:00:00:00:00:02
vlan 1 ethertype 0xffff
Ingress interface: 1/3/47
Source MAC address: 00:00:00:00:00:01
Destination MAC address: 00:00:00:00:00:02
VLAN: 1
Ethertype: 0xffff
Egress interface: lag1 -> 1/3/48
```

Sample Show command - Bridging v4

Sample show command output - Example 1 - Bridging v4

Bridging-v4 : SIP 20.1.1.21 and DIP 20.1.1.31



Showing command output - Bridging v4

```
Leaf2# show forwarding-info mac-ip ingress-interface 1/1/11
source-mac-address 00:12:01:00:00:01 destination-mac-address
00:13:01:00:00:01
source-ip-address 20.1.1.21 destination-ip-address 20.1.1.31 ethertype
0x0800 transport-protocol 61 vlan 1001 vrf VRF_1
Ingress-interface: 1/1/11
Source mac-address: 00:12:01:00:00:01
Destination mac-address: 00:13:01:00:00:01
VLAN: 1001
VRF: VRF_1
```

```

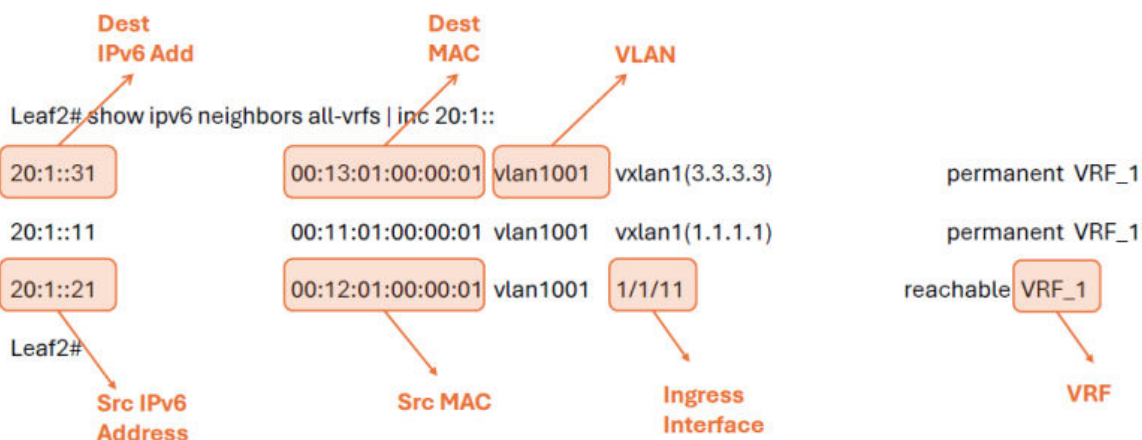
Source IP: 20.1.1.21
Destination IP: 20.1.1.31
Protocol: 61
Ethertype: 0x0800
Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:13:01:00:00:01
Inner Source MAC: 00:12:01:00:00:01
VNI: 1001
VNI Type: 12_vni
Tunnel Source IP: 2.2.2.2
Tunnel Destination IP: 3.3.3.3
UDP Source Port: 11820
UDP Dest Port: 4789
Tunnel Egress Path:
Nexthop L3 Interface      L2 Interface      Destination MAC
72.1.1.1      lag72      lag72->1/1/3      b8d4e7:de:f1:00
Leaf2#

```

Sample Show command - Bridging v6

Sample show command output - Example 2 - Bridging v6

Bridging-v6 - SIP 20:1::21 and DIP 20:1::31



Showing command output - Bridging v6

```

Leaf2# show forwarding-info mac-ip ingress-interface 1/1/11
source-mac-address 00:12:01:00:00:01 destination-mac-address
00:13:01:00:00:01
source-ipv6-address 20:1::21 destination-ipv6-address 20:1::31 ether

```

```

0x86dd vlan 1001 vrf VRF_1
Ingress-interface: 1/1/11
Source mac-address: 00:12:01:00:00:01
Destination mac-address: 00:13:01:00:00:01
VLAN: 1001
VRF: VRF_1
Source IPv6: 20:1::21
Destination IPv6: 20:1::31
Ethertype: 0x86dd
Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:13:01:00:00:01
Inner Source MAC: 00:12:01:00:00:01
VNI: 1001
VNI Type: 12_vni
Tunnel Source IP: 2.2.2.2
Tunnel Destination IP: 3.3.3.3
UDP Source Port: 50828
UDP Dest Port: 4789
Tunnel Egress Path:
Nexthop      L3 Interface    L2 Interface    Destination MAC
62.1.1.1     lag62            lag62->1/1/1    b8:d4:e7:dd:ba:00
Leaf2#

```

Sample Show command - Routing v4

Sample show command output - Example 3 - Routing v4

Routing v4 - SIP 32.1.1.21 and DIP 33.1.1.31

```

Leaf2# show arp all-vrfs | inc 32.1.1.21
32.1.1.21    00:15:01:00:00:01    vlan3201    1/1/11    reachable VRF_1
Leaf2#
Leaf2# sh running-config interface vlan 3201
interface vlan 3201
 vrf attach VRF_1
 ip address 32.1.1.1/16
 active-gateway ip mac 00:00:00:32:32:32
 active-gateway ip 32.1.1.10
 ipv6 address 32:1::1/64
 active-gateway ipv6 mac 00:00:00:32:32:32
 active-gateway ipv6 32:1::10
exit
Leaf2#

```

Showing command output - Routing v4

```

Leaf2# show forwarding-info mac-ip ingress-interface 1/1/11
source-mac-address 00:15:01:00:00:01 destination-mac-address
00:00:00:32:32:32
source-ip-address 32.1.1.21 destination-ip-address 33.1.1.31 ethertype
0x0800 transport-protocol 61 vlan 3201 vrf VRF_1
Ingress-interface: 1/1/11
Source mac-address: 00:15:01:00:00:01
Destination mac-address: 00:00:00:32:32:32
VLAN: 3201
VRF: VRF_1
Source IP: 32.1.1.21
Destination IP: 33.1.1.31
Protocol: 61
Ethertype: 0x0800
Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:00:00:0c:0c:0c
Inner Source MAC: 8c:85:c1:49:00:00
VNI: 9001
VNI Type: 13_vni
Tunnel Source IP: 2.2.2.2
Tunnel Destination IP: 3.3.3.3
UDP Source Port: 41081
UDP Dest Port: 4789
Tunnel Egress Path:
Nexthop   L3 Interface L2 Interface Destination MAC
62.1.1.1  lag62         lag62->1/1/14 b8d4e7dd:ba:00

```

Sample Show command - Routing v6

Sample show command output - Example 4 - Routing v6

Routing v6 - SIP 32:1::21 and DIP 33:1::31

```
Leaf2# show ipv6 neighbors all-vrfs | inc 32:1::21
32:1::21          00:15:01:00:00:01  vlan3201  1/1/11          reachable VRF_1

Leaf2# sh running-config interface vlan 3201
interface vlan 3201
 vrf attach VRF_1
 ip address 32.1.1.1/16
 active-gateway ip mac 00:00:00:32:32:32
 active-gateway ip 32.1.1.10
 ipv6 address 32:1::1/64
 active-gateway ipv6 mac 00:00:00:32:32:32
 active-gateway ipv6 32:1::10
exit
Leaf2#
```

Showing command output - Routing v6

```
Leaf2# show forwarding-info mac-ip ingress-interface 1/1/11
source-mac-address 00:15:01:00:00:01 destination-mac-address
00:00:00:32:32:32
source-ipv6-address 32:1::21 destination-ipv6-address 33:1::31 ethertype
0x86dd vlan 3201 vrf VRF_1
Ingress-interface: 1/1/11
Source mac-address: 00:15:01:00:00:01
Destination mac-address: 00:00:00:32:32:32
VLAN: 3201
VRF: VRF_1
Source IPv6: 32:1::21
Destination IPv6: 33:1::31
Ethertype: 0x86dd
Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:00:00:0c:0c:0c
Inner Source MAC: 8c:85:c1:49:00:00
VNI: 9001
VNI Type: 13 vni
Tunnel Source IP: 2.2.2.2
Tunnel Destination IP: 3.3.3.3
UDP Source Port: 27596
UDP Dest Port: 4789
Tunnel Egress Path:
```

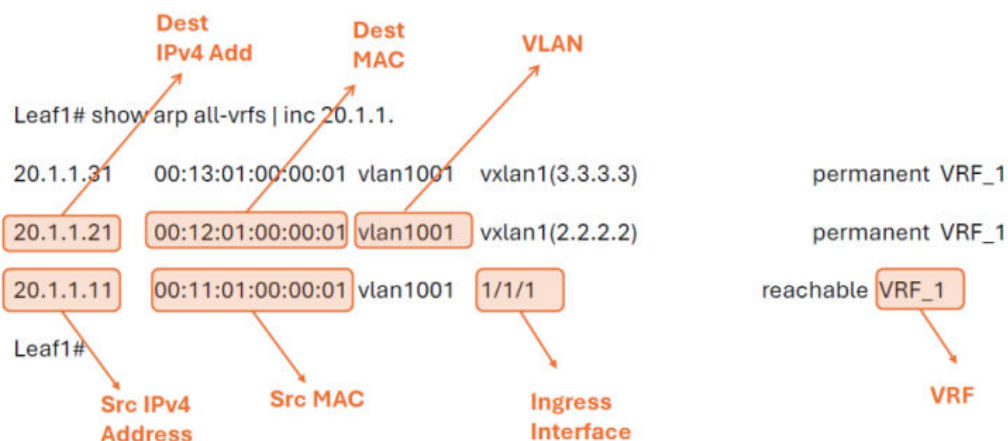
Nexthop L3 Interface	L2 Interface	Destination MAC
72.1.1.1 lag72	lag72->1/1/2	b8:d4:e7:de:f1:00

Leaf2#

Sample Show command - UDP Bridging v4

Sample show command output - Example 5 - UDP Bridging v4

UDP Bridging v4 traffic with Source L4 port 2222, Dest L4 Port 4444, SIP 20.1.1.11 and DIP 20.1.1.21



Showing command output - UDP Bridging v4

```
Leaf1# show forwarding-info mac-ip ingress-interface 1/1/1
source-mac-address 00:11:01:00:00:01 destination-mac-address
00:12:01:00:00:01
source-ip-address 20.1.1.11 destination-ip-address 20.1.1.21
ethertype 0x0800 vlan 1001 vrf VRF_1 transport-protocol 17 source-l4-port
2222
destination-l4-port 4444
Ingress-interface: 1/1/1
Source mac-address: 00:11:01:00:00:01
Destination mac-address: 00:12:01:00:00:01
VLAN: 1001
VRF: VRF_1
Source IP: 20.1.1.11
Destination IP: 20.1.1.21
Protocol: UDP(17)
Source L4 Port: 2222
Destination L4 Port: 4444
Ethertype: 0x0800
```

```

Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:12:01:00:00:01
Inner Source MAC: 00:11:01:00:00:01
VNI: 1001
VNI Type: l2_vni
Tunnel Source IP: 1.1.1.1
Tunnel Destination IP: 2.2.2.2
UDP Source Port: 30424
UDP Dest Port: 4789
Tunnel Egress Path:
Nexthop      L3 Interface      L2 Interface      Destination MAC
61.1.1.1     lag61              lag61->1/1/7      b8:d4:e7:dd:ba:00
Leaf1#

```

Showing forwarding information on a 6300 or 6400 Switch Series:

```

switch# show forwarding-info mac-ip ingress-interface 1/1/26
source-mac-address 00:11:22:33:44:55 destination-mac-address
00:50:56:96:13:b6
source-ip-address 10.1.20.11 destination-ip-address 10.1.10.21 vlan 20
Ingress-interface: 1/1/26
Source mac-address: 00:11:22:33:44:55
Destination mac-address: 00:50:56:96:13:b6
VLAN: 20
Source IP: 10.1.20.11
Destination IP: 10.1.10.21
Egress interface: vxlan1
Tunnel Egress Info:
Inner Destination MAC: 00:00:12:12:12:12
Inner Source MAC: 00:50:56:96:13:b6
VNI: 1010
VNI Type: l2_vni
Tunnel Source IP: 10.0.0.1
Tunnel Destination IP: 10.0.0.2
UDP Source Port: 51432
UDP Dest Port: 4789
Tunnel Egress Path:
Nexthop      L3 Interface      L2
Interface      Destination MAC
192.168.11.10  1/1/35
1/1/35        b8:d4:e7:dd:e2:00

```

Command History

Release	Modification
10.17	Command introduced on the 6400 Switch series.
10.15.1000	Command introduced on the 6300 and 8360 Switch series for the VXLAN feature.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Remote syslog

Remote syslog enables the forwarding of syslog messages to the remote syslog server. The feature supports a maximum of eight remote syslog servers. Only one configuration per remote syslog server is allowed. The remote syslog server supports TCP and UDP transport protocols and TLS to establish a connection. In addition to forwarding logs to the remote server, they can also be preserved in local storage.

When the client certificate associated with the syslog client is updated, the syslog client is restarted and a new TLS connection is established using the updated client certificate.

Syslog over VXLAN support

Syslog message is supported over VxLAN with IPv4 or IPv6 underlay.

Subtopics

Remote syslog commands

Remote syslog commands

Select a command from the list in the left navigation menu.

Subtopics

[clear accounting-logs](#)
[diag persistent-storage](#)
[logging](#)
[logging accounting-format-native](#)
[logging filter](#)
[logging facility](#)
[logging persistent-storage](#)

clear accounting-logs

Syntax

```
clear accounting-logs
```

Description

Use this command to clear accounting logs. Once issued, only logs generated after this command is run will be displayed in the output of the **show accounting log** commands.



NOTE

This command will not clear logs when the [logging accounting-format-native](#) feature is configured. To clear accounting logs on switches with this feature enabled, users should first revert the native accounting format back to the default AOS-CX format by executing the **no logging accounting-format-native** command.

Example

```
switch(config)# clear accounting-logs
```

The following example shows that accounting logs cannot be cleared using the clear accounting-logs command if the **logging accounting-native-format** command has been enabled, and that disabling this option with the **no logging accounting-format-native** command again allows the accounting logs to be cleared.

```
switch# logging audit-format-native
switch# clear accounting-logs
Warning: Clear accounting-logs is not supported for 'audit-format-native'.
switch# no logging audit-format-native
switch# clear accounting-logs
switch# show accounting log last 5
-----
Command logs from current boot
```

No command logs has been logged in the system

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

diag persistent-storage

Syntax

```
diag persistent-storage enable [{severity <LEVEL> | timeout <TIMEOUT> |  
filter <FILTER_NAME> | storage-writes <STORAGE_WRITES> | usb <STORAGE_URL>}]  
diag persistent-storage disable
```

Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.



NOTE

To switch the persistent storage setting from one option (e.g., USB) to another (e.g., storage-writes), you must first disable persistent storage by running the command `diag persistent-storage disable`. that the storage-writes option applies only to local storage and does not work with USB.

Parameter	Description
enable	Enable persistent logging.

Parameter	Description
<code>disable</code>	Disable persistent logging.
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Preserves syslog messages with the severity of alert (6) and emergency (7) • crit: Preserves syslog messages with the severity of critical (5) and above. Default. • debug: Preserves syslog messages with the severity of debug (0) and above. • emer: Preserves syslog messages with the severity of emergency (7) only. • err: Preserves syslog messages with the severity of err (4) and above. • info: Preserves syslog messages with the severity of info (1) and above. • notice: Preserves syslog messages with the severity of notice (2) and above. • warn: Preserves syslog messages with the severity of warning (3) and above.
<code>filter <FILTER_NAME></code>	Preserve logs for specified filter name.
<code>timeout <TIMEOUT></code>	Preserve logs for specified duration. Range: 1 to 60 minutes. Default: 20 minutes.
<code>storage-writes <STORAGE_WRITES></code>	Limit the amount of storage writes.
<code>usb <STORAGE_URL></code>	Preserve logs in the USB storage.

Examples

Enabling storage of logs in storage with severity **err** and timeout 30 minutes:

```
switch(config)#diag persistent-storage enable severity err timeout 30
Logs will be written to storage. Prolonged usage of persistent logging can
affect lifetime of storage. Hence enable persistent logging only for short
duration and also for the right severity.
Do you want to continue (y/n)? y
```

Enabling storage of logs in storage with severity **err** and **timeout** 30 minutes and 512 storage-writes:

```
switch(config)#diag persistent-storage enable severity err timeout 30
storage-writes 512
```

Logs will be written to storage. Prolonged usage of persistent logging can affect lifetime of storage. Hence enable persistent logging only for short duration and also for the right severity.

Do you want to continue (y/n)? y

Disabling log storage:

```
switch(config)# diag persistent-storage disable
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

logging

Syntax

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} {udp [<PORT-NUM>]} |  
{tcp [<PORT-NUM>]} || {tls [<PORT-NUM>]}  
    auth-mode {certificate|subject-name}  
    disable  
    filter <FILTER-NAME>  
    include-auditable-events  
    legacy-tls-renegotiation]  
    rate-limit-burst <BURST>  
    rate-limit-interval <INTERVAL>] ]  
    severity <LEVEL>]  
    vrf <VRF-NAME>]  
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME> }
```

Description

Enables syslog forwarding to a remote syslog server.

The **no** form of this command disables syslog forwarding to a remote syslog server.

Starting with AOS-CX 10.11, payload information is present in accounting logs.

The maximum REST payload that can be sent to RADIUS/TACACS server is 1024 characters, and the maximum of REST payload that can be sent to syslog server is 3500 characters. If this limit is reached, the log will display three dots (...) to indicate that the log exceeded the character limit and is incomplete.

Parameter	Description
{<IPV4-ADDR> <IPV6-ADDR> <HOSTNAME>}	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
[udp [<PORT-NUM>] tcp [<PORT-NUM> tls [<PORT-NUM>]]	Specifies the UDP port, TCP port, or TLS port of the remote syslog server to receive the forwarded syslog messages.
udp [<PORT-NUM>]	Range: 1 to 65535. Default: 514
tcp [<PORT-NUM>]	Range: 1 to 65535. Default: 1470
tls [<PORT-NUM>]	Range: 1 to 65535. Default: 6514
auth-mode	Specifies the TLS authentication mode used to validate the certificate. • certificate: Validates the peer using trust anchor certificate based authentication. Default. • subject - name: Validates the peer using trust anchor certificates as well as subject - name based authentication.
disable	Disable remote syslog configuration. This does not delete the configuration, just disables/pauses the forwarding of syslog messages to the remote server. The config/forwarding can be reenabled (un - paused) again using the no logging <hostname> disable command.
filter <FILTER-NAME>	Specifies the name of the filter to be applied on the syslog messages.
include-auditable-events	Specifies that auditable messages are also logged to the remote syslog server.
legacy-tls-renegotiation	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.

Parameter	Description
<code>rate-limit-burst <BURST></code>	Specifies the rate limit for the messages sent to the remote syslog server.
<code>rate-limit-interval <INTERVAL></code>	Specifies the rate limit interval in seconds. Default: 30 Seconds
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Forwards syslog messages with the severity of alert (6) and emergency (7). • crit: Forwards syslog messages with the severity of critical (5) and above. • debug: Forwards syslog messages with the severity of debug (0) and above. • emerg: Forwards syslog messages with the severity of emergency (7) only. • err: Forwards syslog messages with the severity of err (4) and above • info: Forwards syslog messages with the severity of info (1) and above. Default. • notice: Forwards syslog messages with the severity of notice (2) and above. • warning: Forwards syslog messages with the severity of warning (3) and above.
<code>vrf <VRF-NAME></code>	Specifies the VRF used to connect to the syslog server. Optional. Default: <code>default</code>

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config) # logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of **err (4)** and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF **lab_vrf**:

```
switch(config) # logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config) # no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)#logging example.com tls auth-mode subject-name
```

Applying log filtering for syslog server forwarding:

```
switch(config)# logging 10.0.10.6 severity info filter filter_lldp_logs vrf mgmt
```

Applying log filtering and enabling the rate limit for syslog server forwarding over TCP port:

```
switch(config)# logging 10.0.10.2 tcp 3440 severity err vrf mgmt include-auditable-events filter filter_lldp_logs rate-limit-burst 3 rate-limit-interval 35
```

Command History

Release	Modification
10.12.1000	The disable parameter is introduced
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging accounting-format-native

Syntax

```
logging accounting-format-native  
[no] logging accounting-format-native
```

Description

Change the accounting log message format to native Linux format. (Default: AOS-CX format)

The 'no' form of this command will change the accounting log message format to AOS-CX format.

Usage

This option enables the switch to show all types of accounting records to the user. When configured, the same format will be used while sending messages to syslog servers. When upgrading to this version of from AOS-CX 10.10 or earlier versions, if native accounting logs are preferred, then best practices is to issue this command as a part of the upgrade. If the switch upgrades from AOS-CX 10.10 or earlier without configuring this setting, by default, the accounting log message format will be AOS-CX Format.

Example

This example changes the accounting log message format to native Linux format.

```
switch(config)# logging accounting-format-native
```

The following example returns the accounting log message format to the default AOS-CX format.

```
switch(config)# no logging accounting-format-native
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging filter

Syntax

```
logging filter <FILTER-NAME>
    [{enable | disable}]
    [<SEQUENCE-ID> {permit | deny} [event-id <EVENT-ID-RANGE>] [includes
<REGEX>] [severity <COMPARISON-OPERATOR>
    <LEVEL>]
    no <SEQUENCE-ID>
    resequence <OLD-SEQUENCE-ID>
```

```
<NEW-SEQUENCE-ID>  
no logging filter <FILTER-NAME>
```

Description

Creates a filter to restrict what event or debug logs are logged. A filter can be used to either permit or deny:

- The event logs from being generated on the switch, or
- The event or debug logs generated on the switch from being forwarded to a syslog server.

A filter is identified by a filter name and can have up to 20 rules or entries, each with a different sequence number, matching criteria, and corresponding action (deny or permit). When a filter is applied on a log, the log is matched against the criteria mentioned in the rules or entries in ascending numerical order of their sequence numbers until a matching entry is found. Once a matching entry is found, its corresponding action is applied on the log. If no matching rule is found, the default action (permit) is applied.

The **no** form of this command removes the filter.

Parameter	Description
<code><FILTER-NAME></code>	Specifies the unique name to identify the filter.
<code>enable</code>	Filter event logs generated on the switch.
<code><SEQUENCE-ID></code>	Specifies the filter criteria sequence number. Default: Increments by 10 from the largest sequence - id currently used in this filter.
<code>deny</code>	Prevents the matching log from being logged.
<code>permit</code>	Allows the matching log.
<code><event-id></code>	Matches logs by event ID. Specify an event ID or a range of event IDs. It supports a maximum of 100 event IDs.
<code>includes <REGEX></code>	Matches the log message against a regular expression string.
<code>severity</code>	Matches the logs by severity level. The following options are used to compare the severity: • eq: Match events of severity equal to the specified.

Parameter	Description
	• ge: Match events of severity greater than or equal to the specified. • gt: Match events of severity greater than the specified. • le: Match events of severity lesser than or equal to the specified. • lt: Match events of severity lesser than the specified. The following are the severity levels: • alert: Logs with the severity alert (6). • crit: Logs with the severity critical (5). • debug: Logs with the severity debug (0). • emerg: Logs with the severity emergency (7). • err: Logs with the severity err (4). • info: Logs with the severity info (1). • notice: Logs with the severity notice (2). • warning: Logs with the severity warning (3).

Usage

Filtering event logs on the switch: To permit or deny event logs from being generated on the switch. In this case, the matching event logs are filtered at generation. The denied event logs are neither logged to the switch events nor forwarded to any remote syslog servers. Multiple filters can be configured, but only one filter can be applied to filter the events on the switch. Such a filter can be chosen by adding the **enable** command under its configuration. Configuring the **enable** command under a new filter automatically removes it from the filter where it was previously used.

For example:

```
logging filter low_severity_logs
enable
10 deny severity lt info
```

This configuration denies the event logs which have a severity less than info.



NOTE

If a filter contains **enable** command, it is not recommended to configure this filter in the **logging** command used for remote syslog server configuration. This is because, any event logs denied by the filter are already not available for forwarding to a remote server.

A filter with **enable** command will not affect debug logs. Consider the configuration in the following example of a filter with **enable** command and two rules applied **10 permit severity ge info** and **20 deny**. This implies permit only those event logs which have severity greater than or equal to **info**.

Example:

```
logging filter low_severity_logs
enable
10 permit severity ge info
20 deny
```

Filtering event or debug logs when forwarding to a remote syslog server: The filter name must be configured in the logging command that is used to configure remote syslog server. The logs will be generated on the switch and the filter only decides whether to deny or permit the syslog forwarding for the matching log. For example: **logging 10.0.10.6 filter filter_ldap_logs**



NOTE

The filter affects debug logs only when the command **debug destination syslog** is configured on the switch.



NOTE

The severity mentioned in the remote syslog server configuration using logging command under configuration context has more precedence than the severity mentioned in a filter entry. If a log with warning severity is permitted by a filter, but the remote syslog configuration has severity **err** mentioned in it, the log will not be forwarded to the remote syslog server (since warning(3) is lesser than err(4)). On the other hand, if a log with **err** severity is permitted by a filter and the remote syslog configuration has severity warning mentioned in it, the log will be forwarded to the remote syslog server.

Examples

Configuring a new logging filter:

```
switch(config)# logging filter example_filter
```

To deny logs having event ID 1301 and a range of event IDs from 1305 to 1309:

```
switch(config-logging-filter)# 20 deny event-id 1301,1305-1309
```

To permit logs having event ID 1300:

```
switch(config-logging-filter)# 30 permit event-id 1300
```

To permit logs with severity greater than or equal to `err`:

```
switch(config-logging-filter)# 30 permit severity ge err
```

To deny logs with severity greater than info:

```
switch(config-logging-filter)# 30 deny severity gt info
```

To deny logs with event ID 1024 and a message matching the regular expression LLDP:

```
switch(config-logging-filter)# 40 deny event-id 1024 includes LLDP
```

Denying all logs:

```
switch(config-logging-filter)# 40 deny
```

Changing the sequence ID of an existing rule:

```
switch(config-logging-filter)# resequence 20 70
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config and config-logging-filter	Administrators or local user group members with execution rights for this command.

logging facility

Syntax

```
logging facility {local0 | local1 | local2 | local3 | local4 | local5 |  
local6 | local7}  
no logging facility
```

Description

Sets the logging facility to be used for remote syslog messages. Default: `local7`

The **no** form of this command disables the logging facility to be used for remote syslog messages.

Parameter	Description
<code>{local0 local1 local2 local3 local4 local5 local6 local7}</code>	Selects the logging facility to be used for remote syslog messages. Required. Specifies the severity of the syslog messages: • local0 • local1 • local2 • local3 • local4 • local5 • local6 • local7

Examples

Sets the local5 logging facility to be used for remote syslog messages:

```
switch(config)# logging facility local5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

logging persistent-storage

Syntax

```
logging persistent-storage [severity {alert|crit|debug|emerg|err|info|
notice|warning}]
no logging persistent-storage
```

Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.

The **no** form of this command disables storage of logs in storage.



NOTE

This command is deprecated. Please use "diag persistent-storage" command to preserve logs in storage.

Parameter	Description
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Preserves syslog messages with the severity of alert (6) and emergency (7) • crit: Preserves syslog messages with the severity of critical (5) and above. Default. • debug: Preserves syslog messages with the severity of debug (0) and above. • emerg: Preserves syslog messages with the severity of emergency (7) only. • err: Preserves syslog messages with the severity of err (4) and above. • info: Preserves syslog messages with the severity of info (1) and above. • notice: Preserves syslog messages with the severity of notice (2) and above. • warning: Preserves syslog messages with the severity of warning (3) and above.

Usage

These logs can be copied out by using the **copy support-files all** or **copy support-files previous-boot**.

Examples

Enabling storage of logs in storage with severity **info**:

```
switch(config)#logging persistent-storage severity info  
Logs will be written to storage and made available across reboot.  
Do you want to continue (y/n)?
```

Disabling storage of logs in storage:

```
switch(config)# no logging persistent-storage
```

Command History

Release	Modification
10.17	Command deprecated.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Runtime Diagnostics

Run Time diagnostics framework is intended to monitor and validate the health of different hardware components present in the system. It uses a set of safe hardware diagnostics test cases to validate the health of different hardware components. These diagnostics test cases are run periodically at every predetermined interval. Additionally, these hardware diagnostics test cases can be run on demand.

Subtopics

Runtime diagnostic commands

Runtime diagnostic commands

Select a command from the list in the left navigation menu.

Subtopics

- [diagnostic monitor](#)
- [diag on-demand](#)
- [show diagnostic](#)
- [show diagnostic events](#)

diagnostic monitor

Syntax

```
diagnostic monitor {fan-tray | line-module | management-module} [<SLOT-ID>]
no diagnostic monitor {fan-tray | line-module | management-module} [<SLOT-ID>]
```

For 6400 switches only:

```
diagnostic monitor {fabric <SLOT-ID>}
no diagnostic monitor {fabric <SLOT-ID>}
```

Description

Enables runtime diagnostics for all modules or for a specified module. This feature is enabled by default for all modules.

The **no** form of this command disables runtime diagnostics for all modules or for a specified module.

Parameter	Description
fan-tray	Specifies the enabling of diagnostic monitoring specific to a fan tray.
line-module	Specifies the enabling of diagnostic monitoring specific to a line module.
management-module	Specifies the enabling of diagnostic monitoring specific to a management module.
<SLOT-ID>	Specifies the slot ID of a module. Format: member/slot.

Usage

When no parameters are used in the command (**diagnostic monitor** or **no diagnostic monitor**), the command applies to all modules. This command impacts the diagnostics that run periodically. It does not affect on-demand diagnostics.

Example

Enabling runtime diagnostics for a specified module:

```
switch(config)# diagnostic monitor management-module 1/1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

diag on-demand

Syntax

```
diag on-demand {fan-tray | line-module | management-module} [<SLOT-ID>]
```

For 6400 switches only:

```
diag on-demand {fabric <SLOT-ID>}
```

Description

Runs the diagnostic tests for all modules or for a specified module.

Parameter	Description
[fan-tray line-module management-module]	Selects the options for enabling or disabling run - time diagnostics for a specific module.
fan-tray	Specifies the enabling of diagnostic monitoring specific to a fan tray.
line-module	Specifies the enabling of diagnostic monitoring specific to a line module.
management-module	Specifies the enabling of diagnostic monitoring specific to a management module.
<SLOT-ID>	Specifies the member/slot for management modules (1/1 or 1/2), line modules (1/3 - 1/7, 1/8 - 1/12), fan trays (1/1 - 1/3), and fabric modules (1/1 - 1/2) on a 6400 switch. Specifies the member/slot for management modules (1/1), line modules (1/1), and fan trays (1/1 - 1/2) on a 6300 switch.

Usage

When no parameters are used in the command (**diag on-demand**), the command applies to all modules.

Examples

Running diagnostic tests for all modules on a 6300 switch:

```
switch# diag on-demand
Fetching Test results. Please wait ...
Module           ID      Diagnostics Success
                  ID      Performed
-----
FanTray          1/2          1      100%
FanTray          1/1          1      100%
LineModule       1/1         13      100%
ManagementModule 1/1         13      100%
```

Running diagnostic tests for a specific module on a 6300 switch:

```
switch# diag on-demand management-module 1/1
Performing diagnostic tests. Please wait ...
Fetching Test results. Please wait ...
Module           ID      Diagnostics Success
                  ID      Performed
-----
ManagementModule 1/1         13      100%
```

Running diagnostic tests for all modules on a 6400 switch:

```
switch# diag on-demand
Fetching Test results. Please wait ...
Module                ID      Diagnostics Success
                        ID      Performed
-----
FanTray                1/2          2      100%
LineModule              1/3          24     100%
ManagementModule        1/1          19     100%
LineModule              1/7          12     100%
Fabric                  1/1           6     100%
LineModule              1/5          24     100%
LineModule              1/4          24     100%
FanTray                 1/1           2     100%
LineModule              1/6          24     100%
```

Running diagnostic tests for a specific module on a 6400 switch:

```
switch# diag on-demand management-module
Performing diagnostic tests. Please wait ...
Fetching Test results. Please wait ...
Module                ID      Diagnostics Success
                        ID      Performed
-----
ManagementModule      1/1          19     100%
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

show diagnostic

Syntax

```
show diagnostic {fan-tray | line-module | management-module} [<SLOT-ID>]
{brief | detail} [vsx-peer]
```

Description

Displays the diagnostic test results for all modules or for a specified module.

Parameter	Description
[fan-tray line-module management-module]	Selects the options for enabling or disabling runtime diagnostics for a specific module.
fan-tray	Specifies the enabling of diagnostic monitoring specific to a fan tray.
line-module	Specifies the enabling of diagnostic monitoring specific to a line module.
management-module	Specifies the enabling of diagnostic monitoring specific to a management module.
<SLOT-ID>	Specifies the member/slot for management modules (1/1), line modules (1/1), and fan trays (1/1 - 1/2) on the 6300 switch.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

When no parameters are used in the command (**show diagnostic**), the command applies to all modules.

Example

Showing diagnostic test results in brief format for all modules on a 6300 switch:

```
switch# show diagnostic brief
Module                ID      Diagnostics Success
                   Performed
-----
ManagementModule     1/1          13      100%
LineModule            1/1          13      100%
```

FanTray	1/1	1	100%
FanTray	1/2	1	100%

Showing diagnostic test results in brief format for a specified module on a 6300 switch:

```
switch# show diagnostic line-module brief
Module          ID      Diagnostics Success
                ID      Performed
-----
LineModule      1/1          13      100%
```

Showing diagnostic test results in detail format for all modules on a 6300 switch:

```
switch# show diagnostic detail
Module : ManagementModule 1/1
Diagnostic      Status Error Code History Code Successive      Total Failure
Total          Last Run Timestamp      First Run Timestamp      Failure Count Count
Iteration
-----
ddr_cecount     Pass   0x0      0x0      0
0              109  2019-07-31 16:43:38  2019-07-31 07:44:55
emmc            Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
fan_ctrlr       Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
fepld           Pass   0x0      0x0      0
0              109  2019-07-31 16:43:38  2019-07-31 07:44:54
fru_eeeprom     Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:54
fru_eeeprom_ul  Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:54
mm_lcb          Pass   0x0      0x0      0
0              109  2019-07-31 16:43:37  2019-07-31 07:44:54
pmc             Pass   0x0      0x0      0
0              109  2019-07-31 16:43:37  2019-07-31 07:44:54
rdimm_spd       Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
rdimm_tmp       Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
rtc             Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
tmp1            Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
tmp2            Pass   0x0      0x0      0
0              4   2019-07-31 16:08:04  2019-07-31 07:44:55
```

Module : LineModule 1/1

Diagnostic Total	Status Last Run	Error Timestamp	Code First Run	History Timestamp	Code First Run	Successive Timestamp	Total Failure Count	Failure Count
---------------------	--------------------	--------------------	-------------------	----------------------	-------------------	-------------------------	---------------------------	------------------

Iteration

lc_asic	Pass	0x0	0x0				0	
0	108	2019-07-31 16:43:37	2019-07-31 07:46:03					
poe_ctrlr_1_q1	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:03					
poe_ctrlr_1_q2	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:04					
poe_ctrlr_1_q3	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:04					
poe_ctrlr_2_q1	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:05					
poe_ctrlr_2_q2	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:05					
poe_ctrlr_2_q3	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:05					
poe_ctrlr_3_q1	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:06					
poe_ctrlr_3_q2	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:16	2019-07-31 07:46:06					
poe_ctrlr_3_q3	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:17	2019-07-31 07:46:06					
poe_ctrlr_4_q1	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:17	2019-07-31 07:46:07					
poe_ctrlr_4_q2	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:17	2019-07-31 07:46:07					
poe_ctrlr_4_q3	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:17	2019-07-31 07:46:08					

Module : FanTray 1/1

Diagnostic Total	Status Last Run	Error Timestamp	Code First Run	History Timestamp	Code First Run	Successive Timestamp	Total Failure Count	Failure Count
---------------------	--------------------	--------------------	-------------------	----------------------	-------------------	-------------------------	---------------------------	------------------

Iteration

ft1_eeeprom	Pass	0x0	0x0				0	
0	4	2019-07-31 16:08:33	2019-07-31 07:44:54					

Module : FanTray 1/2

Diagnostic	Status	Error	Code	History	Code	Successive	Total	Failure
Total	Last Run	Timestamp	First Run	Timestamp		Failure Count	Count	
Iteration								

ft2_eeeprom	Pass	0x0		0x0			0	
0	3	2019-07-31 16:07:50	2019-07-31	07:44:54				

Showing diagnostic test results in detail format for a specified module on a 6300 switch:

```
switch# show diagnostic management-module detail
```

Module : ManagementModule 1/1								
Diagnostic	Status	Error	Code	History	Code	Successive	Total	Failure
Total	Last Run	Timestamp	First Run	Timestamp		Failure Count	Count	
Iteration								

ddr_cecount	Pass	0x0		0x0			0	
0	109	2019-07-31 16:43:38	2019-07-31	07:44:55				
emmc	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				
fan_ctrlr	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				
fepld	Pass	0x0		0x0			0	
0	109	2019-07-31 16:43:38	2019-07-31	07:44:54				
fru_eeeprom	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:54				
fru_eeeprom_ul	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:54				
mm_lcb	Pass	0x0		0x0			0	
0	109	2019-07-31 16:43:37	2019-07-31	07:44:54				
pmc	Pass	0x0		0x0			0	
0	109	2019-07-31 16:43:37	2019-07-31	07:44:54				
rdimm_spd	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				
rdimm_tmp	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				
rtc	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				
tmp1	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				
tmp2	Pass	0x0		0x0			0	
0	4	2019-07-31 16:08:04	2019-07-31	07:44:55				

Showing diagnostic test results in brief format for all modules on a 6400 switch:

```
switch# show diagnostic brief
```

Module	ID	Diagnostics Performed	Success
ManagementModule	1/1	19	100%
LineModule	1/3	24	100%
LineModule	1/7	12	100%
LineModule	1/5	24	100%
LineModule	1/4	24	100%
LineModule	1/6	24	100%
Fabric	1/1	6	100%
FanTray	1/2	2	100%
FanTray	1/1	2	100%

Showing diagnostic test results in brief format for a specified module on a 6400 switch:

```
switch# show diagnostic management-module brief
```

Module	ID	Diagnostics Performed	Success
ManagementModule	1/1	19	100%

Showing diagnostic test results in detail format for a specified module on a 6400 switch:

```
switch# show diagnostic management-module detail
```

Module : ManagementModule 1/1

Diagnostic Total	Status Last Run	Error Code Timestp	History Code	Successive Failure Count	Total Failure Count
curr_sensor	Pass	0x0	0x0	0	
0	2	2019-10-14 00:25			
ddr_cecount	Pass	0x0	0x0	0	
0	34	2019-10-14 01:26			
eeeprom	Pass	0x0	0x0	0	
0	2	2019-10-14 00:25			
eeeprom_ul	Pass	0x0	0x0	0	
0	2	2019-10-14 00:25			
emmc	Pass	0x0	0x0	0	
0	2	2019-10-14 00:26			
icbbp	Pass	0x0	0x0	0	
0	34	2019-10-14 01:24			
icbx	Pass	0x0	0x0	0	

0	34	2019-10-14 01:25			
ledpld		Pass	0x0	0x0	0
0	34	2019-10-14 01:24			
mm_mcb		Pass	0x0	0x0	0
0	34	2019-10-14 01:24			
psu1		Pass	0x0	0x0	0
0	2	2019-10-14 00:27			
psu1_eeeprom		Pass	0x0	0x0	0
0	2	2019-10-14 00:26			
psu2		Pass	0x0	0x0	0
0	2	2019-10-14 00:27			
psu2_eeeprom		Pass	0x0	0x0	0
0	2	2019-10-14 00:27			
rdimm_spd		Pass	0x0	0x0	0
0	2	2019-10-14 00:26			
rdimm_tmp		Pass	0x0	0x0	0
0	2	2019-10-14 00:26			
rtc		Pass	0x0	0x0	0
0	2	2019-10-14 00:26			
tmp1		Pass	0x0	0x0	0
0	2	2019-10-14 00:25			
tmp2		Pass	0x0	0x0	0
0	2	2019-10-14 00:25			
tmp3		Pass	0x0	0x0	0
0	2	2019-10-14 00:25			

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

show diagnostic events

Syntax

```
show diagnostic events
```

Description

Displays the diagnostic related event logs.

Example

Showing diagnostic related event logs:

```
switch# show diagnostic events
2019-08-07:17:19:21.214532|hhmd|106001|ERR|
Diagnostic mm_mcbe failed with error code 0x380 on management module 1/1
2019-08-07:17:19:21.214554|hhmd|106001|ERR|
Diagnostic pmc failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.215532|hhmd|106001|ERR|
Diagnostic ledpld failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.353221|hhmd|106001|ERR|
Diagnostic mm_mcbe failed with error code 0x380 on management module 1/1
2019-08-07:17:19:21.354421|hhmd|106001|ERR|
Diagnostic pmc failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.453221|hhmd|106001|ERR|
Diagnostic ledpld failed with error code 0x4 on management module 1/1
```

Showing diagnostic related event logs (Output from a 6400 switch):

```
switch# show diagnostic events
-----
Event logs from current boot
-----
2019-10-17T20:27:04.066486+00:00 6405 hhmd[9237]: Event|3002|LOG_ERR|LC|1/6|
Diagnostic brd_tmp1 failed with error code 0x1000000 on line card 4
2019-10-17T20:27:04.102968+00:00 6405 hhmd[9237]: Event|3002|LOG_ERR|LC|1/3|
Diagnostic brd_tmp1 failed with error code 0x1000000 on line card 1
2019-10-17T20:27:04.117467+00:00 6405 hhmd[9237]: Event|3002|LOG_ERR|LC|1/5|
Diagnostic brd_tmp1 failed with error code 0x1000000 on line card 3
2019-10-17T20:27:04.210276+00:00 6405 hhmd[9237]: Event|3002|LOG_ERR|LC|1/4|
Diagnostic brd_tmp1 failed with error code 0x1000000 on line card 2
2019-10-17T20:27:04.212133+00:00 6405 hhmd[9237]: Event|3002|LOG_ERR|LC|1/7|
Diagnostic brd_tmp1 failed with error code 0x1000000 on line card 5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

Service OS

Service OS is an operating system that the customer only uses to fix filesystem corruption, download and update firmware, and other support related issues. HPE Service OS is a Linux distribution that acts as a standalone bootloader and recovery OS for -based switches. It is only accessible if the user is consoled into the switch. The main high level features provided include:

- Access to file system partitions for retrieval of logs, coredumps, and configuration for supportability purposes.
- Filesystem utilities to format and partition a corrupted storage disk.
- Management interface networking with TFTP to download and update a product image.
- Ability to boot primary and secondary firmware images (.SWI file) on the storage disk.
- Support for clearing the startup-config.
- Ability to not only clear the admin password for , but also change it in SVOS.
- Ability to set the secure mode to enhanced or standard.

This document covers the customer CLI commands available in Service OS, as well as a few non-CLI features.

Subtopics

Service OS CLI login

Service OS user accounts

Service OS boot menu

Console configuration

AOS-CX boot

File system access

Service of OS mount failure

Service OS CLI command list

Service OS CLI features and limitations

Service OS CLI commands

Service OS CLI login

Description

If the user enters 0 at the boot menu prompt, they will be presented with a Service OS CLI login prompt. The user must enter the login account "admin" to log in. By default, Service OS does not require a password.

To reboot without logging in, enter **reboot** as the login user name.

There are two additional login accounts that execute a command without requiring a password: **reboot** and **zeroize**. Enter the login account **reboot** to reboot the management module and **zeroize** to initiate a zeroization process. The zeroize user account helps a user reset the admin user account's password.

Example

```
To reboot without logging in, enter 'reboot' as the login user name.
```

```
ServiceOS login: admin
```

```
SVOS>
```

```
.....
```

```
.....
```

```
To reboot without logging in, enter 'reboot' as the login user name.
```

```
ServiceOS login: reboot
```

```
reboot: Restarting system
```

```
.
```

```
Looking for SVOS.
```

```
Primary SVOS: Checking...Loading...Finding...Verifying...Booting...
```

```
ServiceOS Information:
```

```
Version:          FL.01.07.0002-internal
```

```
Build Date:       2020-09-03 10:38:03 PDT
```

```
Build ID:         ServiceOS:FL.01.07.0002-
```

```
internal:1a017598b673:202009031038
```

```
SHA:              1a017598b6738448ef679175712e022a966eca88
```

```
.....
```

```
.....
```

```
To reboot without logging in, enter 'reboot' as the login user name.
```

```
ServiceOS login: zeroize
```

```
This will securely erase all customer data, including passwords, and
```

```

reset the switch to factory defaults.
This action requires proof of physical access via a USB drive.
* Create a FAT32 formatted USB drive
* Create a file in the root directory of the USB drive named zeroize.txt
* Type the following serial number into the zeroize.txt file: SG9ZKN7050
* Insert the USB drive into the target module
* Confirm the following prompt to continue
Continue (y/n)? y
#####WARNING#####
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####
Continue (y/n)? y
reboot: Restarting system

```

Service OS user accounts

Service OS provides a single admin login account. By default, no password is required to log in. Service OS will require a password if the Service OS admin user account password feature is enabled. This setting can be enabled or disabled in .

Service OS boot menu

Description

On boot, the user is presented with a Service OS version banner with version, build date, build time, build ID, and SHA strings.

The user is then shown the boot image profiles.

- Enter 0 to boot the Service OS login CLI.
- Enter 1 to boot the primary firmware image.
- Enter 2 to boot the secondary firmware image.
- If no input is given within 5 seconds, the default boot profile is selected. Alternatively, press **Enter** to select the default boot profile.

The image selected by the user during boot is a run-time decision only and will not persist across reboots. The default image can be configured using the `boot set-default` command.

Example

```
ServiceOS Information:
  Version:          FL.01.xx.xxxx-internal
  Build Date:       2020-09-03 10:38:03 PDT
  Build ID:         ServiceOS:FL.01.xx.xxxx-
internal:1a017598b673:202009031038
  SHA:             1a017598b6738448ef679175712e022a966eca88
Boot Profiles:
0. Service OS Console
1. Primary Software Image [FL.10.xx.xxxx]
2. Secondary Software Image [FL.10.xx.xxxx-263-g4ac34862e3a]
Select profile(primary):
```



NOTE

The (primary) string in the boot menu displays the default boot profile that will be booted after the timeout period. This string will change to (secondary) or (Service OS) depending on the current default boot option.

Console configuration

During boot, Service OS communicates with the RJ45 serial console with a baud rate of 115200. There is no option to change the baud rate during boot.

Additionally, if a USB console is connected to the management module console port, input will automatically be switched over to use the USB console. Automatic switching to USB is consistent with the USB console behavior.



NOTE

Console output always displays on both the RJ45 console port and the USB console port.

AOS-CX boot

Description

After the user has input a boot profile selection at the boot menu or the 5-second selection timeout has expired, Service OS will boot an AOS-CX image.

Service OS displays the following boot strings embedded in the product image header:

- Image name
- Image version
- Build ID
- Build date

Service OS will then present status and boot the image.

Example

```
Booting primary software image...
Verifying Image...
Image Info:
    Name: AOS-CX
    Version: XL.01.01.0001
    Build Id: AOS-CX:XL.01.01.0001:1a36111da4e0:201707171452
    Build Date: 2017-07-17 14:52:27 PDT
Extracting Image...
Loading Image...
Done.
kexec: Starting new kernel
```

File system access

Description

When the user logs in to the Service OS CLI, they are presented with a limited file system. The user can use standard file system commands of `cd`, `ls`, and `pwd` to view and move through the file system.

On login, the user is first placed in the `/home` directory:

```
(C) Copyright 2017 Hewlett Packard Enterprise Development LP
          RESTRICTED RIGHTS LEGEND
Confidential computer software. Valid license from Hewlett Packard
Enterprise
Development LP required for possession, use or copying. Consistent with FAR
12.211 and 12.212, Commercial Computer Software, Computer Software
Documentation, and Technical Data for Commercial Items are licensed to the
U.S. Government under vendor's standard commercial license.
To reboot without logging in, enter 'reboot' as the login user name.
ServiceOS login: admin
SVOS> pwd
/home
SVOS>
```

The home directory and the USB device (/mnt/usb and any sub directory) are the only writable directories available. These directories can be used as a staging location for downloading product images using TFTP. /home can also be used as temporary storage before copying files from the management module through TFTP or USB. Any changes made to /home will not persist across reboots or after booting an AOS-CX image.

The root / directory displays viewable directories:

```
SVOS> ls /  
bin          coredump    lib          mnt          selftest  
cli          home        logs         nos  
SVOS>
```

The directories coredump, selftest, nos, and logs each provide the user access to an SSD partition mount. The user may read, but not write any file on these partitions.

These mount points allow the user to copy files on the SSD to a USB storage device or upload files using TFTP. Copying files from the SSD is intended to be used under the guidance of a support engineer (to upload logs or coredumps to HPE support).

USB storage device access is provided through the mount at /mnt/usb.

The remaining directories in the root file system bin, cli, and lib are not intended to be used by the customer.

Service of OS mount failure

Description

If the SSD is detected as missing or any of the partitions could not be mounted, Service OS will force the user to boot to the Service OS console and display an error message indicating that recovery should be attempted using the format command.

Example

```
(C) Copyright 2017 Hewlett Packard Enterprise Development LP  
RESTRICTED RIGHTS LEGEND  
Confidential computer software. Valid license from Hewlett Packard  
Enterprise  
Development LP required for possession, use or copying. Consistent with FAR  
12.211 and 12.212, Commercial Computer Software, Computer Software  
Documentation, and Technical Data for Commercial Items are licensed to the  
U.S. Government under vendor's standard commercial license.  
To reboot without logging in, enter 'reboot' as the login user name.  
Error, Could not mount the primary storage device.  
This may be due to filesystem or device corruption.  
Please attempt to recover using the "format" command.  
ServiceOS login:
```

Service OS CLI command list

Description

After login to Service OS CLI, the user may enter the commands help or ? to get a full list of commands and a terse description for each command. The user may also enter <command> followed by --help to get more detailed help and usage for a specific command.

Example

```
SVOS> ?
Available Commands:
    ? - Display help screen
    cd - Change the working directory
    pwd - Print the current working directory
    help - Display help screen
allow-unsafe-updates - Allow non-failsafe updates for a limited amount of
time
    boot - Boot a product image
config-clear - Clears the startup-config
    diag - Run diagnostic commands
    erase - Securely erase storage devices on the management
module
    format - Formats and partitions the primary storage device
identify - Prints hardware identification information
    ip - Sets the OOBM Port Network Configuration
    mount - Mount a storage device
password - Set the admin account password
    ping - Send ICMP ECHO_REQUEST to network hosts (IPv4)
    reboot - Reboots the Management Module
secure-mode - Set or retrieve the secure mode setting
    sh - Launch support shell
    umount - Unmounts a storage device
    update - Update a product image
version - Prints ServiceOS release version information
    cat - Prints files to stdout
    cp - Copy files and directories
    du - Estimate file space usage
    ls - List directory contents
md5sum - Compute and check md5 message digest
    mkdir - Make directories
    mv - Move (rename) files
```

```
rm - Remove files or directories
rmdir - Remove empty directories
tftp - Allows transfer of files to/from a remote machine
exit - Logout
Enter '<command> --help' for more info
```

Service OS CLI features and limitations

The Service OS CLI provides basic shell functionality that allows you to execute commands and pass arguments to those commands only. The following features are not available:

- Input/output redirection (<, >, >>)
- Job control (&, fg, bg)
- Process piping (|)
- File globbing (*)



NOTE

Even though the Service OS CLI does not provide file globbing capabilities, some commands may provide this functionality internally. An example is the `ls` command.

The following common features are available:

- Command history (**Up Arrow**) and search (**Ctrl-R**)
- Tab completion for file and folder names (not CLI commands)
- Command abort using **Ctrl-C**

Service OS CLI commands

Select a command from the list in the left navigation menu.

Subtopics

[boot](#)

[cat](#)

[cd path](#)

[config-clear](#)

[cp](#)

du
erase zeroize
exit
format
identify
ip
ls
md5sum
mkdir
mount
mv
password (svos)
ping
pwd
reboot
rm
rmdir
secure-mode
sh
system serviceos password-prompt
umount
update
tftp
version

boot

Syntax

```
boot
```

Description

Presents you with the boot menu prompt. You can then specify which boot profile: primary, secondary, or Service OS console.

Example

Presenting the boot menu prompt:

```
SVOS> boot
ServiceOS Information:
  Version:          FL.01.07.0002-internal
  Build Date:       2020-09-03 10:38:03 PDT
```

```
Build ID:          ServiceOS:FL.01.07.0002-
internal:1a017598b673:202009031038
SHA:              1a017598b6738448ef679175712e022a966eca88
Boot Profiles:
0. Service OS Console
1. Primary Software Image [FL.10.06.0001]
2. Secondary Software Image [FL.10.08.0000-308-gcfbc0e3]
Select profile(primary):
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

cat

Syntax

```
cat <FILENAME/DIRECTORY-NAME>
```

Description

Prints the contents of a file to the console. The Service OS does not allow command output redirection, so this command is only useful for reading short text files.

Parameter	Description
<code><FILENAME/DIRECTORY-NAME></code>	Shows the contents of the specified file or directory.

Example

Showing the contents of /nos/hosts:

```
SVOS> cat /nos/hosts
127.0.0.1      localhost.localdomain      localhost
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

cd path

Syntax

```
cd path
```

Description

Changes the current working directory.

Example

Changing the current working directory:

```
cd /
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

config-clear

Syntax

```
config-clear
```

Description

Configures the switch to set all configuration settings to factory default when the switch is restarted. The next time the switch starts, the current **startup-config** is renamed to **startup-config-fixme**, and a new **startup-config** is created with factory default settings.



NOTE

Using this command is not the same as performing zeroization, which securely erases the entire primary storage and other devices, and not just the configuration.

Example

Configuring the system to clear the switch configuration:

```
SVOS> config-clear
The switch configuration will be cleared.
Continue (y/n)? y
The system has been configured to clear the startup-config on the next
boot. Please execute the 'boot' command to complete this action.
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (SVOS)	Administrators or local user group members with execution rights for this command.

cp

Syntax

```
cp [options] <SOURCE-FILENAME/SOURCE-DIRECTORY>  
      <DESTINATION-FILENAME/DESTINATION-DIRECTORY>
```

Description

Copies files or directories.

Parameter	Description
[options]	Selects the options for the command.
-d, -P	Specifies the preservation of symlinks (default if - R).
-a	Same as - dpR .
R , -r	Specifies recursiveness, all files, and subdirectories are copied.
-L	Specifies the following of all symlinks.
-H	Specifies the following of symlinks on command line.
-p	Specifies the preservation of file attributes if possible.

Parameter	Description
<code>-f</code>	Specifies the overwriting of a file or directory.
<code>-i</code>	Specifies the prompting before an overwrite.
<code>-l, -s</code>	Specifies the creation of (sym) links.
<code><SOURCE-FILENAME/SOURCE-DIRECTORY></code>	Specifies the name of the source file or directory.
<code><DESTINATION-FILENAME/DESTINATION-DIRECTORY></code>	Specifies the name of the destination file or directory.

Example

Copying /home/customers directory to the /home/clients directory:

```
SVOS> cp /home/customers /home/clients
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

du

Syntax

```
du [options] <FILENAME/DIRECTORY-NAME>...
```

Description

Shows estimated disk space used for each file or directory or both.

Parameter	Description
[options]	Selects the options for the command.
-a	Show file sizes.
-L	Shows all symlinks.
-H	Shows symlinks on a command line.
-d, N	Shows limited output to directories (and files with -a) of depth less than N .
-c	Shows the total disk space usage of all files or directories or both.
-l	Shows the count sizes if hard linked.
-s	Shows only a total for each argument.
-x	Does not show directories on different file systems.
-h	Show sizes in human readable format (1K, 243M, and 2G).
-m	Show sizes in megabytes.
-k	Show sizes in kilobytes (default).
<FILENAME/DIRECTORY-NAME>	Specifies the file or directory or both for displaying a size estimate.

Example

Estimating disk space for the /nos directory:

```
SVOS> du -ah /nos
196.4M  /nos/primary.swi
196.4M  /nos
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

erase zeroize

Syntax

```
erase zeroize
```

Description

Securely erases any user data contained on the SSD or other storage devices on the management module.



NOTE

Back up all data before running this command or all user/config data will be lost.

Usage

Use this command to securely erase all customer data and restore the software environment to factory default. When you issue this command:

- Software images are copied to RAM to be restored on completion.
- All bits undergo a 0>1>0 transition to completely zeroize data. This data is not recoverable.

- This feature can be used to remove all configuration settings or system alterations for debugging or troubleshooting.
- The zeroization process takes approximately two minutes.



NOTE

All logs and data are lost in the zeroization process. Best practices is to collect all applicable data before performing zeroization.

Example

Erasing user data:

```
SVOS> SVOS> erase --help
Usage: erase zeroize
Securely erases storage devices on the management module.
SVOS>
...
...

SVOS> erase zeroize
#####WARNING#####
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####
Continue (y/n)? y
reboot: Restarting system
ServiceOS Information:
  Version: FL.01.07.0002-internal
  Build Date: 2020-09-02 11:53:34 PDT
  Build ID: ServiceOS:FL.01.07.0002-internal:1a017598b673:202009031038
  SHA: 1a017598b6738448ef679175712e022a966eca88
##### Preparing for zeroization #####
##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
##### This should take several minutes #####
##### to one hour to complete #####
##### Restoring files #####
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

exit

Syntax

```
exit
```

Description

Logs the user out from the `SVOS>` prompt.

Example

Logging the user out from the `SVOS>` prompt:

```
SVOS> exit
(C) Copyright  Hewlett Packard Enterprise Development LP
                RESTRICTED RIGHTS LEGEND
Confidential computer software. Valid license from Hewlett Packard
Enterprise
Development LP required for possession, use or copying. Consistent with FAR
12.211 and 12.212, Commercial Computer Software, Computer Software
Documentation, and Technical Data for Commercial Items are licensed to the
U.S. Government under vendor's standard commercial license.
To reboot without logging in, enter 'reboot' as the login user name.
ServiceOS login:
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

format

Syntax

```
format
```

Description

Configures the primary storage device with the correct partition and file system formatting. This command removes all pre-existing data on the primary storage device.

Example

Configuring the primary storage device with the correct partition and file system formatting:

```
SVOS> format
#####WARNING#####
The following action will cause all data on
the primary storage device to be lost. After
formatting has completed, a reboot will be
initiated to complete storage initialization.
#####WARNING#####
Continue? (y/n): y
Working...This may take a few minutes...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

identify

Syntax

```
identify
```

Description

Prints the version and serial number information of hardware devices on the management module (for example, FPGAs, PLDs).

Example

Output from a 6400/6300 switch:

```
SVOS> identify
mc svos_primary           : FL.01.05.0001
mc svos_secondary         : FL.01.05.0001
mc uboot_single            : FL.01.0001
mc uboot_capsule           : FL.01.0001
mc pmc_single              : 0x4
mc pmc_primary             : 0x4
mc pmc_secondary           : 0x4
mc mcb_single              : 0x6
mc mcb_primary             : 0x6
mc mcb_secondary           : 0x6
mc mcb_factory             : 0x3
mc ledpld_single           : 0x4
mc ledpld_primary          : 0x4
```

```
mc ledpld_secondary      : 0x4
mc tpm                   : 0x102420E
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

ip

Syntax

```
ip {show | dhcp | disable | addr <ADDR-NETMASK-GATEWAY>}
```

Description

Shows or configures the port with a static IP address (IPv4 only) or enables the DHCP client on the port. An address is set only if a DHCP server is available to provide one.

Parameter	Description
{show dhcp disable addr <ADDR-NETMASK-GATEWAY>}	Selects the options for the OOBM port.
show	Shows the OOBM port.
dhcp	Configures the port with a DHCP address.
disable	Disables the OOBM port.

Parameter	Description
<code>addr <ADDR-NETMASK-GATEWAY></code>	Configures the port with a static IP address (IPv4 only). Specify address, netmask, and gateway as A.B.C.D.

Example

Configuring the port with a DHCP IP address:

```
SVOS> ip dhcp
SVOS> ip show
Interface : Link Up
IP Address : 10.0.26.17
Subnet Mask: 255.255.252.0
Gateway : 10.0.24.1
SVOS> ip disable
SVOS> ip show
Interface : Disabled
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

ls

Syntax

```
ls [<OPTIONS>] [<FILE-NME>]
```

Description

This command lists directory contents.

Parameter	Description
<code><OPTIONS></code>	Specifies options for the command.
<code>-l</code>	Shows one - column output.
<code>-a</code>	Shows entries which start with a period (.).
<code>-A</code>	Shows output similar to <code>-a</code> , but excludes a period (.) and a double period (..).
<code>-C</code>	Shows output list by columns.
<code>-x</code>	Shows output list by lines.
<code>-d</code>	Shows listing of directory entries instead of contents
<code>-L</code>	Follows symlinks.
<code>-H</code>	Follows symlinks on the command line.
<code>-R</code>	Recurse.
<code>-p</code>	Appends a slash (/) to directory entries.
<code>-F</code>	Appends an indicator to entries. An indicator can be as an asterisk (*) or slash (/) or equal sign (=) or at sign (@) or pipe ().
<code>-l</code>	Shows the output in a long listing format.
<code>-i</code>	Shows the list inode numbers.
<code>-n</code>	Shows a list of numeric UIDs and GIDs instead of names.
<code>-s</code>	Shows a list of allocated blocks.
<code>-e</code>	Shows in one column a list with the full date and time.
<code>-h</code>	Shows list sizes in human readable format (1K, 243M, 2G) with a one - column output.
<code>-r</code>	Shows in one column a sort in reverse order.

Parameter	Description
<code>-S</code>	Shows in one column a sort by size.
<code>-X</code>	Shows in the output sort by extension.
<code>-v</code>	Shows in one column a sort by version.
<code>-c</code>	With <code>-l</code> , it shows a sort in one column by ctime .
<code>-t</code>	With <code>-l</code> , it shows a sort by mtime .
<code>-u</code>	With <code>-l</code> , sort by atime .
<code>-c</code>	With <code>-l</code> , it shows a sort in one column by ctime
<code>-w <N></code>	Assumes that the terminal has the number of columns wide as specified by <code><N></code> .
<code>--color[={always never auto}]</code>	Controls color in the output.
<code><FILE-NAME></code>	Specifies the name of the file to list.

Example

Listing directory contents:

```
SVOS> ls -la /nos
drwxr-xr-x    3 0      0           4096 Nov 21 03:19 .
drwxr-xr-x   11 0      0           220 Nov 21 03:21 ..
drwx-----    2 0      0          16384 Nov 21 03:20 lost+found
-rwxr-xr-x    1 0      0      205957424 Nov 21 03:19 primary.swi
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

md5sum

Syntax

```
md5sum [-c | -s | -w] [<FILE-NAME>]
```

Description

This command computes and checks the MD5 message digest.

Parameter	Description
<code>[-c -s -w]</code>	Selects the options for the command.
<code>-c</code>	Specifies to check the sums against the list in files.
<code>-s</code>	Specifies not output anything, status code shows success.
<code>-w</code>	Specifies to warn about improperly formatted checksum lines.
<code><FILE-NAME></code>	Specifies the file name to run the checksum against.

Example

Computing and checking the MD5 message digest for /nos/primary.swi:

```
SVOS> md5sum /nos/primary.swi
93ffc89e7ec357854704d8e450c4b7ab /nos/primary.swi
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

mkdir

Syntax

```
mkdir [-m | -p] [<DIRECTORY-NAME>]
```

Description

This command makes directories.

Parameter	Description
<code>[-m -p]</code>	Specifies the options for the command.
<code>-m</code>	Specifies the mode.
<code>-p</code>	Specifies to make parent directories as needed with no errors for pre-existing directories.
<code><DIRECTORY-NAME></code>	Specifies the directory to create.

Example

Making the dir directory:

```
SVOS> mkdir dir
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

mount

Syntax

```
mount <DEVICE>
```

Description

This command mounts the SSD partitions to the following locations: **/coredump**, **/logs**, **/nos**, **/selftest**, and mounts the USB device to **/mnt/usb**.

Users can mount USB flash drives formatted as either FAT16 or FAT32 with a single partition.

Parameter	Description
<code><DEVICE></code>	Specifies the device to be mounted. Supported device options include <code>all</code> and <code>usb</code> .

Examples

Mounting all of the SSD partitions:

```
SVOS> mount all
SVOS> mount usb
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (SVOS>)	Administrators or local user group members with execution rights for this command.

mv

Syntax

```
mv [-f | -i | -n] <TARGET-DIRECTORY>
```

Description

This command moves (renames) files.

Parameter	Description
-f	Specifies not to prompt before overwriting.
-i	Specifies to prompt before overwriting.
-n	Specifies to not overwrite an existing file.

Example

Moving the file named myfile:

```
SVOS> mv myfile
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

password (svos)

Syntax

```
password
```

Description

Sets the admin user account password for both Service OS and once the user boots into and saves the configuration. This will overwrite the previous password if one exists. User input is masked with asterisks.

This command is not available if enhanced secure mode is set.

Example

Setting the admin account password:

```
SVOS> password
Enter password:*****
Confirm password:*****
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

ping

Syntax

```
ping <HOST-IP-ADDRESS>
```

Description

Pings network hosts for debug purposes.

Parameter	Description
<code><HOST-IP-ADDRESS></code>	Specifies the host IP address.

Example

Pinging a network host:

```
SVOS> ping 10.0.8.10
PING 10.0.8.10 (10.0.8.10): 56 data bytes
64 bytes from 10.0.8.10: seq=0 ttl=63 time=3.496 ms
64 bytes from 10.0.8.10: seq=1 ttl=63 time=0.367 ms
64 bytes from 10.0.8.10: seq=2 ttl=63 time=0.380 ms
64 bytes from 10.0.8.10: seq=3 ttl=63 time=0.282 ms
```

```
64 bytes from 10.0.8.10: seq=4 ttl=63 time=0.669 ms
^C
--- 10.0.8.10 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.282/1.038/3.496 ms
SVOS>
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
6300 6400	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

pwd

Syntax

```
pwd
```

Description

Displays the current working directory.

Example

Displaying the current working directory:

```
SVOS> pwd
/home
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

reboot

Syntax

```
reboot
```

Description

Reboots the Management Module.

Example

Rebooting the management module:

```
SVOS> reboot  
reboot: Restarting system
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (SVOS)	Administrators or local user group members with execution rights for this command.

rm

Syntax

```
rm [-f | -i | -R | -r] <FILE-NAME>
```

Description

Removes files or directories.

Parameter	Description
<code>[-f -i -R -r]</code>	Selects the options for removing files or directories.
<code>-f</code>	Never prompt before removing files or directories.
<code>-i</code>	Always prompt before removing files or directories.
<code>-R -r</code>	Recursive.

Example

Removing the file named **foo**:

```
SVOS> rm foo
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

`rmdir`

Syntax

```
rmdir [-p] <DIRECTORY-NAME>
```

Description

Removes empty directories.

Parameter	Description
<code>-p</code>	Specifies to remove parent directories.

Example

Removing the empty **foo** directory:

```
SVOS> rmdir foo  
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

secure-mode

Syntax

```
secure-mode <enhanced | standard | status>
```

Description

Sets the secure mode to enhanced or standard secure mode. Also can display the current secure mode. A zeroization is required before switching between enhanced and standard secure modes.

The command also displays a message notifying the user that they are already in the targeted secure mode.

Example

Setting the secure mode to enhanced or standard:

```
SVOS> secure-mode --help
Usage: secure-mode <enhanced | standard | status>
Set or retrieve the secure mode setting. Requires a zeroization to change modes.
SVOS>
...
...

SVOS> secure-mode enhanced
#####WARNING#####
This will set the switch into enhanced secure mode. Before enhanced secure mode is enabled, the switch must securely erase all customer data and reset the switch to factory defaults. This will initiate a reboot and render the switch unavailable until the zeroization is complete. This should take several minutes to one hour to complete.
#####WARNING#####
Continue (y/n)? y
reboot: Restarting system
...
```

```

...
SVOS> secure-mode standard
#####WARNING#####
This will set the switch into standard secure mode. Before
standard secure mode is enabled, the switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####
Continue (y/n)? y
reboot: Restarting system
...
...

SVOS> secure-mode standard
#####WARNING#####
Secure mode is already set to standard. Setting it again will
repeat the zeroization process. The switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####
Continue (y/n)? y
reboot: Restarting system
...
...

SVOS> secure-mode status
enhanced secure mode is set.
SVOS>

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

sh

Syntax

```
sh
```

Description

Launches a bash shell for support purposes. To quit bash, enter **exit**.

This command is not available if enhanced secure mode is set.

Example

Launching a bash shell:

```
SVOS> sh
switch:/cli/fs/home#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

system serviceos password-prompt

Syntax

```
system serviceos password-prompt
no system serviceos password-prompt
```

Description

Use this command to enable password authentication for ServiceOS. By default, the ServiceOS shell (accessible only from the local switch console port) requires no password to login as an admin use.

When this setting is enabled, the same password used to authenticate the admin user in the AOS-CX CLI or WebUI can be used to log in to the ServiceOS shell. If this setting is enabled, a forgotten admin user password cannot be reset using ServiceOS; if there are no other local or RADIUS/TACACS user accounts with administrator-level access, the switch must be zeroized by entering the **username zeroize** command at the ServiceOS login prompt to restore administrator access.

serviceos password-prompt cannot be enabled if the built-in admin user has been disabled.

The admin cannot be disabled while **serviceos password-prompt** is enabled.

Example

Enabling password authentication for ServiceOS

```
switch(config)# system serviceos password-prompt
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

umount

Syntax

```
umount <DEVICE>
```

Description

Unmounts the SSD partitions mounted to the following locations: **/coredump**, **/logs**, **/nos**, **/selftest**, and unmounts the USB device mounted to **/mnt/usb**.

Parameter	Description
<code><DEVICE></code>	Specifies the device to be unmounted. Supported device options include all and usb .

Examples

Unmounting all devices:

```
SVOS> umount all
SVOS> umount usb
```

Unmounting a USB device:

```
SVOS> umount all
SVOS> umount usb
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

update

Syntax

```
update {primary | secondary} <IMAGE>
```

Description

Verifies and installs a product image. The user can select the primary or secondary boot profile to update and the location of the file.

Parameter	Description
{primary secondary}	Selects either the primary or secondary image.
<IMAGE>	Specifies the image name.

Examples

Updating the software image using TFTP:



NOTE

The OOBM port is disabled on first boot and must be enabled using the **ip** command.

```
SVOS> ip dhcp
SVOS> ip show
Interface   : Link Up
IP Address  : 192.0.2.22
Subnet Mask: 255.255.200.20
Gateway     : 10.0.24.1
SVOS> tftp -g -r XL.10.00.0001.swi -l image.swi 192.4.8.10
XL.10.00.0001.swi 100% |*****| 178M 0:00:00 ETA
SVOS> ls
image.swi
SVOS> update primary image.swi

Updating primary software image...
Verifying image...
Done
```

Update the software image using USB:



NOTE

This example assumes that the user has preloaded a USB flash drive with the image to be updated. The image name on the flash drive is not important.

```
SVOS> mount usb
SVOS> ls /mnt/usb
image.swi
SVOS> update primary /mnt/usb/image.swi
Updating primary software image...
Verifying image...
Done
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

tftp

Syntax

```
tftp {-b | -g | -l <LOCAL-FILE> | -p | -r <REMOTE-FILE>} host [<PORT>]
```

Description

Transfers files to and from a remote machine (TFTP a file).

Parameter	Description
{-b -g -l -p -r <REMOTE-FILE>}	Selects the options for transferring a file.
-b	Specifies the transfer blocks of size octets. The default block size is set to 1468, which can be overridden with the -b option.
-g	Specifies to get a file.
-l	Specifies a local file.
-p	Specifies to put a file in remote location.
-r <REMOTE-FILE>	Specifies a remote file.
<PORT>	Specifies the port for transfer. If no port option is specified, TFTP uses the standard UDP port 69 by default.

Example

Transferring files:

```
SVOS> tftp -b 65464 -g -r XL.10.00.0002.swi.swi 192.0.2.1
XL.10.00.0002 100% |*****| 178M 0:00:00 ETA
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

version

Syntax

```
version
```

Description

Displays the following build strings:

- Version.
- Build date.
- Build time.
- Build ID.
- SHA.

Example

Displaying version build strings:

```
SVOS> version
ServiceOS Information:
  Version:          GT.01.01.0001
  Build Date:       2017-07-19 14:52:31 PDT
  Build ID:         ServiceOS:GT.01.01.0001:461519208911:201707191452
  SHA:              46151920891195cdb2267ea6889a3c6cbc3d4193
SVOS>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	ServiceOS (<code>SVOS></code>)	Administrators or local user group members with execution rights for this command.

In-System Programming

The ISP (In-System Programming) feature provides an automated way to roll out updates to various programmable devices in a network switch, after the product has shipped. ISP is intended to run automatically either at boot time or as new modules are inserted into the chassis at runtime.

Subtopics

[Show tech command list for the ISP feature](#)

[In-System Programming commands](#)

Show tech command list for the ISP feature

Task	Command
Displaying versions of all present programmable devices.	<code>show tech isp</code>
Displaying stored log files from any ISP updates on the system.	<code>show tech update-log</code>

See the for additional information about the `show tech` commands.

In-System Programming commands

Select a command from the list in the left navigation menu.

Subtopics

[clear update-log](#)

[show needed-updates](#)

clear update-log

Syntax

```
clear update-log
```

Description

Clears stored log files of any In-System Programming updates on the system.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show needed-updates

Syntax

```
show needed-updates [next-boot [primary|secondary]]
```

Description

Displays whether any programmable devices are in need of an update.

Without the **next-boot** parameter, this command displays needed updates relative to the currently running image.

With the **next-boot** parameter, this command displays needed updates relative to an image file in the persistent storage of the switch, which might be different from the currently running image. If either the **primary** or **secondary** parameter is specified, this command queries that specific image file. Otherwise, it queries the default image file as set by the most recent **boot system** or **boot set-default** command.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Selftest

Diagnostics involve four important sub-categories.

- Boot-up Diagnostics (Power On Selftest aka POST)
- Run-time Diagnostics (Hardware Health Monitoring)
- Online Diagnostics
- Offline Diagnostics

Power On Self Test (POST) is the first task which verifies the hardware functionality of various modules during boot-up. Based on the criticality of the test, the selftest module decides whether to go ahead with the boot-up sequence of a particular subsystem or interface during a POST failure.

POST comprises of the following:

- **Memory BISTs (Built-in Self Test)**

This is to verify the memory block(s) present in the module and involves both internal and external memories.

The memory tables are critical for proper functionality of the system, so any failures in these tests will result in the corresponding subsystem to be marked as "Failed" and thus that subsystem will not be available for use.

- **Front-end Port Loopback tests**

This is to verify the physical port front-end interface.

These tests check if a particular interface can function properly. A test failure would mean that the particular interface is marked as "Failed" and thus it would become unavailable for use.

This test is run when "no fastboot" is configured.

Subtopics

Selftest commands

Selftest commands

Select a command from the list in the left navigation menu.

Subtopics

[fastboot](#)

[show selftest](#)

fastboot

Syntax

```
fastboot
no fastboot
```

Description

Enables fastboot for the system.

The **no** form of this command disables fastboot for the system.

Usage

When fastboot is enabled, most tests under a Power On Self Test (POST) are skipped. By default, fastboot is enabled.

After disabling fastboot, save switch configurations and then reboot for POST to run. POST verifies the hardware functionality of various modules during boot-up. Based on the criticality of the test, the selftest module decides whether to go ahead with the boot-up sequence of a particular subsystem or interface during a POST failure.

POST runs memory built-in selftest (BISTs) and front-end port loopback tests. Memory BISTs verify the internal and external memory blocks present in the module. The memory tables are critical for proper functionality of the system so any failures in these tests results in the corresponding subsystem to be marked as "Failed" and thus that subsystem is not available for use.

Front-end port loopback tests verify the physical port front-end interface. These tests check if a particular interface can function properly. A test failure means that a particular interface has been marked as "Failed" and is now unavailable for use.

On 6300 and 6400 switches, the line-module and fabric-module selftest is run regardless of fastboot setting. The interface selftest is only run when fastboot is disabled.

Examples

Enabling fastboot:

```

switch# configure terminal
switch(config)# fastboot
switch(config)# end
switch# show running-config
Current configuration:
!
!Version

                ML.10.06.0001
module 1/1 product-number j1726a!Version

                FL.10.06.0001
module 1/1 product-number j1661a!Version

                XL.10.00.0002
module 1/1 product-number j1363a!Version

                PL.10.06.0001
module 1/1 product-number j1677a
!
!
!
!
!
!
!
vlan 1
interface 1/1/1
    no shutdown

```

Disabling fastboot:

```

switch# configure terminal
switch(config)# no fastboot
switch(config)# end
switch(config)# write mem
Configuration changes will take time to process, please be patient.
switch# show running-config
Current configuration:
!
!Version

                ML.10.06.0001

```

```

module 1/1 product-number j1726a!Version

        FL.10.06.0001
module 1/1 product-number j1661a!Version

        XL.10.00.0002
module 1/1 product-number j1363a!Version

        PL.10.06.0001
module 1/1 product-number j1677a
!
!
!
no fastboot
!
!
!
!
vlan 1
interface 1/1/1
    no shutdown

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show selftest

Syntax

```
show selftest [brief] [vsx-peer]
show selftest line-module <SLOT-ID>
show selftest line-module <SLOT-ID> interface [brief] [vsx-peer]
show selftest interface [<PORT-NUM>] [vsx-peer]
```

For 8400 and 6400 switches only:

```
show selftest {line-module | fabric-module} [<SLOT-ID>] [brief] [vsx-peer]
```

Description

Displays selftest results.

Parameter	Description
[brief]	Shows the selftest results as a brief description. Default.
line-module	Shows the selftest results for a line module.
fabric-module	Shows the selftest results for a fabric module. Applicable only for 8400 and 6400 switches.
<SLOT-ID>	Shows the selftest results for the slot ID of the line or fabric module.
<PORT-NUM>	Shows the selftest results for the port number.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Displaying the output when fastboot is disabled on an 8400 or a 6400 switch:

```
switch# show selftest
Name      Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/1  passed      0x0        2016-10-15 10:10:09
LineModule 1/2  failed      0x09       2016-10-15 10:10:56
```

```

Fabric 1/1      passed      0x0      2016-10-15 10:10:09
Fabric 1/2      failed      0x1E     2016-10-15 10:10:56
switch# show selftest line-module
Name          Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/1  passed      0x0      2016-10-15 10:10:09
LineModule 1/2  failed      0x09     2016-10-15 10:10:56
switch# show selftest fabric-module
Name    Id      Status      ErrorCode  LastRunTime
-----
Fabric 1/1      passed      0x0      2016-10-15 10:10:09
Fabric 1/2      failed      0x1E     2016-10-15 10:10:56
switch# show selftest fabric-module 1/2
Name    Id      Status      ErrorCode  LastRunTime
-----
Fabric 1/2      failed      0x11     2016-10-15 10:10:56
switch# show selftest line-module 1/10
Name          Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/10 failed      0x1A     2016-10-15 10:10:56
switch# show selftest interface 1/2/2
Name    Status      ErrorCode  LastRunTime
-----
1/2/2    passed      0x0      2016-11-19 05:10:11
switch# show selftest line-module 1/3 interface
Name    Status      ErrorCode  LastRunTime
-----
1/3/1    passed      0x0      2016-11-19 05:10:11
1/3/2    passed      0x0      2016-11-19 05:10:11
1/3/3    passed      0x0      2016-11-19 05:10:11
1/3/31   failed      0x20     2016-11-19 05:10:11

```

Displaying the output when fastboot is disabled on a 6300 switch:

```

switch# show selftest
interface

```

```

Name          Status      ErrorCode
LastRunTime

```

```

-----
1/1/2          skipped

```

```

0x0

1/1/44      skipped
0x0

1/1/46      skipped      0x0
switch# show selftest interface 1/1/1
Name        Status        ErrorCode        LastRunTime
-----
1/1/1       skipped        0x0

```

Displaying the output when fastboot is enabled on a 6400 switch:

```

switch# show selftest
Name        Id    Status        ErrorCode    LastRunTime
-----
LineModule  1/1  passed        0x0
LineModule  1/2  passed        0x0
Fabric      1/1  passed        0x0
Fabric      1/2  passed        0x0
switch# show selftest line-module
Name        Id    Status        ErrorCode    LastRunTime
-----
LineModule  1/1  passed        0x0
LineModule  1/2  passed        0x0
switch# show selftest fabric-module
Name        Id    Status        ErrorCode    LastRunTime
-----
Fabric      1/1  passed        0x0
Fabric      1/2  passed        0x0
switch# show selftest fabric-module 1/2
Name    Id        Status        ErrorCode    LastRunTime
-----
Fabric  1/2        passed        0x0
switch# show selftest line-module 1/1
Name        Id    Status        ErrorCode    LastRunTime
-----
LineModule  1/1  passed        0x0

```

Displaying the output when fastboot is enabled:

```

switch# show selftest interface 1/1/2
Name        Status        ErrorCode    LastRunTime
-----
1/1/2       skipped        0x0
switch# show selftest line-module 1/1 interface
Name        Status        ErrorCode    LastRunTime

```

```

-----
1/1/1      skipped      0x0
1/1/2      skipped      0x0
1/1/3      skipped      0x0
1/1/31     skipped      0x0

```

Displaying the output when fastboot is disabled:

Testing to register read/write:



NOTE

This test is run irrespective of fastboot being enabled or disabled.

```
switch# show selftest
```

```

Name      Id      Status      ErrorCode  LastRunTime
-----
LineModule 1/1    passed      0x0        2018-02-16 18:15:53

```

Command History

Release

Modification

10.07 or earlier

- -

Command Information

Platforms

Command context

Authority

All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
---------------	---------------	--

Zeroization

Device zeroization lets you remove all user files from flash storage, including solid-state drives (SSDs). User files cannot be retrieved after the zeroization is complete.



NOTE

Zeroization can occur in both and Service OS. This section covers zeroization and . For information about zeroization and Support OS, see [erase zeroize](#).

Zeroization preserves the primary and secondary software images on the SSD . Zeroization also preserves manufacturing information.

The sensitive user files stored on an SSD or SPI flash/EEPROM storage or both include:

- Switch configurations.
- System generated private keys.
- User installed private keys.
- Admin/operator password files.

Using CLI, you can change the switch settings from standard secure mode to enhanced secure mode. Setting the switch back to standard secure mode can only be performed through Service OS. For more information on how to change switch settings using Server OS, see [Service OS](#).

Enhance secure mode is used to enhance the switch security. In enhanced security mode, the switch (Product OS) `start-shell` command is disabled for security purpose except through ServiceOS.

Subtopics

Zeroization commands

Zeroization commands

Select a command from the list in the left navigation menu.

Subtopics

erase all zeroize

erase all zeroize

Syntax

```
erase [all] zeroize
```

Description

Restores the switch to its factory default configuration. You will be prompted before the procedure starts. Once complete, the switch will restart from the primary image with factory default settings.

Usage

The **erase all** command is always available in the CLI. On running the **erase all** command, the switch is restored to a factory default settings, but retains the enhanced secure mode settings.

The **erase all zeroize** command is not available in the CLI when enhanced secure mode is enabled. This command restore the switch to a factory default settings. On running the **erase all zeroize** command in enhanced secure mode, displays a notification stating that the command is unavailable in enhanced secure mode.



NOTE

Back up all data before running this command as all configuration settings will be lost.

Example

Restoring the switch to factory default configuration, except for the enhance secure mode settings:

```
switch# erase all
This command will erase all data and reset the switch to factory
defaults, with the exception of the secure mode setting. This process
will take several minutes to an hour to complete and the switch will
be unavailable during that time.
Continue (y/n)?
ServiceOS Information:
Version: GT.01.01.0007
Build Date: 2017-12-07 11:48:44 PST
Build ID: ServiceOS:GT.01.01.0007:42c7d15cf7e5:201712071148
SHA: 42c7d15cf7e5af5bf1c7d8764ff673471084c2a4
##### Preparing for zeroization #####
##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
##### This should take several minutes #####
##### to one hour to complete #####
##### Restoring files #####
```

Restoring the switch to factory default configuration only when enhance secure mode settings is disabled.

```
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y
The system is going down for zeroization.
...
##### Preparing for zeroization #####
##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
```

```
##### This should take several minutes #####
##### to one hour to complete #####
##### Restoring files #####
...
We'd like to keep you up to date about:
  * Software feature updates
  * New product announcements
  * Special events
Please register your products now at: https://networkingsupport.hpe.com
```



NOTE

When you log in after zeroization, you get a prompt to create a password for the administrator account. You can set the password as blank (to set the password as blank, hit enter at the prompt) or type 1 to 32 printable ASCII characters, excluding spaces and question marks (?). For more information on password requirements, see Password requirements in the .

```
switch login: admin
Password:
Please configure the 'admin' user account password.
Enter new password: *****
Confirm new password: *****
```

Command History

Release	Modification
10.11.1010	Introduced erase all CLI command
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Terminal Monitor

The terminal monitor is used to display selective logs dynamically on the VTYSH session. When the terminal monitor feature is enabled on the switch, it displays only the live or active logs. These logs are displayed on the SSH session or console session. If required, you can enable the terminal monitoring on multiple sessions.

It is important to monitor the logs dynamically while debugging, so that you can co-relate the issues. The logs can be filtered by type (event or debug), severity, or keyword. The terminal monitor runs in synchronous mode, where the user enters any command, the log display pauses until the command execution is complete. This ensures that the logs will not appear in between other CLI outputs or while the user is typing.



NOTE

Terminal monitoring is not persistent in the SSH session. If the SSH session is terminated, the terminal monitor is no longer valid. However, logging console is persistent and is added to the switch configuration, so it will persist between telnet sessions.

Subtopics

Terminal monitor commands

Terminal monitor commands

Select a command from the list in the left navigation menu.

Subtopics

logging console {notify | severity | filter}

show terminal-monitor

terminal-monitor {notify | severity | filter}

```
logging console {notify | severity | filter}
```

Syntax

```
logging console{notify <event|debug|all> | severity <level> | filter  
keyword}  
no logging console
```

Description

Enables the logging console feature in the console session. It display all debug log or event log or both debug and event log messages. Monitoring can be filtered with the severity options or with the help of keywords. Enabling terminal monitor without options displays both debug and event log with a severity error. This command is persistent across reboot.

The **no** form of this command disables the terminal monitor configuration.

Parameter	Description
<code>notify <event debug all></code>	Specifies the type of log notification. • Event: Displays the event log messages. (Default) • Debug: Displays the debug log messages. • All: Displays both event and debug log messages.
<code>severity <level></code>	Specifies the severity level for the logs. The different severity levels are emergency, critical, error, warning, notice, information (default), alert, and debug (shows all severities).
<code>filter <keyword></code>	Specifies the filter by applying keyword for the logs.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring console logging in the console session:

```
switch(config)# logging console
Terminal-monitor is enabled successfully
switch(config)# logging console notify all
Terminal-monitor is enabled successfully
switch(config)# logging console notify event severity info
Terminal-monitor is enabled successfully
switch(config)# logging console filter lldp
Terminal-monitor is enabled successfully
```

Command History

Release	Modification
10.08	Feature introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show terminal-monitor

Syntax

```
show terminal-monitor
```

Description

Shows whether the terminal monitoring is enabled or disabled.



NOTE

This command will not show any information about console logging.

Examples

Displaying terminal monitor when enabled:

```
switch# show terminal-monitor
Terminal-monitor is enabled
-----
Notify      | Severity  | Filter
-----
event       | debug     | lldp
-----
```

Displaying terminal monitor when disabled:

```
switch# show terminal-monitor
Terminal-monitor is disabled
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
terminal-monitor {notify | severity | filter}
```

Syntax

```
terminal-monitor {notify <event|debug|all> | severity <level> | filter  
<keyword>}  
no terminal-monitor
```

Description

Enables and saves the terminal monitor feature in the switch configuration. It displays all debug log or event log or both debug and event log messages. Terminal monitoring can be filtered with the severity options or with the help of keywords. Enabling terminal monitor without options displays both debug and event log with a severity error.

The **no** form of this command removes the terminal monitor feature from the switch configuration and the command will not persist.

Parameter	Description
notify <event debug all>	Specifies the type of log notification. • Event: Displays the event log messages. (Default) • Debug: Displays the debug log messages. • All: Displays both event and debug log messages.
severity <level>	Specifies the severity level for the logs. The different severity levels are emergency, critical, error, warning, notice, information (default), alert, and debug (shows all severities).
filter <keyword>	Specifies the filter by applying keyword for the logs.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling terminal monitor:

```
switch# terminal-monitor
Terminal-monitor is enabled successfully
switch# terminal-monitor notify all
Terminal-monitor is enabled successfully
switch# terminal-monitor notify event severity info
Terminal-monitor is enabled successfully
switch# terminal-monitor filter lldp
Terminal-monitor is enabled successfully
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Troubleshooting Web UI and REST API Access Issues

The following section describes symptoms, causes and corrective actions for 401 or 404 errors.

Subtopics

- [HTTP 404 error when accessing the switch URL](#)
- [HTTP 401 error "Login failed: session limit reached"](#)

HTTP 404 error when accessing the switch URL

Symptom

The switch is operational and you are using the correct URL for the switch, but attempts to access the REST API or Web UI result in an HTTP 404 "Page not found" error.

Cause

REST API access is not enabled on the VRF that corresponds to the access port you are using. For example, you are attempting to access the REST API or Web UI from the management (OOBM) port, and access is not enabled on the `mgmt` VRF. By default, **https-server** is enabled on the `mgmt` VRF for the 6300 and 6400 switches.

Action

Use the **https-server vrf** command to enable REST API access on the specified VRF.

For example:

```
switch(config)# https-server vrf mgmt
```

HTTP 401 error "Login failed: session limit reached"

Symptom

A REST request or Web UI login attempt returns response code 401 and the response body contains the following text string:

```
Login failed: session limit reached
```

Cause

A user attempted to log into the REST API or the Web UI, but that user already has the maximum number of concurrent sessions running.

Action

1. Log out from one of the existing sessions.
Browsers share a single session cookie across multiple tabs or even windows. However, scripts that POST to the login resource and later do not POST to the logout resource can easily create the maximum number of concurrent sessions.
2. If the session cookie is lost and it is not possible to log out of the session, then wait for the session idle time limit to expire.
When the session idle timeout expires, the session is terminated automatically.
3. If it is required to stop all HTTPS sessions on the switch instead of waiting for the session idle time limit to expire, you can stop all HTTPS sessions using the **https-server session close all** command.

This command stops and starts the hpe-restd service, so using this command affects all existing REST sessions, Web UI sessions, and real-time notification subscriptions.

Feature and Proactive Troubleshooting

AOS-CX switches include an intelligent, intuitive and distributed troubleshooting framework that makes network troubleshooting simpler, inexpensive and efficient. Use the CLI commands in this section to troubleshoot individual features of AOS-CX software, or issue the **troubleshoot proactive** command to set up proactive system troubleshooting to automatically troubleshoot issues based on any of the following.

1. Events
2. Log messages
3. CoPP or other database changes
4. CPU levels exceeding a threshold
5. Memory exceeding a threshold
6. Daemon crashes



NOTE

Feature-specific troubleshooting information can also be viewed via the **Deepshoot > Troubleshoot** page of the Switch WebUI. For more details, refer to the latest WebUI Reference Guide for your switch model.

Subtopics

show troubleshoot

troubleshoot aaa-mgmt

troubleshoot application

troubleshoot classifier

troubleshoot client-insight

troubleshoot l3

troubleshoot event

troubleshoot multicast

troubleshoot flow

troubleshoot flow-telemetry

troubleshoot proactive

troubleshoot port-access authentication

troubleshoot radius

troubleshoot system

troubleshoot tunnel

show troubleshoot

Syntax

```
show troubleshoot {details instance <1-10000>}|history
```

Description

Displays the history of past troubleshooting sessions. Include the **history** parameter to display an overview of all troubleshooting commands executed on the switch within the past 30 minutes, including the instance IDs of each troubleshooting session, the name of the feature, and timestamps for the start and end of each session. Use the **details** parameter to display details for an individual session.

Parameter	Description
details instance <1-1000>	Display details for an individual troubleshooting session. To obtain the instance ID for a particular session, issue the command show troubleshoot history to display instance IDs for sessions within the previous 30 minutes.
history	Display a list of troubleshooting sessions completed within the past 30 minutes.

Examples

The following example displays basic flow telemetry troubleshooting details. The output of this command can vary depending upon switch model and configuration.

```
switch# show troubleshoot history
Instance Command                               Start timestamp      End
timestamp
-----
5603      troubleshoot 13 ospfv3                2026:04:20 05:28:11
2026:04:20 05:28:15      manual
3241      troubleshoot 13 ospf                          2026:04:20 05:27:56
2026:04:20 05:28:00      manual
3782      troubleshoot 13                               2026:04:20 05:27:44
2026:04:20 05:27:48      manual
3184      troubleshoot 12 stp                          2026:04:20 05:27:01
2026:04:20 05:27:05      manual
4933      troubleshoot 12                               2026:04:20 05:26:14
2026:04:20 05:26:18      manual
6095      troubleshoot app-recognition ar             2026:04:20 05:25:56
2026:04:20 05:26:00      manual
```

```

2059      troubleshoot event 2611      2025:06:25 16:09:32
2025:06:25 16:09:32      manual
7021      troubleshoot system      2026:04:20 05:32:10
2026:04:20 05:32:12      proactive

```

```

switch# show troubleshoot details instance 1
-----Troubleshoot Instance ID :
1-----
Troubleshoot Type      : manual
Feature Name          : multicast
Component Name        : igmp
Progress Status       : feature_troubleshoot_complete
Elapse Time           : 31 seconds
Config Status         : igmp - No_config_error
-----
---

switch# show troubleshoot details instance 7021
----- Troubleshoot Instance ID : 7021
-----
Troubleshoot Type      : proactive
Feature Name          : system
Component Name        : resources
Progress Status       : feature_troubleshoot_basic_complete
Elapsed Time          : 2 seconds
Action Type           : Support files
Action Result         : pass
Event ID              : 1208
Event Log              : High system memory usage detected
Results               : Timestamp: 2025-05-20 05:32:09
                      : Error ID: 402
                      : Severity: ERROR
                      : Alert: High memory utilization
-----
---

```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	manager	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

troubleshoot aaa-mgmt

Syntax

```
troubleshoot aaa-mgmt
  config [parameters]
  console [{health[verbose]}|operations] [parameters]
  health [verbose]
  operations [parameters]
  radius health [verbose]
  radsec health [verbose]
  rbac health [verbose]
  ssh [{health[verbose]}|operations] [parameters]
  tacacs [config|{health [verbose]}]
  telnet [{health[verbose]}|operations]
```

Description

Troubleshoot configuration or operational issues that can impact management security. Issue this command without any additional parameters to display troubleshooting information in all categories, or choose a specific category to limit information to that category type.

Parameter	Description
console health operations	Troubleshoot AAA management console issues. By default, the troubleshoot aaa-mgmt console command displays both feature health and operations troubleshooting information. Include the health or operations parameter after the command to display information of that type only.
health [verbose]	Display health information for AAA management errors on the network, including the error message, message ID, severity level, and when the message was last reported. By default, the troubleshoot aaa - mgmt health command displays basic troubleshooting information for all AAA management features. Include the

Parameter	Description
	e health parameter after a AAA management feature name to display troubleshooting information for that feature only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
operations	By default, the troubleshoot aaa-mgmt operations command displays comprehensive feature diagnostics with a detailed analysis for the console, ssh and telnet operations. Include the operations parameter after the troubleshoot aaa-mgmt console, troubleshoot aaa-mgmt ssh, or troubleshoot aaa-mgmt telnet commands to display operations troubleshooting information for that feature only.
radius health [verbose]	Troubleshoot issues with a RADIUS server for AAA management. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
radsec health [verbose]	Troubleshoot issues with a RADSEC server for AAA management. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
rbac health [verbose]	Troubleshoot issues with Role - Based Access Control (RBAC). Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
ssh [{health[verbose]} operations]	Troubleshoot issues with Secure Shell (SSH). By default, the troubleshoot aaa-mgmt ssh command displays both feature health and operations troubleshooting information. Include the health or operations parameter after the command to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
tacacs [config]{health [verbose]}	Troubleshoot issues with telnet. By default, the troubleshoot aaa-mgmt tacacs command displays both feature configuration and feature health troubleshooting information. Include the config or health parameter after the command to display information of that type only.

Parameter	Description
telnet [{health[verbose]}] operations]	Troubleshoot telnet issues.. By default, the troubleshoot aaa-mgmt telnet command displays both feature health and operations troubleshooting information. Include the health or operations parameter after the command to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
parameters "<word>"	Reserved for future use.

Examples

The following examples displays all available AAA management troubleshooting information.

```
(config)# troubleshoot aaa-mgmt
-----
Health troubleshoot
-----
Message ID
Message
Last-Reported          Severity
-----
119          RADSEC server host 172.16.1.4 on port 2083 in VRF default is
unreachable
2025-10-14 06:06:09    ERROR
Message ID
Message
Last-Reported          Severity
-----
101          TACACS server host 172.16.1.5 on port 49 in VRF default is
unreachable
2025-10-14 06:06:07    ERROR
Message ID
Message
Last-Reported          Severity
-----
```

```

102          RADIUS server host 172.16.1.4 on port 1812 in VRF default is
unreachable  2025-10-14 06:06:06    ERROR
-----
Configuration troubleshoot
-----
tacacs   : No_config_error
=====
Troubleshoot aaa-mgmt executed successfully
=====
=====

```

The following example uses the optional **verbose** parameter to include detailed log information for a RADIUS server in the output of this command.

```

switch# troubleshoot aaa-mgmt radius health verbose
-----
Starting verbose troubleshoot logs:
-----
[2025-08-28 05:41:34.133] Starting basic health check for feature 'aaa-
mgmt', component
'radius'
[2025-08-28 05:41:34.356] [OVSDB] Dumping table 'Radius_Server' with all
columns:
reachability_status port_type address vrf port
[2025-08-28 05:41:34.587] OVSDB check for component 'radius' found 1 errors
[2025-08-28 05:41:34.588] [SYSLOG] Checking pattern 'Failed to get current
namespace'
[2025-08-28 05:41:34.662] [SYSLOG] Checking pattern 'Failed to switch to
namespace
{namespace}'
[2025-08-28 05:41:34.730] [SYSLOG] Checking pattern 'Failed to set mount
point to
namespace {namespace}'
[2025-08-28 05:41:34.800] [SYSLOG] Checking pattern 'packet from RADIUS
server {server}
failed verification: The shared secret is probably incorrect'
-----
-----
Health troubleshoot
-----
-----
Message ID      Message                                          Last-
Reported
Severity
-----

```

```

-----
102          RADIUS server host 10.1.1.2 on port is unreachable  2025-08-28
05:41:34  ERROR
-----

```

```

End of verbose troubleshoot logs
-----
=====

```

```

Troubleshoot RADIUS executed successfully
=====

```

The following example displays configuration troubleshooting information for a TACACS+ server correctly configured for management user authentication.

```

(config)# troubleshoot aaa-mgmt tacacs
-----
-----
Health troubleshoot
-----
-----
Message ID
Message
Last-Reported          Severity
-----
-----
101          TACACS server host 172.16.1.5 on port 49 in VRF default is
unreachable
2025-10-14 06:13:22    ERROR
-----
Configuration troubleshoot
-----
tacacs  : TACACS server group 'group1' used in AAA group 'authentication'
is empty.
=====
Troubleshoot TACACS executed successfully
=====

```

The following example displays configuration troubleshooting information for a TACACS+ server with an invalid configuration:

```

(config)# troubleshoot aaa-mgmt tacacs config
-----
Configuration troubleshoot
-----
tacacs  : TACACS server group 'group1' used in AAA group 'authentication'
is empty.
=====

```

Troubleshoot TACACS executed successfully

=====

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	manager	Administrators or local user group members with execution rights for this command.

troubleshoot application

Syntax

```
troubleshoot application operations parameters "<query><word>"
```

Description

Troubleshoot congested application traffic for the specified parameters

Parameters

Parameter	Description
parameters "<query> <word>"	The following parameters are available to troubleshoot applications: " - - client - ip <ip>": identify application issues for the specified client IP. " - - app - name <name>": identify issues for the specified application name. " - - vrf <vrf - name>": identify application issues on the specified VRF.

Parameter	Description
	Note that the "<query><word>" string must be enclosed in double quotation marks (""). If you include more than one set of "<query> <word>" parameters, they should all be included within the same set of double quotation marks. For example, parameters "--client-ip 172.21.18.170 --vrf red"

Example

The following example displays application operations troubleshooting details. The output of this command can vary depending upon switch model and configuration.

```
switch# troubleshoot application operations parameters "--client-ip
10.10.10.20 --app-name ssh"
-----
Operations troubleshoot
-----

application  : Interface 1/1/1 and queue 2 has queue congestion
                Flow 10.10.10.20:45126->10.10.10.30:22 (Proto:6,
VRF:default)
                Flow 10.10.10.20:45130->10.10.10.35:22 (Proto:6,
VRF:default)
                Interface 1/1/2 and queue 3 has queue congestion
                Flow 10.10.10.20:45140->10.10.10.40:22 (Proto:6,
VRF:default)

=====
Troubleshoot APPLICATION executed successfully
=====
...
```

Troubleshoot application operations parameters with a VRF:

```
...

switch# troubleshoot application operations parameters "--client-ip
192.168.1.10 --app-name http --vrf mgmt"
-----
Operations troubleshoot
-----
```

```

application : Interface 1/1/3 and queue 1 has queue congestion
              Flow 192.168.1.10:8080->192.168.1.20:80 (Proto:6,
VRF:mgmt)
              Flow 192.168.1.10:8081->192.168.1.25:80 (Proto:6,
VRF:mgmt)
              Flow 192.168.1.10:8082->192.168.1.30:80 (Proto:6,
VRF:mgmt)
              Interface 1/1/4 and queue 2 has queue congestion
              Flow 192.168.1.10:8090->192.168.1.40:80 (Proto:6,
VRF:mgmt)

=====
Troubleshoot APPLICATION executed successfully
=====

```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300, except f or S3L75A, S3 L76A, S3L77 A 6400	manager	Administrators or local user group members with execution rights for this command.

troubleshoot classifier

Syntax

```

troubleshoot classifier
acl
counter
health [verbose]

```

```
policer
policy
range-checker
tcam
```

Description

Troubleshoot classifier features, including ACLs, counters, policies, policers, range-checkers and TCAM values.

Parameters

Parameter	Description
acl	Troubleshoot Access Control Lists (ACLs).
counter	Troubleshoot TCAM counter allocation.
health [verbose]	Troubleshoot classifier feature health. Issue the command troubleshoot classifier health to display health information for all classifier features, or include the health parameter after a feature name (for example, troubleshoot classifier acl health) to display health information for that feature only. Include the optional verbose parameter to display detailed troubleshooting progress.
policer	Troubleshoot issues with policer resource allocation.
policy	Troubleshoot issues with classifier policies.
range-checker	Troubleshoot issues with range-checker resources.
tcam	Troubleshoot issues with TCAM resource allocation.

Example

```
=====
6300(config)# troubleshoot classifier acl health
-----
Health troubleshoot
-----
```

Component: acl

```
-----  
-  
Message ID  Message                                Last-Reported(UTC)  Severity  
-----  
3001        ACL application failure.  2026-03-27 02:38:32  ERROR
```

```
=====  
Troubleshoot classifier executed successfully  
=====
```

Release

Modification

10.18	Command introduced
-------	--------------------

Command Information

Platforms

Command context

Authority

6300	manager	Administrators or local user group members with execution rights for this command.
6400		

troubleshoot client-insight

Syntax

```
troubleshoot client-insight clientmac  
  health [verbose] [[parameters "<mac-addr>"]  
  operations [parameters "<mac-addr>"]
```

Description

Troubleshoot any client configuration or operational issues found via Client Insight.

Parameter	Description
health [verbose]	Display health information for client-insight errors on the network, including the error message, message ID, severity level, and when the message was last reported. Include the optional verbose parameter to display all available troubleshooting details from client insight log files.
operations	Display comprehensive client insight feature diagnostics with a detailed analysis.
parameters "<mac-addr>"	Display detailed diagnostics by including the MAC address of the client to troubleshoot. Note that the MAC address must be enclosed in double quotation marks ("").

Examples

Showing advanced troubleshooting information for a client with the MAC address **00:50:56:96:16:dd**:

```
switch# troubleshoot client-insight clientmac health parameters
"00:50:56:96:16:dd"
client 00:50:56:96:16:92 auth failed as there is no response from
authentication server
```

Showing client-insight troubleshooting information for all clients:

```
switch# troubleshoot client-insight
-----
Health troubleshoot
-----
Message ID
Message

Last-Reported          Severity
-----
701      Client 00:50:56:11:22:aa authentication failed during onboarding
stage (L2)
2026-04-04 09:20:37    ERROR
702      Client 00:50:56:96:75:ab L2 authentication failed during onboarding
due to
2026-04-04 09:25:41    ERROR
RADIUS server issues
703      Client 00:50:56:96:75:ac L3 onboarding failed due to DHCP related
issues
2026-04-04 09:01:32    ERROR
```

```

704      Client 00:50:56:96:75:ad L4 onboarding failed for DNS or ARP L4
onboarding
2026-04-04 09:30:00      ERROR
failed for DNS or ARP
705      Client 00:50:56:96:e4:ae ongoing L2 authentication failure
detected
2026-04-04 09:15:30      ERROR
706      Client 00:50:56:96:54:af DNS server response failure counters
detected
2026-04-04 09:20:37      ERROR
707      Client 00:50:56:96:46:ba onboarding DHCP failure on specific
VLAN
2026-04-04 09:26:46      ERROR
708      Client 00:50:56:96:ee:bb ongoing L3 DHCP failure on specific
VLAN
2026-04-04 09:45:38      ERROR
-----
Operations troubleshoot
-----
clientmac   : No result
=====
Troubleshoot CLIENTMAC executed successfully
=====

```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
all platforms	manager	Administrators or local user group members with execution rights for this command.

troubleshoot 13

Syntax

```
troubleshoot l3
  arp [{health[verbose]}|operations]
  config
  dhcp-relay [config|{health[verbose]}|operations]
  dhcp-server [config|{health[verbose]}|operations]
  dhcpv6-relay [config|{health[verbose]}|operations]
  dhcpv6-server [config|{health[verbose]}|operations]
  external-storage health [verbose]
  health [verbose]
  operations
  ospf [config|{health[verbose]}|operations]
  ospfv3 [config|{health[verbose]}|operations]
```

Description

Troubleshoot issues with Layer-3 features and protocols.

Parameter	Description
arp health [{health[verbose]} operations]	Troubleshoot the Address Resolution Protocol (ARP). By default, the troubleshoot l3 arp command displays both feature health and feature operations troubleshooting information. Include the optional health or operations parameters to display information for that category only.
config	Issue the troubleshoot l3 Config command to display the results of configuration validation checks for all L3 features. Include the config parameter after a specific L3 component to display configuration validation information for that component only.
dhcp-relay [config {health[verbose]} operations]	By default, the troubleshoot l3 dhcpv-relay command displays feature configuration, health, and operations troubleshooting information. Include the optional Config, health, or operations parameter to display information for that category only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
dhcp-server [config {health[verbose]} operations]	By default, the troubleshoot l3 dhcpv-server command displays feature configuration, health, and operations troubleshooting information. Include the optional Config , health , or operations parameter to display information for that category only.

Parameter	Description
	Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
<pre> dhcpv6-relay [config {health[verbose]} operations] </pre>	By default, the troubleshoot l3 dhcpv6-relay command displays feature configuration, health, and operations troubleshooting information. Include the optional Config, health, or operations parameter to display information for that category only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether other errors were suppressed.
<pre> dhcpv6-server [config {health[verbose]} operations] </pre>	Troubleshoot DHCPv6 server settings. By default, the troubleshoot l3 dhcpv6-server command displays feature configuration, health, and operations troubleshooting information. Include the optional Config, health, or operations parameter to display information for that category only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
<pre> external-storage [config {health[verbose]}] </pre>	Troubleshoot issues with external storage. By default, the troubleshoot l3 external-storage command displays feature health, and configuration troubleshooting information. Include the optional health or Config parameter to display information for that category only.
<pre> health [verbose] </pre>	The troubleshoot l3 health command displays basic health information for all L3 features, including error messages, message IDs, severity levels, and when the messages were reported. Include the health parameter after an L3 feature name (for example, troubleshoot l3 ospf health) to display health troubleshooting information for that feature only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
<pre> operations </pre>	Display comprehensive L3 feature diagnostics with a detailed analysis.
<pre> ospf [config {health[verbose]} operations] </pre>	Troubleshoot issues with Open Shortest Path First version 2 (OSPF). By default, the troubleshoot l3 ospf command displays feature configuration, health, and operations troubleshooting information.

Parameter	Description
	ation. Include the optional Config, health, or operations parameter to display information for that category only.
ospfv3 [config {health[verbose]} operations]	Troubleshoot issues with Open Shortest Path First version 3 (OSPFv3). By default, the troubleshoot l3 ospfv3 command displays feature configuration, health, and operations troubleshooting information. Include the optional Config, health , or operations parameter to display information for that category only.

Usage

Filter the output of a troubleshooting command by including a **parameters "<query><word>"** option after the **health, config**, or **operations** parameters for some troubleshooting L3 commands. The "**<query><word>**" string must be enclosed in double quotation marks ("). More than one set of filter parameters can be used in a single command, but all filter parameters must be enclosed within a single set of quotation marks, for example, "interface <interface> vrf <vrf - name>".

Table 1. Filter examples for L3 Troubleshooting Commands

Command	Filter options
troubleshooting l3 dhcp-relay config health operations	parameters "interface <interface>": Troubleshoot issues with the specified interface.
troubleshooting l3 dhcp-server config health operations	- parameters "--pool <pool>": Troubleshoot issues with the specified address pool. - parameters "--vrf <vrf>": Troubleshoot issues with the specified VRF.
troubleshooting l3 dhcpv6-relay config health operations	parameters "interface <interface>": Troubleshoot issues with the specified interface.
troubleshooting l3 dhcpv6-server config health operations	- parameters "--pool <pool>": Troubleshoot issues with the specified address pool. - parameters "--vrf <vrf>": Troubleshoot issues with the specified VRF.
troubleshooting l3 ospf config health operations	- parameters "neighbor <router - id>": Troubleshoot issues with the specified neighbor. - parameters "neighbor <router - id> interface <interface>": Troubleshoot neighbor issues on a specific interface.

Command

Filter options

- parameters "neighbor <router - id> interface <interface> vrf <vrf - name>": Troubleshoot neighbor issues on a specific interface in a VRF.
- parameters "interface <interface>": Troubleshoot issues with the specified interface.
- parameters "interface <interface> vrf <vrf - name>": Troubleshoot interface issues in a specific VRF.
- parameters "route <ipv6 - prefix>": Troubleshoot issues with the specified IPv6 route.
- parameters "interface vlan{vlan_id}": Troubleshoot issues on a specific VLAN.
- parameters "area <area-id>": Troubleshoot issues within a specific OSPF area.

troubleshooting I3 ospfv3 config|health|operations

- parameters "neighbor <router - id>": Troubleshoot issues with the specified neighbor.
- parameters "neighbor <router - id> interface <interface>": Troubleshoot neighbor issues on a specific interface.
- parameters "neighbor <router - id> interface <interface> vrf <vrf - name>": Troubleshoot neighbor issues on a specific interface in a VRF.
- parameters "interface <interface>": Troubleshoot issues with the specified interface.
- parameters "interface <interface> vrf <vrf - name>": Troubleshoot interface issues in a specific VRF.
- parameters "route <ipv6 - prefix>": Troubleshoot issues with the specified IPv6 route.
- parameters "interface vlan{vlan_id}": Troubleshoot issues on a specific VLAN.
- parameters "area <area-id>": Troubleshoot issues within a specific OSPF area.

Example

Displaying display the results of configuration validation checks for an OSPFv3 configuration

```

switch# troubleshoot l3 ospfv3 config parameters "neighbor 1.1.1.1
interface 1/1/1"
-----
Configuration troubleshoot
-----
ospfv3 : error: RCVD_BAD_UNKNOWN.1/1/1
Possible errors:
1. DUT and Peer Not in Same Network
error: RCVD_DEAD_MISMATCH1/1/1
error: 1/1/1 TX_RX_MISMATCH
error: Neighbor not found. Troubleshoot interface for any issues
with the ospfv3 interface.
=====
Troubleshoot OSPFv3 executed successfully

```

The following command displays troubleshooting information for the health of the DHCP-relay feature.

```

DocTeam-8325# troubleshoot l3 dhcp-relay health
-----
Health troubleshoot
-----
Component: dhcp-relay
-----
-----
Message ID
Message
      Last-Reported(UTC)    Severity
-----
-----
1710      Atleast one helper-address should be configured for DHCPv4
Relay to be effective.  2026-03-27 02:38:32  ERROR
=====
Troubleshoot dhcp-relay executed successfully
=====

```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	manager	Administrators or local user group members with execution rights for this command.

troubleshoot event

Syntax

```
troubleshoot event <event-id>
```

Description

Troubleshooting a specific event by its event ID. The output of this command will display the event message and the outcome of the troubleshooting operation.

Examples

Displaying troubleshooting information for event ID **2611**.

```
switch# troubleshoot event 2611
-----
Event troubleshoot:
-----
Event|2611|LOG_INFO|CDTR|1|IGMP snooping is enabled on VLAN 1
multicast : IGMP snooping is enabled on VLAN 1
=====
Troubleshoot event "2611" executed successfully
=====
```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	<code>manager</code>	Administrators or local user group members with execution rights for this command.

troubleshoot multicast

Syntax

```
troubleshoot multicast
  config
  health [verbose]
  igmp [config|{health[verbose]}|operations]
  mld [config|{health[verbose]}|operations]
  pim [config|{health[verbose]}|operations]
  pim6 [config|{health[verbose]}|operations]
  operations
```

Description

Use the `troubleshoot multicast` commands to troubleshoot multicast networks while focusing on specific multicast settings. This targeted approach to troubleshooting helps identify and fix issues in large, complex networks that include multiple virtual routing domains (VRFs) and interfaces. AOS-CX troubleshooting works with both IPv4 and IPv6 multicast protocols, and can address multicast group membership problems with IGMP and MLD, and verify the routing paths for multicast traffic using PIM or PIM6.

Parameter	Description
<code>config</code>	Issue the troubleshoot multicast config command to display the results of configuration validation checks for all multicast features, or include this parameter after a specific multicast component (for example, troubleshoot multicast igmp config) to display configuration validation information for that component only.
<code>health [verbose]</code>	The troubleshoot multicast health command displays basic health information for all multicast features, including error messages, message IDs, severity levels, and when the messages were reported. Include the health parameter after an multicast feature name (for example, troubleshoot multicast igmp health) to display health troubleshooting information for that feature only.

Parameter	Description
	Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
<pre>igmp [config]{health[verbose]} operations]</pre>	Troubleshoot IGMP multicast traffic. By default, the troubleshoot multicast igmp command displays information for the feature configuration, health and operations. Include the config, health, or operations parameters to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
<pre>mld [config]{health[verbose]} operations]</pre>	Troubleshoot MLD multicast traffic By default, the troubleshoot multicast mld command displays information for the feature configuration, health and operations. Include the config, health, or operations parameters to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
<pre>operations]</pre>	Display comprehensive multicast feature diagnostics with a detailed analysis. Issue the troubleshoot multicast operations command to display comprehensive multicast feature diagnostics for all multicast features, or include this parameter after a specific multicast component (for example, troubleshoot multicast igmp operations) to display comprehensive feature diagnostics for that component only.
<pre>pim [config]{health[verbose]} operations]</pre>	Troubleshoot PIM multicast traffic. By default, the troubleshoot multicast pim command displays information for the feature configuration, health and operations. Include the config, health, or operations parameters to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.



NOTE

This parameter is not supported on 4100i, 6000, 6100, and 6300L Switch series.

Parameter	Description
<pre>pim6 [config {health[verbose]} operations]</pre>	Troubleshoot PIM6 multicast traffic. By default, the troubleshoot multicast pim6 command displays information for the feature configuration, health and operations. Include the config, health, or operations parameters to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.  NOTE This parameter is not supported on 4100i, 6000, 6100, and 6300L Switch series.

Usage

If you do not specify a protocol, the **troubleshoot multicast** command will display troubleshooting information for all protocols. If you do not specify whether you want the output of this command to display **health**, **configuration** or **operations** troubleshooting, the output of this command will display all three types of information.

Filter the output of a multicast troubleshooting command by including a **parameters "<query><word>"** option after the **health**, **config**, or **operations** parameters for some multicast troubleshooting commands. The "**<query><word>**" string must be enclosed in double quotation marks ("). More than one set of filter parameters can be used in a single command, but all filter parameters should be enclosed within a the same set of quotation marks, for example, " - - group <group - ip> - - source <source - ip>".

- **parameters " - - group <group - ip>":** When troubleshooting IGMP, MLD, PIM or PIM6 traffic, use these optional parameters to pinpoint problems affecting a specific multicast group.
- **parameters " - - source <source - ip>":** When troubleshooting IGMP, MLD, PIM or PIM6 traffic, use these optional parameters to identify issues related to a specific source sending multicast traffic.
- **parameters " - - vrf <vrf - name>":** For PIM and PIM6 traffic, narrow down troubleshooting to a particular virtual routing domain.
- **parameters " - - vlan <downstream-vlan - id>":** When troubleshooting IGMP or MLD issues, use this optional parameter to check how multicast behaves in a specific VLAN on the network.
- **parameters " - - rp <group - ip>":** When troubleshooting PIM and PIM6 traffic, use this optional parameter to ensure that Rendezvous Point (RP) settings and operations are correct for a specific multicast group.

Examples

The following example displays advanced multicast troubleshooting details for a multicast group with the IP address **192.0.2.1**.

```
switch(config)# troubleshoot multicast igmp operations parameters "--group 192.0.2.1"
```

The following example displays multicast troubleshooting operations details for a multicast group with the IP address **192.0.2.1** on VLAN **10**. This example contains more than one parameter, so both the **--group** and **--vlan** parameter options are contained within the same set of double quotation marks.

```
switch(config)# troubleshoot multicast igmp operations parameters "--group 192.0.2.1 --vlan 10"
```

The following example displays multicast troubleshooting operations details for a multicast group with the IP address **192.0.2.1** with traffic originating from a source with the IP address **10.1.1.100**. This example contains more than one parameter, so both the **--group** and **--source** parameter options are contained within the same set of double quotation marks.

```
switch(config)# troubleshoot multicast igmp operations parameters "--group 192.0.2.1 --source 10.1.1.100"
```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

troubleshoot flow

Syntax

```
troubleshoot flow operations parameter "--source-ip-address <ip> --destination-type ipv4|ipv6 --destination-ip-address <ip> --transport-protocol <protocol> --source-l4-port <port> --destination-l4-port <port> --vrf <vrf>"
```

Description

Add in description here

Parameters

Parameter	Description
<code>--source-ip-address <ip></code>	Troubleshoot issues for traffic from the specified source IP address.
<code>--destination-type ipv4 ipv6</code>	Specify whether the traffic to be analyzed is going to an IPv4 or IPv6 destination.
<code>source-ip-address <ip></code>	Troubleshoot issues for traffic going to the specified destination IP address.
<code>--transport-protocol <protocol></code>	Specify a transport protocol ID to troubleshoot issues for the specified protocol.
<code>--source-l4-port</code>	Troubleshoot issues for traffic from the specified port
<code>--destination-l4-port</code>	Troubleshoot issues for traffic going to the specified port
<code>--vrf <vrf></code>	Troubleshoot issues for traffic on the specified VRF.

Usage

Delete this section if there is no usage info

Example

```
6300# troubleshoot flow operations parameters "--source-ip-address
100.0.0.9 --destination-type ipv4 --destination-ip-address 200.0.0.9 --
transport-protocol 17 --source-l4-port 45126 --destination-l4-port 22 --vrf
default"
3077 -----
3078 Operations troubleshoot
3079 -----
3080
3081 flow : Interface 1/1/1 and queue 2 has queue congestion
3082         Egress interface: 200.0.0.1 -> 1/1/1
3083
3084 =====
3085 Troubleshoot FLOW executed successfully
3086 =====
```

Release	Modification
release	description

Command Information

Platforms	Command context	Authority
6300 6400	manager	Administrators or local user group members with execution rights for this command.

troubleshoot flow-telemetry

Syntax

```
troubleshoot flow-telemetry
  app-recognition [ config | {health [verbose]} ]
  ipfix [health [verbose]]
```

Description

Perform troubleshooting on flow telemetry issues for basic health, incorrect configurations, application performance, application recognition failures, and ipfix flow exporters and flow records. When you issue the **troubleshoot flow-telemetry** command for switches that support both application recognition and IPFIX, the output of the command displays information for both features.

Parameter	Description
app-recognition [config [health [verbose]]]	Troubleshoot the application recognition feature. By default, the troubleshoot flow-telemetry app-recognition command displays information for the feature configuration and health. Include the config or health parameters to display information of that type only. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.

Parameter	Description
<code>ipfix health [verbose]</code>	Display basic health information for the IP Flow Information Export (IPFIX) feature, including the error message, message ID, severity level, and when the message was last reported. Include the optional verbose keyword after the health parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.

Examples

The following example displays basic flow telemetry troubleshooting details. The output of this command can vary depending upon switch model and configuration.

```
switch# troubleshoot flow-telemetry

Troubleshoot CLI execution is in progress, please wait...
-----
Health troubleshoot
-----
Component: ipfix
-----
-----
Message ID
Message                                     Last-
Reported(UTC)    Severity
-----
-----
410              The configuration of flow exporter 'test' is not
accepted.        2026-04-29 19:46:36  WARNING
-----
-----
Configuration troubleshoot
-----
-----
app-recognition : App-recognition is not configured on any ports.

=====
Troubleshoot flow-telemetry executed successfully
=====
```

Showing IPFIX health information:

```
switch# troubleshoot flow-telemetry ipfix
-----
Health troubleshoot
```

```
-----  
Component: ipfix  
-----
```

```
-----  
Message ID
```

```
Message
```

```
                Last-Reported(UTC)    Severity  
-----
```

```
-----  
420          STRICT_MODE: IPFIX Flow Congestion Monitor is blocked due to  
invalid or missing feature pack.  2026-04-29 01:00:56  INFO
```

```
=====  
Troubleshoot ipfix executed successfully  
=====
```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
6300	manager	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6400		

troubleshoot proactive

Syntax

```
troubleshoot proactive {enable system}|{action collect support-files tftp://  
<IP>/<FILE> [vrf <vrf>]}  
no troubleshoot proactive enable|{action collect support-files}
```

Description

This command enables and configures the proactive system troubleshooting mechanism. If this feature is enabled and the switch detects anomaly conditions (for example, the switch's capacity thresholds are exceeded, protocol error counts are increasing beyond normal limits, or resource alerts are triggered.) the switch will automatically launch a system troubleshooting session.

Use the **collect** parameter to allow a proactive troubleshooting session to collect a bundle of support files support and send them to a remote server via TFTP.



NOTE

To prevent excessive troubleshooting sessions for a reoccurring event, if a second anomaly event occurs within five minutes of a previous event of the same type, the second event logs an info message but does not launch a new troubleshooting event.

Parameters

Parameter	Description
<code>enable system</code>	Enable proactive system troubleshooting.
<code>action collect support-files tftp://<IP>/<FILE></code>	Collect a set of support-files and send them to a remote server via TFTP when a proactive troubleshooting session is launched . Include the IP address of the remote server and the name to give the file bundle.
<code>[vrf <vrf-name>]</code>	Optional. Include the vrf parameter and a vrf name to send troubleshooting files for the selected VRF.

Usage

The proactive troubleshooting feature can collect the following system failure event IDs.

Table 1.

Event ID	Name	Description
1201	SUPPORTABILITY_DAEMON_CRASH	Daemon crash detected
1206	REBOOT_LOG	Device reboot recorded in reboot log
1207	SYS_MEM_NORMAL	System memory usage returned to normal
1208	SYS_MEM_HIGH	High system memory usage detected

Event ID	Name	Description
1209	SYS_MEM_CRITICAL	Critical system memory usage detected
6310	SYSMON_SYS_CPU_HIGH	High system CPU utilization detected
6311	SYSMON_SYS_CPU_CRITICAL	Critical system CPU utilization detected

Example

The following example enables the proactive troubleshooting features and defines the path to where support files should be sent.

```
switch (config)# troubleshoot proactive
switch(config-troubleshoot-proactive)# enable
switch(config-troubleshoot-proactive)# action collect support-files tftp://
10.0.0.2/support_files.tar.gz vrf default
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400	config-troubleshoot-proactive-context	

troubleshoot port-access authentication

Syntax

```
troubleshoot port-access authentication
    config
    health [verbose]
    operations
```

```
parameters "<word>"
```

Description

Display detailed troubleshooting information for port - access authentication, including errors and the reasons those errors were triggered.

Parameter	Description
config	Troubleshoot issues with an incomplete or incorrect configuration
health [verbose]	troubleshoot port-access health issues. Include the optional verbose parameter to display verbose troubleshoot logs with additional troubleshooting information.
operations	Troubleshoot advanced port access operation issues
parameters "<word>"	Reserved for future use.

Usage

Port-access features are only supported on bridged ports. Note that when the an interface is changed from a layer-2 interface to a layer-3 interface or added as a member of a LAG interface, all port-access configurations are cleared.

Examples

Troubleshooting the port-access configuration:

```
switch(config)# troubleshoot port-access config
-----
Configuration troubleshoot
-----

Component: authentication
-----
-----
Message ID
Message
Severity
-----
-----
304          802.1X is enabled on the interface 1/1/17 but not enabled
globally.    WARNING
304          802.1X is enabled on the interface 1/1/21 but not enabled
globally.    WARNING
```

```
304      802.1X is enabled on the interface 1/1/22 but not enabled
globally.  WARNING
```

Troubleshooting the health of the port-access feature:

```
switch (config)# troubleshoot port-access health
-----
Health troubleshoot
-----
Component: authentication
-----
Message ID  Message                               Last-Reported(UTC)  Severity
-----
314         Daemon port-accessd crashed.    2026-04-23 05:20:04  ERROR

Component: authentication - No issues found

=====
Troubleshoot port-access executed successfully
=====
```

Performing port-access operations troubleshooting:

```
switch# troubleshoot port-access authentication operations
-----
Operations troubleshoot
-----
authentication  : The port-access enabled port(s) 1/1/24,1/1/19 are linked
up without any MAC learns. The
clients connected are likely silent clients. Try solutions such as Wake on
LAN and client probe to trigger the clients to authenticate on the port.
802.1X authentication failed for client(s) with MAC address
00:50:56:96:aa:aa likely due to a large EAP
frame. Adjust the EAP-TLS fragment size in the RADIUS message to avoid
oversized RADIUS frames.

=====
Troubleshoot AUTHENTICATION executed successfully
=====
```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	manager	Administrators or local user group members with execution rights for this command.

troubleshoot radius

Syntax

```
troubleshoot radius tracking health [verbose]
```

Description

Perform diagnostics to identify RADIUS tracking issues.

Parameters

Parameter	Description
verbose	Include the optional verbose keyword after the health parameter to inspect detailed diagnostic progress and confirm whether additional errors were suppressed.

Example

```
switch# troubleshoot radius tracking health

-----
Health troubleshoot
-----

-----
Message ID      Message                               Last-Reported
Severity
-----
2601            Daemon radius-srv-trkd crashed.      2026-04-04 10:32:26  ERROR#
```

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authorit
all platforms	config	Administrators or local user group members with execution rights for this command.

troubleshoot system

Syntax

```
troubleshoot system
  capacity-check|interfaces|resources|services
  health [verbose]
  operations [parameters "<word>"]
```

Description

Perform basic troubleshooting for system-level components.

Parameter	Description
capacity-check	Perform diagnostic checks on system capacities and their values.
interfaces	Perform diagnostic checks on physical interfaces.
resources	Perform diagnostic checks on system CPU and memory utilization.
services	Perform diagnostic checks on the status of system daemons/services.
health [verbose]	Display basic information for system errors, including the error message, message ID, severity level, and when the message was last reported. The troubleshoot system health command displays troubleshooting information for all system health components. Include the health parameter after a specific system component.

Parameter	Description
	nent name to display basic troubleshooting information for that component only. Include the optional verbose keyword after the health parameter to inspect detailed diagnostic progress and confirm whether additional errors were suppressed.
operations	Display a comprehensive system operations diagnostic with detailed analysis
parameters "<word>"	Reserved for future use.

Examples

Troubleshooting system capacity:

```
6300# troubleshoot system capacity-check health
-----
Health troubleshoot
-----
Component: capacity-check
-----
-----
Message ID
Message
Last-Reported(UTC)    Severity
-----
-----
1506      System capacity vxlan_interfaces has reached maximum
threshold      2026-03-27 00:30:04  ERROR
1506      System capacity traffic_insight_instance has reached maximum
threshold 2026-03-27 00:30:04  ERROR

Component: resources - No issues found
Component: interfaces - No issues found
Component: services - No issues found
Component: capacity-check - No issues found
=====
Troubleshoot capacity-check executed successfully
=====
```

Troubleshooting system operations status:

```
6300(config)# troubleshoot system operations
-----
Operations troubleshoot
```

```

-----
system : Summary:
        Active Alerts           : 1
        Severity Status         : 0 Errors,1 Warnings,0 Critical

Alerts:
  [WARNING][1507] High CPU usage detected: 115% (threshold: 60%)

Root Cause:
  Error ID: 1507
    - CPU usage exceeded threshold
    - Last Updated: Wed Mar 25 01:44:36 2026
    - Top 5 Consumers: journalctl (115.00%), ovssdb-server
(80.00%), switchd_agent0 (30.00%), poe-hald (25.00%), portd (5.00%)

```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config manager	Administrators or local user group members with execution rights for this command.

troubleshoot tunnel

Syntax

```

troubleshoot tunnel
  ubt health [verbose]
  vxlan [{health [verbose]}|config|operations] [parameters]
"<query><word>"

```

Description

Diagnose and resolve issues related to User-Based Tunneling (UBT) or VXLAN tunnel configuration and operation, including daemon health issues, application performance degradation, and incorrect, unsupported or non-recommended configurations.

Parameter	Description
health [verbose]	Display basic health information for UBT or VXLAN tunnel errors on the network, including the error message, message ID, severity level, and when the message was last reported. Include the optional verbose parameter to display detailed troubleshooting progress.
config	Display information about any configuration errors that might be impacting VXLAN tunnels.
operations	Display comprehensive feature diagnostics with detailed analysis information for VXLAN tunnels, including checks for: • VXLAN interfaces • VNIs • VTEPs • VTEP - VNI links • route reflector (RR) & ECMP groups
parameters "<query><word>"	Reserved for future use.

Examples

The following example shows basic health troubleshooting for a VXLAN tunnel in a **down** state.

```
switch# troubleshoot tunnel vxlan health
-----
Health troubleshoot
-----
vxlan  : 202: Vxlan Interface is down
=====
Troubleshoot tunnel executed successfully
=====
```

The following example shows troubleshooting details for a VXLAN tunnel that does not have a configured VNI.

```
switch# troubleshoot tunnel vxlan operations
-----
Operations troubleshoot
-----
vxlan  : -----VXLAN INTERFACE CHECK-----
        VXLAN interface is up and configured with local_ip : 1.1.1.1
```

```
-----VNI CHECK-----
VNIs are not configured
=====
Troubleshoot VXLAN executed successfully
=====
```

The following example shows that there are currently no VXLAN configuration errors that need to be addressed:

```
switch# troubleshoot tunnel config
-----
Configuration troubleshoot
-----
=====
Troubleshoot tunnel executed successfully
=====
```

The following example shows health information for user-based tunnels.

```
config# troubleshoot tunnel ubt health
-----
Health troubleshoot
-----
Component: ubt
-----
-----
Message ID
Message
Last-Reported(UTC)    Severity
-----
-----
505      Operational state of zone8 zone is DOWN due to Controller is
unreachable.  2026-04-29 01:51:55  ERROR
=====
Troubleshoot ubt executed successfully
=====
```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
all platforms	manager	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

[Accessing HPE Aruba Networking Support](#)

[Accessing Updates](#)

[Warranty Information](#)

[Regulatory Information](#)

[Documentation Feedback](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	<u>https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html</u>
AOS - CX Switch Software Documentation Portal	<u>https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html</u>
HPE Aruba Networking Support Portal	<u>https://networkingsupport.hpe.com/home</u>
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	<u>https://www.hpe.com/psnow/doc/a50011948enw</u>

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end - of - life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.